



CNS Module - Part 1

Questions

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NERD

CNS Module - Part 1

Questions

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PREFACE

Dear colleagues,

Presenting you with the third edition of **NERD | CNS Module (Questions)**.
This edition's content is made out to help you solve MCQs & Short Essay easily.

Hoping this book improves your studying experience to be as enjoyable and as fruitful as possible

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NERD | CNS Module - Part 1 (Questions)

By: Omar Elazab

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First Edition 2023

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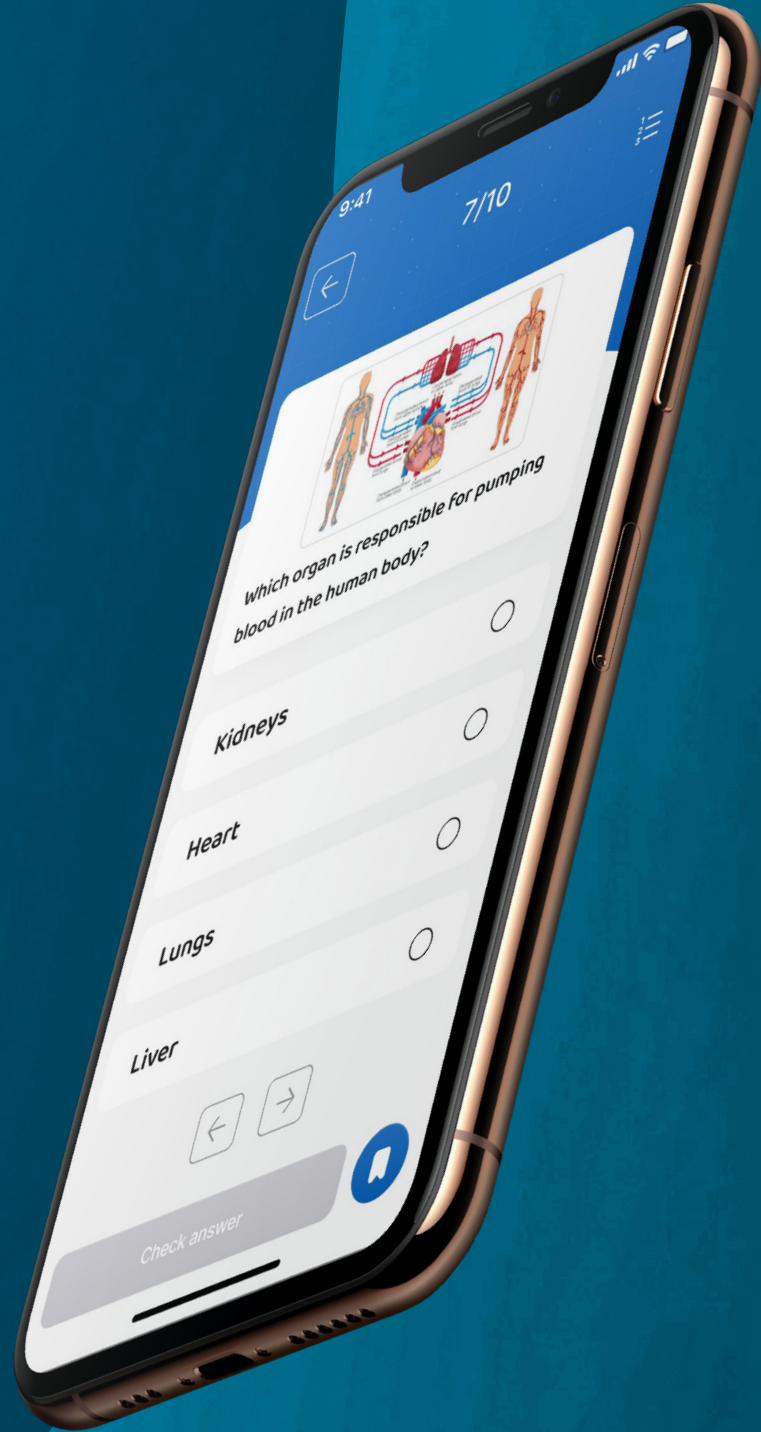


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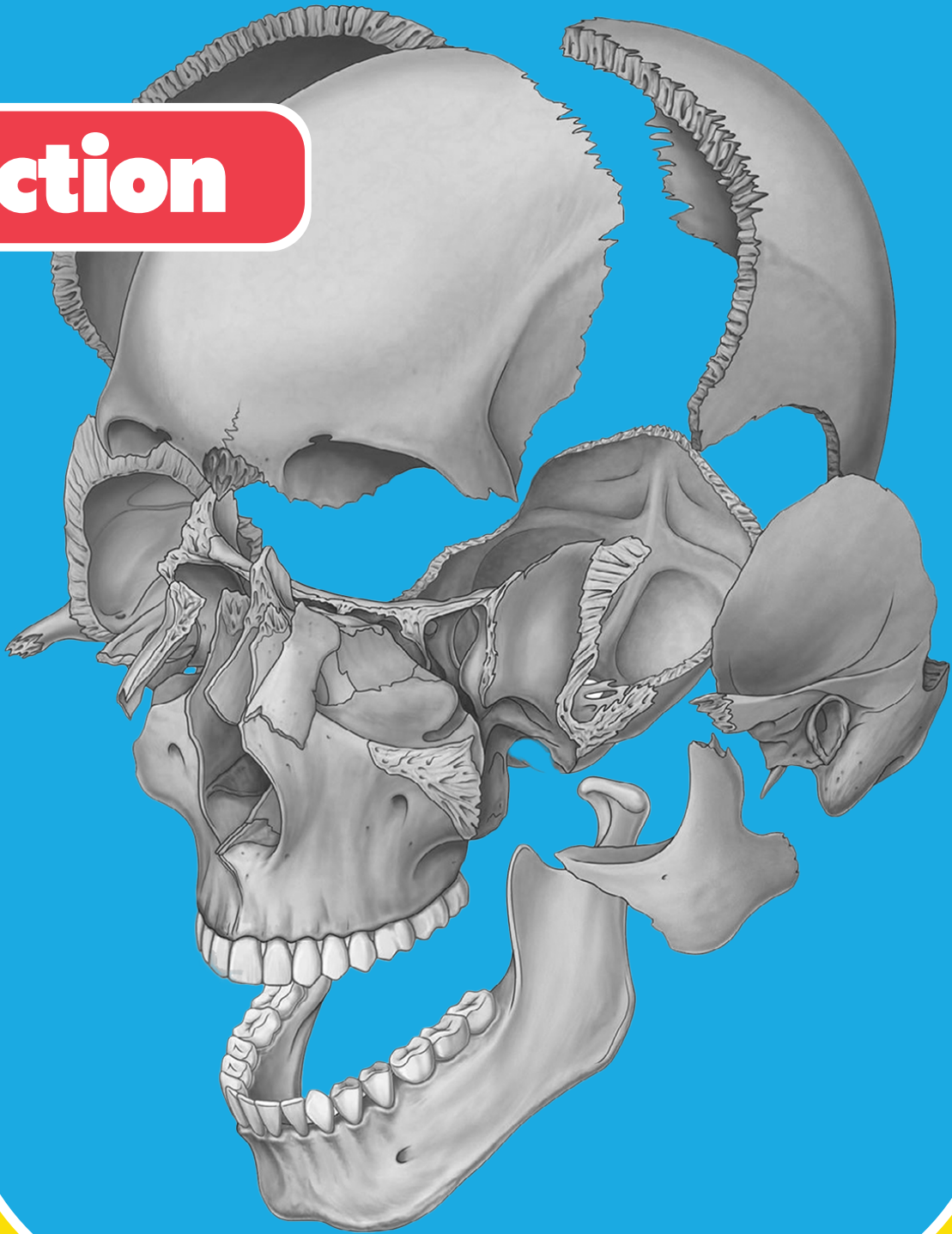
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MCQ

1

Section



Anatomy



1. Pain from ala of nose is carried by which nerve?

- a. Infratrochlear
- b. Infraorbital
- c. External nasal
- d. Buccal
- e. Auriculotemporal

2. Regarding the scalp the following statements are true Except:

- a. Infection from scalp wounds can spread to superior sagittal sinus.
- b. Its blood vessels run in the loose subaponeurotic space.
- c. Its skin, superficial fascia and aponeurosis move as one layer.
- d. Frontal bellies of occipito-frontalis can corrugate skin of forehead.
- e. Occipital bellies are supplied by facial nerve.

3. The large blood vessels of scalp run into

- a. Skin.
- b. Subcutaneous connective tissue
- c. Epicranial aponeurosis.
- d. Loose areolar tissue.
- e. Pericranium.

4. Which of the following veins unite with a posterior a branch of the retromandibular vein to form external jugular vein?

- a. Posterior Auricular
- b. Superficial Temporal.
- c. Facial.
- d. Transverse facial.
- e. Deep facial.

5. The plane of movement of scalp is between:

- a. Skin & epicranial aponeurosis
- b. Epicranial aponeurosis & pericranium.
- c. Skin & subcutaneous fat
- d. Pericranium & skull
- e. Connective tissue layer & aponeurosis.

6. Which of the following nerves supplying scalp and face is not derived from trigeminal nerve?

- a. Great auricular
- b. Auriculotemporal.
- c. Zygomaticotemporal.
- d. Supraorbital
- e. Supratrochlear.

7. Loss of sensation from the lower eye lid indicates injury of which nerve?

- a. Lacrimal
- b. Infratrochlear
- c. Infraorbital
- d. Zygomaticofacial
- e. Zygomaticotemporal

8. Which of the following muscles can produce transverse wrinkles in the forehead?

- a. Corrugator supercilii.
- b. Frontalis.
- c. Orbicularis oculi.
- d. Zygomaticus minor.
- e. None of the above.

9. Retromandibular vein:

- a. Is formed by union of maxillary & superficial temporal veins.
- b. Begins inside parotid gland.
- c. Is deep to facial nerve inside parotid gland.
- d. Divides into 2 branches.
- e. All the above.

10. Bleeding beneath the pericranium of the scalp:

- a. Spreads underneath all of the scalp.
- b. Makes a swelling that takes the shape of the underlying bone.
- c. Reaches to the upper eyelids.
- d. Reaches to the back of neck.
- e. Reaches the extradural space.

11. The motor nerve of the scalp behind the auricle is:

- a. Buccal branch of facial nerve.
- b. Posterior auricular branch of facial nerve.
- c. Zygomatic branch of facial nerve:
- d. Temporal branch of facial nerve.
- e. Buccal branch of facial nerve.

12. Buccinator:

- a. Is one of the muscles of mastication.
- b. Is supplied by buccal branch of mandibular nerve.
- c. Arises from both mandible and maxilla.
- d. Is covered by investing layer of fascia.
- e. Accumulates food in vestibule of mouth.

13. Internal carotid artery supplies:

- a. Chin
- b. Forehead
- c. Cheek
- d. Non-hairy temple
- e. Hairy temple

14. The upper lip:

- a. Is supplied by facial artery.
- b. Is supplied by maxillary nerve.
- c. Lies in the dangerous area of face.
- d. Its philtrum is obliterated in infections.
- e. All the above is true.

15. Which of the following contains parasympathetic postganglionic fibers?

- a. Lesser petrosal nerve.
- b. Glossopharyngeal nerve.
- c. Tympanic nerve.
- d. Auriculotemporal nerve.
- e. Great auricular nerve.

16. Pulse felt at antero-inferior angle of masseter is from which artery?

- a. Facial artery.
- b. Lingual artery.
- c. Superficial temporal artery.
- d. Submental artery.
- e. External carotid artery.

17. Lymph from central part of upper lip drains directly to:

- a. Submandibular lymph nodes.
- b. Submental lymph nodes.
- c. Parotid lymph nodes.
- d. Mastoid lymph nodes.
- e. Deep cervical lymph nodes.

18. The branch of the mandibular nerve that supplies the scalp is the:

- a. Zygomaticotemporal nerve
- b. Lesser occipital nerve
- c. Supraorbital nerve
- d. Third occipital nerve
- e. Auriculotemporal nerve

19. The scalp is:

- a. Supplied by the great auricular nerve in front of the auricle.
- b. Supplied by the supratrochlear and supraorbital arteries branches of the facial artery.
- c. Supplied posteriorly by the greater occipital nerve from the ventral ramus of C2.
- d. Supplied behind the auricle by the lesser occipital nerve from the ventral ramus of C2.
- e. Drained by supratrochlear and supraorbital veins, which unite to form the superior ophthalmic vein.

20. Facial artery:

- a. Is a branch of internal carotid artery
- b. Enters face at antero-inferior angle of Buccinator
- c. Has a tortuous course
- d. Ends at side of nose.
- e. Ends by anastomosing with supraorbital artery.

21. The nerve supply of the muscle of the scalp is: .

- a. Trigeminal nerve.
- b. Great auricular nerve.
- c. Glossopharyngeal nerve
- d. Facial nerve.
- e. Great occipital nerve.

22. Infection from dangerous area of face may be transmitted to the following venous sinus:

- a. Superior petrosal.
- b. Sigmoid.
- c. Inferior sagittal.
- d. Superior sagittal.
- e. Cavernous.

23. The lowermost soft part of the auricle of ear is called:

- a. Promontory
- b. Tragus
- c. Pyramid
- d. Lobule
- e. Concha

24. In which of the following layers of the scalp does infection tend to spread:

- a. Skin.
- b. Connective tissue layer.
- c. Aponeurotic layer
- d. Loose areolar tissue layer.
- e. Periosteal layer.

25. Pain from upper lip is carried by.

- a. Buccal
- b. Mental
- c. Infraorbital
- d. Zygomaticofacial
- e. Marginal mandibular

26. Infection of lateral part of the upper eye lid leads to enlargement of this group of lymph nodes:

- a. Submental group
- b. Mastoid group
- c. Submandibular group
- d. Parotid group
- e. Occipital group

27. Bleeding in which layer of scalp causes black eye?

- a. Skin.
- b. Connective tissue.
- c. Aponeurosis.
- d. Loose areolar tissue.
- e. Pericranium.

28. The nerve that carries the sensation for the lower eyelid is:

- a. Supraorbital
- b. Infraorbital
- c. Buccal
- d. Mental
- e. Auriculotemporal

29. Regarding Scalp, all are true Except:

- a. Its wounds bleed profusely.
- b. Its wounds gape if the aponeurosis is cut.
- c. It has no sebaceous glands.
- d. Its veins communicate with the intracranial venous sinuses.
- e. It contains a muscle of facial expression.

30. The non-hairy region of the temple is supplied by which of the following nerves?

- a. Infratrochlear
- b. Infraorbital
- c. Auriculotemporal
- d. Zygomaticotemporal
- e. Zygomaticofacial

31. The main sensory nerve to the lower lip is the:

- a. Buccal.
- b. Cervical of facial.
- c. Mental.
- d. Great auricular.
- e. None of the above.

32. Infection of scalp area behind the ear affects which group of LNs

- a. Submental group
- b. Submandibular group
- c. Deep parotid group
- d. Superficial parotid group
- e. Occipital group

33. Which of the following is sensory to the scalp behind the auricle?

- a. Auriculotemporal nerve.
- b. Superficial temporal nerve.
- c. Great auricular nerve.
- d. Zygomaticotemporal nerve.
- e. Buccal nerve.

34. All of the following regarding orbicularis oris muscle is correct Except:

- a. It is a muscle of facial expression.
- b. It is supplied by facial nerve.
- c. Its fibers form complete circles around the mouth.
- d. It can close the oral (labial) fissure.
- e. It helps in speech.

35. The nerve supplying the buccinator muscle is:

- a. Mandibular
- b. Facial
- c. Glossopharyngeal
- d. Vagus
- e. Cranial accessory

36. The following is a branch of the superficial temporal artery in the face:

- a. Supraorbital.
- b. Supratrochlear.
- c. Infraorbital.
- d. Transverse facial.
- e. Inferior labial.

37. The 2 frontal bellies:

- a. Originate from skin of eyebrows.
- b. Are separated from each other.
- c. Are supplied by supraorbital nerve.
- d. Form the only muscle of face.
- e. Are attached to superior temporal line.

38. Regarding buccinator muscle, all the following statements are true Except:

- a. It arises from both maxilla and mandible.
- b. The pterygomandibular raphe is a common origin for this muscle and the middle constrictor.
- c. Its middle fibers decussate at the modiolus.
- d. It is covered by buccopharyngeal fascia.
- e. It compresses the cheeks against the teeth.

39. Lacrimal part of orbicularis oculi

- a. Is attached to medial palpebral ligament
- b. Closes the eye gently.
- c. Is attached to lateral palpebral raphe
- d. Is supplied by auriculotemporal nerve.
- e. Is a part of orbicularis oris.

40. Regarding lips all are true Except:

- a. The main sensory supply to upper lip is facial nerve.
- b. The main sensory supply to lower lip is mental nerve.
- c. Their muscles are formed mainly by orbicularis oris.
- d. Their blood supply is through facial artery.
- e. In facial palsy lips can't be dilated.

41. The branch of facial nerve that supplies the platysma is the

- a. Temporal
- b. Zygomatic
- c. Buccal
- d. Mandibular
- e. Cervical

42. Mandibular nerve is sensory to:

- a. Upper eyelid.
- b. Tip of nose.
- c. Lower lip.
- d. Lower eyelid.
- e. Upper lip.



1. Leptomeninges refer to the:

- a. Dura mater.
- b. Dura mater and arachnoid,
- c. Pia and arachnoid.
- d. Superficial glial membrane.
- e. Periosteum

2. Arachnoid granulations are:

- a. Prolongations of the dura mater.
- b. Sites of fluid transfer from the venous sinuses to the subarachnoid space.
- c. Sites of cerebrospinal fluid formation.
- d. Sites of fluid transfer from the subarachnoid space to the venous sinuses.
- e. A pathological feature of the arachnoid.

3. All the following statements regarding pia mater are true Except:

- a. It is an avascular membrane.
- b. It closely invests the brain.
- c. It forms the denticulate ligament.
- d. It extends along the spinal nerve roots.
- e. It ends as the filum terminale.

4. At birth the spinal cord ends at the level of:

- a. Lower border of the 1st lumbar vertebra.
- b. Upper border of the 2nd lumbar vertebra.
- c. Upper border of the 3rd lumbar vertebra.
- d. The 5th sacral vertebra.
- e. None of the above.

5. The spinal dura extends from the level of the foramen magnum to the level of which vertebra?

- a. First lumbar.
- b. Third lumbar.
- c. Second sacral.
- d. Fifth sacral.
- e. First coccygeal.

6. The spinal cord is anchored to the dura mater by the:

- a. Coccygeal ligament.
- b. Ligamentum flavum.
- c. Anterior longitudinal ligament.
- d. Denticulate ligaments.
- e. Cauda equina.

7. To do lumbar puncture in an adult a needle is introduced:

- a. Below or above tip of 4th lumbar spine
- b. Below tip of 1st lumbar spine
- c. Below tip of 2nd lumbar spine
- d. Below tip of thoracic 12 spine
- e. None of the above

8. Regarding posterior column tracts all statements are true Except:

- a. Their first neuron lies in the posterior root ganglion
- b. Fibres from lower limb are close to the middle line
- c. Axons of gracile and cuneate nuclei decussate in the medulla forming the lateral lemniscus
- d. Their lesion produces astereognosis
- e. They reach the postcentral gyrus

9. The spinal dura mater ends at the vertebral level of:

- a. S1.
- b. S2.
- c. L1.
- d. L2.
- e. L3.

10. In adults, the spinal cord ends at which vertebral level?

- a. 1st lumbar.
- b. 3rd lumbar.
- c. 1st sacral.
- d. 3rd sacral.
- e. 1st coccygeal.

11. Compression of which segment of the spinal cord would occur by a tumor at 4th thoracic vertebra?

- a. T3
- b. T4
- c. T5
- d. T6
- e. T7

12. Which of the following is not present in the sacral canal?

- a. Filum terminale.
- b. Cauda equina.
- c. Cerebrospinal fluid.
- d. Ligamentum denticulatum.
- e. Tube of dura & arachnoid mater.

13. The spinal dura extends from the level of the foramen magnum to the level of which vertebra?

- a. First lumbar.
- b. Third lumbar.
- c. Second sacral.
- d. Fifth sacral.
- e. First coccygeal.

14. In adults the spinal cord usually terminates at the:

- a. Upper border of the 12th thoracic vertebra.
- b. Lower border of the 1st lumbar vertebra.
- c. Upper border of the 4th lumbar vertebra.
- d. Level of the 5th lumbar vertebra.
- e. End of sacral canal.

15. At birth the spinal cord ends at the level of:

- a. Lower border of the 1st lumbar vertebra.
- b. Upper border of the 2nd lumbar vertebra.
- c. Upper border of the 3rd lumbar vertebra.
- d. The 5th sacral vertebra.
- e. None of the above.

16. At birth, the conus medullaris ends at which vertebral level?

- a. 1st lumbar.
- b. 3rd lumbar.
- c. 1st sacral.
- d. 3rd sacral.
- e. 1st coccygeal.

17. A disc prolapsed between T5 and T6 vertebrae would exert pressure upon:

- a. T4 nerve.
- b. T5 nerve.
- c. T6 nerve.
- d. T7 nerve.
- e. T8 nerve.

18. As regards the spinal cord; choose the INCORRECT statement:

- a. In adults, it ends at lower border of L1.
- b. It is anchored to the dura mater by denticulate ligaments.
- c. Its covering dura mater below the level of L1 forms filum terminale.
- d. It has eight cervical segments.
- e. Lumbar puncture may be done between L3 and L4.

19. The upper 9 thoracic vertebrae contain:

- a. Upper 6 thoracic segments of spinal cord.
- b. Upper 8 thoracic segments of spinal cord.
- c. Upper 10 thoracic segments of spinal cord.
- d. Lower 10 thoracic segments of spinal cord.
- e. All thoracic segments of spinal cord.

20. Which of the following is purely sensory?

- a. Ventral ramus of spinal nerve.
- b. Ventral root of spinal nerve.
- c. Dorsal root of spinal nerve.
- d. Dorsal ramus of spinal nerve.
- e. Spinal nerve trunk.

21. As regards the spinal nerves, choose the CORRECT statement:

- a. Its posterior root contains autonomic fibers.
- b. Its ventral root carries a root ganglion.
- c. The dendrites of its posterior root pass centrally.
- d. Its autonomic fibers arise from anterior horn.
- e. Cells of spinal ganglion are pseudounipolar.

1. Lesion of which cerebral area in left hemisphere causes motor aphasia?

- a. Inferior parietal lobule.
- b. Superior temporal gyrus.
- c. Medial frontal gyrus.
- d. Inferior frontal gyrus.
- e. Superior frontal gyrus.

2. Persistent pain in big toe is received in:

- a. Lower part of post central gyrus.
- b. Upper part of post central gyrus.
- c. Anterior part of paracentral lobule.
- d. Posterior part of paracentral lobule.
- e. Superior lip of lateral sulcus.

3. Motor aphasia is due to lesion of the following area:

- a. Insula.
- b. Broca's area.
- c. Cuneus.
- d. Wernike's area.
- e. Uncus.

4. Changes in personality and judgment occur in lesion of:

- a. Frontal lobe.
- b. Parietal lobe.
- c. Occipital lobe.
- d. Insula.
- e. Temporal lobe.

5. Injury to the left paracentral lobule leads to paralysis of:

- a. Left hand.
- b. Right hand.
- c. Left foot.
- d. Right foot.
- e. Right face.

6. Injury to the right paracentral lobule leads to paralysis of:

- a. Left side of face.
- b. Right side of face.
- c. Left leg.
- d. Right upper limb.
- e. Right foot.

7. Changes in Judgment & intelligence occur in lesion of:

- a. Frontal lobe.
- b. Parietal lobe.
- c. Temporal lobe.
- d. Occipital lobe.
- e. Limbic lobe.

8. A lesion in the Broca's area of the cerebral hemisphere leads to:

- a. Motor aphasia.
- b. Sensory aphasia.
- c. Hemiplegia.
- d. Return of grasp reflex.
- e. Astereognosis.

9. Superior parietal lobule.

- a. Return of grasp reflex.
- b. Loss of conjugate eye deviation.
- c. Astereognosis.
- d. Motor aphasia.
- e. Personality changes.

10. Broca's area:

- a. Is the sensory speech center.
- b. It is located in middle frontal gyrus.
- c. It is present in the dominant hemisphere.
- d. All of the above.
- e. None of the above.

11. Astereognosis & receptive aphasia characterize lesions in:

- a. Dominant parietal lobe.
- b. Frontal lobe.
- c. Temporal lobe.
- d. Occipital lobe.
- e. Non dominant parietal lobe.

12. Cortical areas in temporal lobe include all the following Except:

- a. Part of facial recognition area.
- b. Part of Wernicke's area.
- c. Primary auditory area.
- d. Secondary auditory area.
- e. Area for control of involuntary conjugate lateral movements of eye.

13. The facial recognition areas are found in the undersurface of:

- a. Frontal lobes.
- b. Temporal lobes only.
- c. Occipital lobes only.
- d. Temporal & occipital lobes.
- e. None of the above.

14. A patient presents with inability to understand words and speaks non understandable words. Lesion is most likely in Brodmann areas number:

- a. 1,2,3.
- b. 44,45.
- c. 17,18.
- d. 41,42.
- e. 39,40.

15. Pain in right thumb is received in:

- a. Lower part of right post central gyrus.
- b. Upper part of right post central gyrus.
- c. Lower part of left post central gyrus.
- d. Upper part of left post central gyrus.
- e. Posterior part of left paracentral lobule.



**1. All about the gracile tract is true Except:**

- a. Conveys tactile discrimination
- b. Present in the whole length of the spinal cord
- c. Carries proprioception from the urinary bladder
- d. Cervical fibres are most medial
- e. Its 2nd neuron is the gracile nucleus

2. The following tracts cross to the opposite side Except:

- a. Rubrospinal
- b. Posterior spinocerebellar
- c. Lateral spinothalamic
- d. Tectospinal
- e. Ventral spinothalamic

3. Select the correct answer about the gracile tract:

- a. Conveys proprioceptive sensation
- b. Starts at mid thoracic region
- c. Its sacral fibres are most lateral
- d. Originates from neurons of lamina I
- e. Terminates in nucleus ventralis posterior medialis (V.P.M) of thalamus

4. Select the correct statement about the cuneate tract:

- a. Carries crude touch
- b. Starts in the midthoracic region
- c. Its main 2nd neuron is the accessory cuneate nucleus
- d. Passes in the anterior limb of the internal capsule
- e. Terminates in the precentral gyrus

5. The right medial lemniscus carries sensations of:

- a. Pain from the left side of the body
- b. Conscious proprioception from the left side of the body
- c. Muscle tone from right side of the body
- d. Crude touch from the left side of the body
- e. Fine touch from right side of body

6. Regarding the ventral spinocerebellar tract select the correct answer:

- a. Found in all segments of spinal cord
- b. Passes in inferior cerebellar peduncle
- c. Arises mainly from ipsilateral nucleus dorsalis
- d. Lies in ventral white column
- e. Carries conscious proprioception to cerebellum

7. Conscious proprioceptive sensations from right lower limb are carried by:

- a. left lateral spinothalamic tract
- b. Right cuneate tract
- c. Tracts of left posterior white column
- d. Right gracile tract
- e. None of the above

8. Select the correct answer about the rubrospinal tract:

- a. Arises from the red nucleus of the same side
- b. Descends in the anterior white column of the spinal cord
- c. Relays impulses from corpus striatum to lower motor neurons
- d. Has a facilitatory effect on extensor muscles
- e. None of the above

9. Select one tract whose fibers usually remain uncrossed:

- a. Spinotectal
- b. Ventral spinocerebellar
- c. Spinoreticular
- d. Vestibulospinal
- e. Spino-olivary

10. Unconscious proprioception is carried by:

- a. Gracile and cuneate tracts
- b. Spinocerebellar tracts
- c. Ventral spinothalamic tract
- d. Lateral spinothalamic tract
- e. None of the above is true

11. The inferior cerebellar peduncle includes the following tracts Except:

- a. Reticulocerebellar
- b. Vestibulocerebellar
- c. Olivocerebellar
- d. Ventral (anterior) spinocerebellar
- e. Cuneocerebellar

12. Lateral spinothalamic tract carries the sensations of:

- a. Touch
- b. Vibration
- c. Position.
- d. Pain and temperature
- e. All of the above

13. Regarding the ascending tracts of spinal cord select the correct answer:

- a. Coccygeal fibres lie most laterally in gracile tract
- b. Ventral spinothalamic carries pain and temperature
- c. Anterior spinocerebellar ascends in inferior cerebellar peduncle
- d. Spinotectal tract ends in ipsilateral superior colliculus
- e. Cuneate tract carries sense of movement of fingers

14. The right spinal lemniscus carries sensations of:

- a. Pain and temperature from left side of the body
- b. Conscious proprioception from left side of the body
- c. Muscle tone from right side of body
- d. All of the above
- e. None of the above

15. Select the correct statement about lateral spinothalamic tract:

- a. Conveys impulses of crude touch
- b. Is located in the anterior white column
- c. Is laminated so that sacral fibers are superficial
- d. All of the above
- e. None of the above

16. All statements about ventral spinothalamic tract are true Except:

- a. Is concerned with crude touch and pressure
- b. Its fibers decussate in the medulla oblongata
- c. Terminates in ventral posterolateral nucleus of thalamus
- d. Shares in formation of spinal lemniscus
- e. Lies in the anterior white column

17. The lateral most fibers of the dorsal column tract:

- a. Cervical fibers.
- b. Upper thoracic fibers.
- c. Lower thoracic fibers.
- d. Lumbar fibers.
- e. Sacral fibers.

18. Crude touch is carried on which tract?

- a. Lateral spinothalamic T.
- b. Ventral spinothalamic T.
- c. Spinocerebellar T.
- d. Spinotectal T.
- e. Cuneate T.

19. The second order neuron in the pathway of pain from the face is:

- a. Dorsal root ganglion.
- b. Geniculate ganglion.
- c. Trigeminal ganglion.
- d. Spinal nucleus of Trigeminal nerve.
- e. Substantia gelatinosa of Rolandi.

20. The ability to recognize an unseen familiar object placed in the hand depends on the integrity of which tract?

- a. Gracile T.
- b. Cuneate T.
- c. Lateral spinothalamic T.
- d. Anterior spinothalamic T.
- e. Spinotectal T.

21. The lateral most fibers of the spinothalamic tract:

- a. Cervical fibers.
- b. Upper thoracic fibers.
- c. Lower thoracic fibers.
- d. Lumbar fibers.
- e. Sacral fibers.

22. Gracile tract:

- a. Conveys proprioceptive sensation.
- b. Starts at mid thoracic region.
- c. Its sacral fibers are most lateral.
- d. Originates from neurons of lamina I.
- e. Terminates in ventral posterior lateral nucleus of thalamus (VPMN).

23. Cuneate tract:

- a. Carries crude touch.
- b. Starts in the midthoracic region.
- c. Its main 2nd neuron is the accessory cuneate nucleus.
- d. Its 3rd neuron passes in the anterior limb of the internal capsule.
- e. Its 3rd neuron terminates in the precentral gyrus.

24. The right medial lemniscus carries sensations of:

- a. Pain from the left side of the body.
- b. Conscious proprioception from the left side of the body.
- c. Muscle tone from right side of the body.
- d. Crude touch from the left side of the body.
- e. Fine touch from right side of body.

25. Conscious proprioception from right lower limb are carried by:

- a. Left lateral spinothalamic tract.
- b. Right cuneate tract.
- c. Tracts of left posterior white column.
- d. Right gracile tract.
- e. None of the above.

26. The postcentral gyrus receives all the following sensations Except:

- a. Tactile discrimination.
- b. Touch.
- c. Muscle tone.
- d. Pain and temperature.
- e. Form.

27. All statements about ventral spinothalamic tract are true Except:

- a. Is concerned with crude touch and pressure.
- b. Its fibers decussate in the medulla oblongata.
- c. Terminates in ventral posterolateral nucleus of thalamus.
- d. Shares in formation of spinal lemniscus.
- e. Lies in the anterior white column.

28. Sense of movement and sense of position are lost after lesion of:

- a. Spinoreticular tract.
- b. Lateral spinothalamic tract.
- c. Ventral spinothalamic tract.
- d. Gracile and cuneate tracts.
- e. Spinotectal tract.

29. The cuneate tract carries which of the following sensations?

- a. Crude touch from upper limb.
- b. Fine touch from upper limb.
- c. Crude touch from upper & lower limbs.
- d. Fine touch from upper & lower limbs.
- e. Unconscious proprioception from upper and lower limbs.

30. The left spinal lemniscus carries sensations of:

- a. Pain from the left side of the body.
- b. Conscious proprioception from the left side of the body.
- c. Muscle tone from right side of the body.
- d. Crude touch from the right side of the body.
- e. Fine touch from right side of body.

31. The medial most fibers of the gracile tract are:

- a. Cervical fibers.
- b. Upper thoracic fibers.
- c. Lower thoracic fibers.
- d. Lumbar fibers.
- e. Sacral fibers.

32. A lesion in the right gracile tract leads to loss of:

- a. Sense of position of left big toe.
- b. Vibration sense of right tibia.
- c. Sense of movement of right elbow joint.
- d. Feeling of a pencil put in the right hand.
- e. Site of touch on the right iliac fossa.

33. The first order neuron of pain pathway from face is:

- a. Trigeminal ganglion.
- b. Geniculate ganglion.
- c. Gracile and cuneate nuclei.
- d. Main sensory nucleus.
- e. Mesencephalic nucleus.



1. Descending tracts in the posterior white column of the spinal cord include:

- a. Gracile tract
- b. Cuneate tract
- c. Descending autonomic fibers
- d. All of the above
- e. None of the above

2. Regarding the extrapyramidal tracts all statements are true Except:

- a. Reticulospinal tracts are present in the anterior and lateral white funiculi
- b. Tectospinal arises from inferior colliculus of the opposite side
- c. Rubrospinal arises from red nucleus of the opposite side
- d. Their lesion results in exaggerated deep reflexes
- e. Vestibulospinal arises from lateral vestibular nucleus of same side

3. Regarding lamination of lateral corticospinal select the most medial fibers:

- a. Cervical fibers
- b. Lumbar fibers
- c. Sacral fibers
- d. Coccygeal fibers
- e. Thoracic fibers

4. The corticospinal fibers are broken into bundles in:

- a. Basis pontis
- b. Basis pedunculi
- c. Tegmental part of pons
- d. Tegmentum of midbrain
- e. None of the above

5. Select the correct answer about lateral corticospinal tract:

- a. 60% of its fibers arise from precentral gyrus
- b. Present in anterior funiculus
- c. Passes in anterior limb of internal capsule
- d. Decussates in upper medulla
- e. Its injury produces a lower motor neuron lesion

6. The following areas contribute to the origin of pyramidal tract:

- a. Primary motor and premotor areas
- b. Supplementary motor area
- c. First and 2nd somatosensory areas
- d. All of the above
- e. A & B only

7. All statements about pyramidal tract are true Except:

- a. Arises from motor and premotor areas of cerebral cortex
- b. Gives corticonuclear fibers to all ipsilateral and contralateral cranial nerve nuclei
- c. Controls voluntary movements
- d. Its fibers decussate in the medulla
- e. Descends in anterior 2/3 of posterior limb of internal capsule

8. Regarding the descending tracts of the spinal cord, select the correct answer:

- a. Cortico-spinal tracts are concerned with skilled movements
- b. Vestibulo-spinal is a crossed tract
- c. Rubro-spinal arises from ipsilateral red nucleus
- d. Medial reticulospinal tract arises from medulla
- e. Olivospinal arises from superior olive

9. Injury of the right cortico-bulbar tract would affect muscles of:

- a. Right upper quadrant of face.
- b. Left upper quadrant of face.
- c. Right lower quadrant of face.
- d. Left lower quadrant of face.
- e. Right ½ of tongue.

10. The corticonuclear tract has unilateral connections with:

- a. Medial part of oculomotor nucleus.
- b. Lateral part of oculomotor nucleus.
- c. Upper ½ of facial nucleus.
- d. Lower ½ of facial nucleus.
- e. Motor nucleus of trigeminal nerve.

11. The crossed corticospinal tract occupies the:

- a. Anterior limb of internal capsule.
- b. Lateral white column of spinal cord.
- c. Pyramid.
- d. Basis pedunculi.
- e. Basis pontis.

12. Select the correct answer about lateral corticospinal tract:

- a. Most of its fibers arise from precentral gyrus.
- b. Present in anterior funiculus.
- c. Passes in anterior limb of internal capsule.
- d. Decussates in upper medulla.
- e. Its injury produces a lower motor neuron lesion.

13. Regarding lamination of lateral corticospinal tract in brain stem. select the most medial fibers:

- a. Cervical fibers.
- b. Lumbar fibers.
- c. Sacral fibers.
- d. Coccygeal fibers.
- e. Thoracic fibers.

14. The corticospinal fibers are

- a. Basis pontis.
- b. Basis pedunculi.
- c. Tegmental part of pons.
- d. Tegmentum of midbrain.
- e. None of the above.

15. As regards the motor decussation, choose the CORRECT statement:

- a. It occurs in caudal part of medulla.
- b. It is formed by internal arcuate fibers.
- c. It shows the crossing of 20% of the fibers.
- d. All of the above is true.
- e. None of the above is true.

16. As regards the pyramidal tract, choose the INCORRECT statement:

- a. It arises mainly from motor & premotor areas.
- b. It passes in posterior limb of internal capsule.
- c. It descends as a compact bundle in basis pontis.
- d. Its crossed fibers form lateral corticospinal tract.
- e. In spinal cord, its cervical fibers are medial.

17. As regards the pyramidal tract, choose the CORRECT statement:

- a. It arises from motor area 4 only.
- b. It passes through anterior limb of internal capsule.
- c. It descends within the medial 3/5 of basis pedunculi.
- d. It descends as scattered bundles in basis pontis.
- e. Its crossed fibers in medulla form anterior corticospinal tract.

18. As regards corticobulbar tract, choose the INCORRECT statement:

- a. It ends in brain stem.
- b. It ends on contralateral cranial nerve nuclei only.
- c. The part of facial nucleus that controls right half of orbicularis oris muscle receives left corticonuclear tract.
- d. The left genioglossus muscle is controlled by right corticonutlear tract.
- e. It forms a part of upper motor neuron.



1. **The spinal cord region that has the richest blood supply is:**
 - a. Cervical region.
 - b. Lumbar region.
 - c. Sacral region.
 - d. Lower thoracic region.
 - e. Upper thoracic region.
2. **A patient having paraplegia with loss of pain sensation in spite of preservation of proprioception in both lower limbs most likely has:**
 - a. Completer transection of the spinal cord.
 - b. Anterior spinal artery occlusion.
 - c. Hemisection of the spinal cord.
 - d. Tabes dorsalis.
 - e. Poliomyelitis
3. **Regarding the posterior spinal artery select correct answer:**
 - a. Is a single trunk
 - b. Descends in front of attachment of dorsal root
 - c. Supplies the base of posterior horn
 - d. Arises from vertebral or posterior inferior cerebellar artery
 - e. Gives posterior radicular arteries
4. **Regarding the anterior spinal artery select the correct answer:**
 - a. Arises from vertebral artery
 - b. Is a single trunk
 - c. Supplies medulla oblongata
 - d. All of the above
 - e. A & B only
5. **Regarding the blood supply of the spinal cord select the correct answer:**
 - a. Posterior spinal artery is a single trunk
 - b. Anterior spinal artery supplies posterior horn
 - c. Six longitudinal veins drain it
 - d. All of the above
 - e. None of the above
6. **The gracile and cuneate tracts of the spinal cord are supplied by:**
 - a. Anterior spinal artery
 - b. Posterior spinal artery
 - c. Radicular arteries
 - d. All of the above
 - e. B & C only
7. **Radicular arteries arise from the following arteries Except:**
 - a. Posterior intercostal
 - b. Deep cervical
 - c. Lumbar
 - d. Posterior spinal
 - e. Vertebral
8. **Anterior spinal artery supplies:**
 - a. Anterior and lateral horns
 - b. Anterior and lateral funiculi
 - c. Anterior 2\3 of the surface area of spinal cord
 - d. All of the above
 - e. A & B only
9. **Hemisection in the cervical spinal cord may cause one of the following .**
 - a. Contralateral loss of proprioception from the lower limb
 - b. Contralateral flaccid weakness in the upper limb.
 - c. Contralateral flaccid weakness in the lower limb
 - d. Contralateral pain sensory loss in the trunk and lower limb
 - e. Ipsilateral temperature sensory loss in the lower limb
10. **The posterior spinal artery:**
 - a. Is a single trunk.
 - b. Descends in front of attachment of dorsal root.
 - c. Supplies the base of posterior horn.
 - d. Arises from vertebral or posterior inferior cerebellar artery.
 - e. Gives posterior radicular arteries.

11. Complete section of spinal cord above C5 results in:

- a. Quadriplegia.
- b. Death.
- c. Paraplegia.
- d. None of the above

12. Regarding the anterior spinal artery select the correct answer:

- a. Arises from vertebral artery.
- b. Is a single trunk.
- c. Supplies medulla oblongata.
- d. All of the above.
- e. A & B only.

13. Anterior spinal artery supplies:

- a. Anterior and lateral horns.
- b. Anterior and lateral white columns.
- c. Anterior 2/3 of the surface area of spinal cord.
- d. All of the above.
- e. A & B only

14. Complete section of spinal cord above C5 results in:

- a. Quadriplegia.
- b. Death.
- c. Paraplegia.
- d. Hemiplegia.
- e. None of the above.

15. Complete section of spinal cord at level of C6 results in:

- a. Death.
- b. Quadriplegia.
- c. Hemiplegia.
- d. Paraplegia.
- e. Monoplegia.

16. Hemisection of spinal cord gives the following symptoms on the same side below the level of lesion:

- a. Proprioception loss.
- b. Upper motor neuron lesion.
- c. Loss of pain and temperature.
- d. All of the above.
- e. a & B only.



1. Which of the following cerebellar parts has mainly spinal connections:
 - a. Archicerebellum
 - b. Paleocerebellum
 - c. Neocerebellum
 - d. All of the above
 - e. None of the above
2. The following are contained in the superior cerebellar peduncle Except:
 - a. Anterior spinocerebellar
 - b. Tectocerebellar
 - c. Dentatothalamic
 - d. Cerebellorubral
 - e. Cerebellovestibular
3. The following include afferent fibers in the superior cerebellar peduncle:
 - a. Paraolivocerebellar
 - b. Reticulocerebellar
 - c. Tectocerebellar
 - d. Cuneocerebellar
 - e. Vestibulocerebellar
4. Arterial supply to the cerebellum includes:
 - a. Superior cerebellar artery
 - b. Anterior inferior cerebellar artery
 - c. Posterior inferior cerebellar artery
 - d. All of the above
 - e. None of the above
5. The following are afferent fibers in the inferior cerebellar peduncle Except:
 - a. Olivocerebellar
 - b. Posterior spinocerebellar
 - c. Anterior external arcuate fibers
 - d. Corticopontocerebellar
 - e. Vestibulocerebellar
6. Which of the following fibers carry proprioception from the upper limb to the cerebellum?
 - a. Ventral spinocerebellar tract.
 - b. Dorsal spinocerebellar tract.
 - c. Cuneocerebellar tract.
 - d. Internal arcuate fibers.
 - e. Vestibule-cerebellar fibers.
7. Afferent fibers in the superior cerebellar peduncle include:
 - a. Dentatothalamic fibers.
 - b. Fastigiorubral fibers.
 - c. Posterior spinocerebellar fibers.
 - d. Dorsal external arcuate fibers.
 - e. Anterior spinocerebellar fibers.
8. The inferior cerebellar peduncle contains:
 - a. Dentatothalamic fibers.
 - b. Gracile tract.
 - c. Tectocerebellar tract.
 - d. Posterior spinocerebellar tract.
 - e. Pontocerebellar tract.
9. Superior cerebellar peduncle contains which of the following tracts?
 - a. Dorsal spinocerebellar tract.
 - b. Ventral spinocerebellar tract.
 - c. Olivocerebellar tract.
 - d. Cuneocerebellar tract.
 - e. Ponto-cerebellar tract.
10. The most lateral nucleus in the interior of the cerebellum is:
 - a. Dentate nucleus.
 - b. Fastigial nucleus.
 - c. Emboliform nucleus.
 - d. Globose nucleus.
 - e. Vestibular nucleus.

11. The nucleus that projects neocerebellar efferent is the:

- a. Dentate nucleus.
- b. Fastigial nucleus.
- c. Emboliform nucleus.
- d. Globose nucleus.
- e. Vestibular nucleus.

12. The neocerebellum is anatomically located in the following part of the cerebellum:

- a. Vermis.
- b. Flocculo-nodular lobe.
- c. Posterior lobe only.
- d. Lateral parts of cerebellar hemispheres.
- e. Anterior lobe only.

13. Which of the following structures have vestibular connections:

- a. Paleocerebellum.
- b. Archicerebellum
- c. Neocerebellum.
- d. All of the above.
- e. None of the above.

14. The following are afferent fibers in inferior cerebellar peduncle Except:

- a. Olivocerebellar tract.
- b. Posterior spinocerebellar tract.
- c. Anterior external arcuate fibers.
- d. Cortico-ponto-cerebellar tract.
- e. Vestibulo-cerebellar tract.

15. The following are contained in the superior cerebellar peduncle Except:

- a. Anterior spinocerebellar tract.
- b. Tectocerebellar tract.
- c. Dentatothalamic fibers.
- d. Cerebello-rubral tract.
- e. Cerebello-vestibular tract.

16. The superior cerebellar peduncle contains:

- a. Vestibule-cerebellar tract.
- b. Cuneo-cerebellar tract.
- c. Anterior spinocerebellar tract.
- d. Posterior spinocerebellar tract.
- e. Cortico-ponto-cerebellar fibers.

17. The inferior cerebellar peduncle contains:

- a. Dentalothalamic tract.
- b. Gracile tract.
- c. Anterior spinocerebellar tract.
- d. Posterior spinocerebellar tract.
- e. Cortico-ponto-cerebellar fibers.

18. The flucculo-nodular lobe of the cerebellum is concerned with:

- a. Voluntary movements.
- b. Equilibrium.
- c. Reflexes.
- d. Muscle tone.
- e. Involuntary movements.

19. The basal ganglia are supplied by central branches of:

- a. Anterior cerebral artery.
- b. Middle cerebral artery.
- c. Posterior cerebral artery.
- d. Middle & posterior cerebral arteries.
- e. Anterior & middle cerebral arteries.

20. As regards the caudate nucleus, choose the **CORRECT statement:**

- a. Its head bulges into floor of body of lateral ventricle.
- b. Its body lies in roof of body of lateral ventricle.
- c. Its tail tip is continuous with mammillary body.
- d. Its head is continuous with globus pallidus.
- e. Its tail is related to inferior horn of lateral ventricle.

21. As regards the lentiform nucleus, choose the **CORRECT statement:**

- a. It is C-shaped.
- b. It is related to external capsule medially.
- c. It is related to internal capsule laterally.
- d. Globus pallidus forms its lateral part.
- e. Its putamen appears white due to its rich myelin content.

22. As regards the cerebellum, choose the **CORRECT statement:**

- a. It is a part of the midbrain.
- b. It is roofed by tentorium cerebelli.
- c. Its posterior notch lodges pons & medulla.
- d. Its fissure prima separates posterior lobe from flocculo-nodular lobe.
- e. Superior surface shows vallecule.



1. Hind brain (Rhomben cephalon) includes the following Except:

- a. Medulla oblongata
- b. Cerebellum
- c. Pons
- d. Crus cerebri
- e. Fourth ventricle

2. Rootlets of glossopharyngeal nerve lie:

- a. Between pyramid and olive
- b. Between olive and inferior cerebellar peduncle
- c. Behind the inferior cerebellar peduncle
- d. Between lower border of pons and medulla
- e. None of the above

3. Anterior spinal artery supplies the following structures Except:

- a. Nucleus ambiguus
- b. Hypoglossal nucleus
- c. Pyramid
- d. Medial lemniscus -
- e. Medial longitudinal bundle

4. Brain stem includes the following Except:

- a. Mid brain
- b. Pons
- c. Medulla oblongata
- d. Cerebellum
- e. Cerebral aqueduct

5. Rootlets of hypoglossal nerve lie:

- a. Between pyramid and olive
- b. Between olive and inferior cerebellar peduncle
- c. Behind the inferior cerebellar peduncle
- d. Between lower border of pons & pyramid
- e. None of the above

6. Which of the following nerves lie close to the median plane in the transverse furrow between pons and medulla?

- a. Facial
- b. Vestibulocochlear
- c. Nervus intermedius
- d. Abducent
- e. Trigeminal

7. Which of the following structures lie between the limbs of inferior fovea?

- a. Hypoglossal trigone
- b. Vagal trigone
- c. Vestibular trigone
- d. Facial colliculus
- e. Stria medullaris

8. Facial colliculus overlies:

- a. Facial motor nucleus
- b. Superior olive
- c. Locus coeruleus
- d. Vestibular area
- e. Abducent nucleus

9. Branches of basilar artery include the following Except:

- a. Internal auditory artery
- b. Superior cerebellar artery
- c. Posterior inferior cerebellar artery
- d. Anterior inferior cerebellar artery
- e. Posterior cerebral artery

10. Which of the following nerves is the last to be affected in tumors of the cerebellopontine angle?

- a. Vestibulocochlear
- b. Nervus intermedius
- c. Abducent
- d. Trigeminal
- e. Facial

11. Arterial blood supply to mid brain comes from:

- a. Basilar artery
- b. Posterior cerebral artery
- c. Superior cerebellar artery
- d. All of the above
- e. B & C only

12. Nerve attached to the lower border of pons above pyramid is:

- a. Trigeminal nerve.
- b. Facial nerve.
- c. Abducent nerve.
- d. Vestibular nerve.
- e. Cochlear nerve.

13. Which of the following cranial nerves is attached to the back of the brain stem?

- a. Oculomotor nerve.
- b. Trochlear nerve.
- c. Maxillary nerve.
- d. Abducent nerve.
- e. Facial nerve.

14. One of the following is not a content of the interpeduncular fossa:

- a. Tuber cinereum.
- b. Mammillary bodies.
- c. Anterior perforated substance.
- d. Posterior perforated substance.
- e. Oculomotor nerve.

15. A nerve attached to the cerebello-pontine angle:

- a. Glossopharyngeal nerve.
- b. Hypoglossal nerve.
- c. Abducent nerve.
- d. Vestibulo-cochlear nerve.
- e. Trigeminal nerve.

16. Which of the following is a recess of the fourth ventricle?

- a. Lateral recess
- b. Pineal recess.
- c. Suprapineal recess.
- d. Optic recess.
- e. Infundibular recess.

17. Medial eminence in floor of fourth ventricle overlies the nucleus of:

- a. Hypoglossal nerve.
- b. Vagus nerve.
- c. Abducent nerve.
- d. Vestibular nerve.
- e. Facial nerve.

18. The superior angle of the fourth ventricle leads to:

- a. Optic recess.
- b. Cerebral aqueduct.
- c. Lateral recess.
- d. Central canal.
- e. Foramen of Luschka.

19. A pontine cranial nerve that is not attached to lower border of pons:

- a. Abducent nerve.
- b. Facial nerve.
- c. Vestibular nerve.
- d. Cochlear nerve.
- e. Trigeminal nerve.

20. Brain stem includes all the following Except:

- a. Mid brain.
- b. Pons.
- c. Medulla oblongata.
- d. Cerebellum.
- e. Cerebral aqueduct.

21. Which of the following nerves lies close to the median plane in the transverse furrow between pons and medulla?

- a. Facial nerve.
- b. Vestibulo-cochlear nerve.
- c. Nervus intermedius.
- d. Abducent nerve.
- e. Trigeminal nerve.

22. Facial colliculus overlies:

- a. Facial motor nucleus.
- b. Superior olive.
- c. Locus coeruleus.
- d. Vestibular area.
- e. Abducent nucleus.

23. The facial colliculus is formed by:

- a. Facial nerve.
- b. Abducent nucleus.
- c. Facial motor nucleus.
- d. Abducent nucleus and facial nerve fibers.
- e. Facial motor nucleus and abducent nerve fibers.

24. Posterolateral boundary of the interpeduncular fossa is formed by:

- a. Optic chiasma.
- b. Optic tract.
- c. Pons.
- d. Cerebral peduncles.
- e. None of the above.

25. The anterolateral boundary of interpeduncular fossa is formed by:

- a. Optic chiasma.
- b. Optic tract.
- c. Crus cerebri.
- d. Pons.
- e. Posterior perforated substance.

26. The cranial nerve that has the most medial attachment to the pons:

- a. Abducent nerve.
- b. Facial nerve.
- c. Vestibular nerve.
- d. Cochlear nerve.
- e. Trigeminal nerve.

27. As regards cranial nerve attachment to brain stem, choose the CORRECT matching:

- a. Hypoglossal nerve - posterolateral sulcus of medulla.
- b. Vagus nerve - groove between pyramid & olive.
- c. Vestibulo-cochlear nerve - cerebello-pontine angle.
- d. Facial nerve - lower border of pons above pyramid.
- e. Trochlear nerve - interpeduncular fossa.

28. Posterior external arcuate fibers are axons of:

- a. Arcuate nucleus.
- b. Accessory cuneate nucleus.
- c. Gracile and cuneate nuclei.
- d. Inferior olivary nucleus.
- e. Medial accessory olive.

1. The cranial nerve nucleus that supplies muscles of the larynx is the:

- a. Solitary nucleus.
- b. Ambiguus nucleus.
- c. Dorsal motor vagal nucleus.
- d. Motor facial nucleus.
- e. Spinal nucleus of V (trigeminal).

2. Which of the following nuclei represents the visceral afferent column (GVA & SVA)?

- a. Abducent nucleus.
- b. Ambiguus nucleus.
- c. Solitarius nucleus.
- d. Dorsal vagal nucleus.
- e. Motor nucleus of V.

3. The common motor nucleus of IX, X and XI cranial nerves is:

- a. Solitarius nucleus.
- b. Nucleus ambiguus.
- c. Spinal nucleus of V.
- d. Inferior salivatory nucleus.
- e. Dorsal vagal nucleus.

4. Which of the following nuclei represents the visceral afferent column?

- a. Nucleus Solitarius.
- b. Superior salivatory nucleus.
- c. Motor facial nucleus.
- d. Edinger-Westphal nucleus.
- e. Lacratory nucleus.

5. The nucleus ambiguus does not supply the following muscle(s):

- a. Laryngeal muscles.
- b. Stylopharyngeus.
- c. Pharyngeal constrictors.
- d. Palatoglossus.
- e. Tensor palati.

6. The mesencephalic nucleus of trigeminal nerve has the following functional component:

- a. Special visceral afferent.
- b. General somatic afferent.
- c. General visceral afferent.
- d. Special visceral efferent.
- e. General somatic efferent.

7. The parasympathetic nucleus of the ninth cranial nerve is the:

- a. Lacratory nucleus.
- b. Dorsal vagal nucleus.
- c. Inferior salivatory nucleus.
- d. Superior salivatory nucleus.
- e. Ambiguus nucleus.

8. Somatic efferent column of nuclei includes:

- a. Abducent nucleus.
- b. Hypoglossal nucleus.
- c. Trochlear nucleus.
- d. Oculomotor nucleus.
- e. All of the above.

9. General visceral efferent column includes the following nuclei Except:

- a. Dorsal vagal nucleus.
- b. Edinger Westphal nucleus.
- c. Nucleus ambiguus.
- d. Superior salivatory nucleus.
- e. Inferior salivatory nucleus.

10. Special visceral afferent nuclei include:

- a. Spinal nucleus of trigeminal nerve.
- b. Vestibulocochlear nuclei.
- c. Nucleus solitarius.
- d. Dorsal vagal nucleus (lateral part).
- e. None of the above.

11. Special visceral efferent (branchiomotor) column includes:

- a. Trigeminal motor nucleus.
- b. Facial motor nucleus.
- c. Nucleus ambiguus.
- d. All of the above.
- e. None of the above.

12. Which of the following nuclei represents the somatic efferent column?

- a. Abducent nucleus.
- b. Nucleus ambiguus.
- c. Nucleus solitarius.
- d. Dorsal vagal nucleus.
- e. Motor nucleus of V.

13. All the following are nuclei of facial nerve EXCEPT:

- a. Nucleus Solitarius.
- b. Superior salivatory nucleus.
- c. Spinal nucleus of V.
- d. Inferior salivatory nucleus.
- e. Motor nucleus.

14. The nucleus that receives taste fibers is:

- a. Nucleus Solitarius.
- b. Nucleus Ambiguus.
- c. Spinal nucleus of V.
- d. Inferior salivatory nucleus.
- e. Dorsal vagal nucleus.

15. Choose the INCORRECT matching:

- a. Dorsal vagal nucleus - muscles of bronchial tree.
- b. Edinger Westphal nucleus - ciliary muscle.
- c. Nucleus ambiguus - palatine muscles.
- d. Mesencephalic nucleus - proprioception of head.
- e. Inferior salivatory nucleus - submandibular gland.

16. Connects the vestibular nuclei with cranial nuclei of III, IV, & VI:

- a. Medial longitudinal bundle.
- b. Medial forebrain bundle.
- c. Trapezoid body.
- d. Stria terminalis.
- e. Fronto-occipital bundle.

17. The pharyngeal arch supplied by the only pontine cranial nerve that is NOT attached to the lower border of pons is:

- a. First arch.
- b. Second arch.
- c. Third arch.
- d. Fourth arch.
- e. Sixth arch.



1. The falx cerebri:

- a. Is attached anteriorly to anterior clinoid process.
- b. Contains superior sagittal sinus.
- c. Separates pituitary gland from cerebrum.
- d. Separates cerebrum from cerebellum.
- e. Contains superior petrosal sinus.

2. Extradural hemorrhage is caused by rupture of:

- a. Anterior cerebral artery.
- b. Internal carotid artery.
- c. Middle meningeal artery.
- d. Superior cerebral vein.
- e. None of the above.

3. Regarding pituitary gland, all the following statements are true Except:

- a. It lies in anterior part of the neck.
- b. It is controlled by hypothalamus.
- c. It has a portal system of veins.
- d. Axons of nerve cells in hypothalamus carry hormones to the neurohypophysis.
- e. The adenohypophysis secretes at least six different hormones.

4. All the following drain into cavernous sinus Except:

- a. Ophthalmic veins.
- b. Central retinal vein.
- c. Sphenoparietal sinus.
- d. Great cerebral vein.
- e. Superficial middle cerebral vein.

5. The falx cerebri contains which of the following sinuses in its concave lower margin?

- a. Superior sagittal.
- b. Inferior sagittal.
- c. Sigmoid.
- d. Straight.
- e. Sphenoparietal.

6. The straight sinus:

- a. Lies at the line of attachment of falx cerebri to skull.
- b. Is formed by union of great cerebral vein and inferior sagittal sinus.
- c. Is commonly absent.
- d. Drains the superior sagittal sinus.
- e. Is a paired venous sinus.

7. One of the following nerves are not in relation to the cavernous sinus:

- a. Oculomotor.
- b. Trochlear.
- c. Abducent.
- d. Ophthalmic.
- e. Mandibular.

8. The cavernous sinus communicates with all the following Except:

- a. Ophthalmic veins.
- b. Facial vein.
- c. Pterygoid venous plexus.
- d. Superior sagittal sinus.
- e. Inferior petrosal sinus.

9. The superior sagittal sinus:

- a. Is invaginated by aggregations of arachnoid villi.
- b. Contains few valves.
- c. Drains cerebellar veins.
- d. Joins the inferior sagittal sinus to form straight sinus.
- e. Is connected with scalp through mastoid emissary vein.

10. Regarding tentorium cerebelli: All are true Except:

- a. It roofs over middle cranial fossa.
- b. Its free border is attached to anterior clinoid process.
- c. Its attached border is attached to post clinoid process.
- d. It includes transverse venous sinus in its attached border.
- e. Oculomotor and trochlear nerves pierce dura at the point of crossing of its free & attached borders.

11. The following dural venous sinus lies in the junction of the falx cerebri with the tentorium cerebelli: •

- a. Sphenoparietal.
- b. Inferior petrosal.
- c. Straight.
- d. Inferior sagittal.
- e. Occipital.

12. Regarding meninges; all are true Except:

- a. Outer layer of dura forms folds to separate different parts of brain.
- b. Subdural space is between dura & arachnoid.
- c. Pia dips into the sulci of brain.
- d. Subarachnoid space contains C.S.F..
- e. Dura receives nerves from all divisions of the trigeminal.

13. Which of the following arteries supply the pituitary gland?

- a. Common carotid.
- b. Internal carotid.
- c. External carotid.
- d. Subclavian.
- e. Vertebral,

14. All the following venous sinuses lie between outer (endosteal) and inner (meningeal) layers of dura Except:

- a. Superior sagittal.
- b. Inferior sagittal.
- c. Sigmoid.
- d. Right transverse.
- e. Left transverse.

15. The following are tributaries of the cavernous sinus Except:

- a. Sphenoparietal sinus.
- b. Straight sinus.
- c. Central retinal vein.
- d. Superior ophthalmic vein.
- e. Inferior ophthalmic vein.

16. Regarding the superior sagittal sinus, all the statements are true Except:

- a. May share in confluence of sinuses.
- b. Has lacunae lateralis.
- c. Receives diploic veins.
- d. May communicate with veins of nose.
- e. Receives the superficial middle cerebral vein.

17. All the following are communications of cavernous sinus, Except:

- a. Cavernous sinus of the opposite side.
- b. Pharyngeal venous plexus through foramen lacerum.
- c. Pterygoid venous plexus through foramen ovale.
- d. Transverse venous sinus via superior petrosal sinus.
- e. Sigmoid sinus by straight sinus.

18. One of the following structures is embedded in the wall of the cavernous sinus:

- a. Mandibular nerve.
- b. Internal carotid artery.
- c. Ophthalmic nerve.
- d. Abducent nerve.
- e. Trigeminal ganglion.

19. The nerve that runs in the floor of the cavernous sinus is:

- a. Oculomotor nerve.
- b. Trochlear nerve.
- c. Ophthalmic
- d. Maxillary nerve.
- e. Abducent nerve.

20. The sinus that runs along the lesser wing of sphenoid is:

- a. Sphenoparietal sinus.
- b. Superior sagittal sinus.
- c. Inferior sagittal sinus.
- d. Straight sinus.
- e. Sigmoid sinus.

21. Which of the following foramina gives passage to emissary vein of the superior sagittal sinus?

- a. Foramen caecum.
- b. Carotid canal.
- c. Foramen lacreum.
- d. Mastoid foramen.
- e. Condylar canal.

22. The inferior petrosal sinus connects the cavernous sinus to the:

- a. Straight sinus.
- b. Transverse sinus.
- c. Inferior sagittal sinus.
- d. Internal jugular vein.
- e. Sigmoid sinus.

23. Sensory innervation to the dura above the tentorium cerebelli is from which cranial nerve?

- a. Trigeminal nerve.
- b. Facial nerve.
- c. Glossopharyngeal nerve.
- d. Vagus nerve.
- e. Hypoglossal nerve.

24. Which of the following foramina gives passage to emissary veins of the cavernous sinuses?

- a. Foramen caecum.
- b. Jugular foramen.
- c. Foramen lacerum.
- d. Mastoid foramen.
- e. Condylar foramen.

25. Infection in the mastoid air cells may spread to which sinus?

- a. Inferior sagittal sinus.
- b. Straight sinus.
- c. Occipital sinus.
- d. Sigmoid sinus.
- e. Superior petrosal sinus.

26. The most superior nerve that runs in lateral wall of cavernous sinus:

- a. Oculomotor nerve.
- b. Trochlear nerve.
- c. Ophthalmic nerve.
- d. D. Maxillary nerve.
- e. Abducent nerve.

27. The sinus that runs along the junction between the flax cerebri and the tentorium cerebelli is the:

- a. Sphenoparietal sinus.
- b. Superior sagittal sinus.
- c. Inferior sagittal sinus.
- d. Straight sinus.
- e. Sigmoid sinus.

28. Which of the following cranial nerves is not related to cavernous sinus?

- a. Oculomotor nerve.
- b. Trochlear nerve.
- c. Maxillary nerve.
- d. Abducent nerve.
- e. Facial nerve.

29. Cavernous sinus is related laterally to:

- a. Pituitary gland.
- b. Temporal lobe of brain.
- c. Foremen lacerum.
- d. Optic chiasma.
- e. Sphenoidal air sinus.

30. As regards the cavernous sinus. choose the INCORRECT statement:

- a. Internal carotid artery runs in its floor.
- b. Abducent nerve is embedded in 1 its lateral wall.
- c. It receives the ophthalmic veins anteriorly.
- d. It is connected with pterygoid plexus of veins.
- e. Sphenoidal sinus lies below it.

31. In case of fracture of pterion & formation of extradural hemorrhage, which blood vessel is most likely injured?

- a. Internal carotid artery.
- b. Middle meningeal artery.
- c. Middle cerebral artery.
- d. Superficial middle cerebral vein.
- e. Superior cerebral veins.

32. Rupture of middle meningeal artery causes which type of hemorrhage?

- a. Extradural.
- b. Subdural.
- c. Subarachnoid.
- d. Cerebral.
- e. Ventricular.

33. A tributary of cavernous sinus is:

- a. Central retinal vein.
- b. Pterygoid plexus of veins.
- c. Superior petrosal sinus.
- d. Inferior petrosal sinus.
- e. Great cerebral vein.

34. A branch from cavernous part of internal carotid artery is:

- a. Caroticotympanic artery.
- b. Pterygoid artery.
- c. Hypophyseal artery.
- d. Anterior choroidal artery.
- e. Ophthalmic artery.

35. Extradural hematoma occurs due to the rupture of the vessel that passes through:

- a. Foramen rotundum.
- b. Foramen ovale.
- c. Foramen spinosum.
- d. Carotid canal.
- e. Inferior orbital fissure.

36. Middle meningeal artery supplies dura of:

- a. Anterior cranial fossa.
- b. Middle cranial fossa.
- c. Posterior cranial fossa.
- d. Anterior & middle cranial fossae.
- e. Middle & posterior cranial fossae.

37. Cranial dura is supplied by all the following Except:

- a. Maxillary nerve.
- b. Mandibular nerve.
- c. C1-C3 spinal nerves.
- d. Vagus nerve.
- e. Glossopharyngeal nerve.

38. Dural venous sinuses have all the following characters Except:

- a. They are lined by endothelium.
- b. They are rich in valves.
- c. They drain brain & meninges.
- d. They are connected to extracranial veins.
- e. They lie between 2 dural layers.

39. All the following are paired sinuses Except:

- a. Sphenoparietal sinus.
- b. Cavernous sinus.
- c. Straight sinus.
- d. Sigmoid sinus.
- e. Inferior petrosal sinus.

40. Which of the following is considered as a multiple venous sinus?

- a. Basilar plexus of sinuses.
- b. Cavernous sinus.
- c. Transverse sinus.
- d. Straight sinus.
- e. Occipital sinus.

41. As regards superior sagittal sinus, choose the **INCORRECT statement:**

- a. Its anterior end communicates with veins of nose.
- b. Usually it ends by forming left transverse sinus.
- c. Its posterior end is called confluence of sinuses.
- d. It receives superior cerebral veins.
- e. It helps in drainage of CSF through arachnoid villi & granulations

42. As regards relations of cavernous sinus, choose the **INCORRECT statement:**

- a. It lies in posterior cranial fossa.
- b. The endosteal dura forms its roof.
- c. It lies between inferior orbital fissure & apex of petreous temporal bone.
- d. It lies on either side of greater wing of sphenoid.
- e. Expansile pulsation of external carotid artery helps the passing of the blood outside the sinus.

43. As regards relations of cavernous sinus, choose the **INCORRECT matching:**

- a. Anteriorly - apex of orbit.
- b. Posteriorly - crus cerebri of midbrain.
- c. Medially - pituitary gland.
- d. Laterally - sphenoidal air sinus.
- e. Superiorly - Internal carotid artery.

44. Which one of the following structures is not embedded in lateral wall of cavernous sinus?

- a. Ophthalmic nerve.
- b. Abducent nerve.
- c. Oculomotor nerve.
- d. Trochlear nerve.
- e. Maxillary nerve.

45. Which one of the following is considered as an emissary vein?

- a. Sphenoparietal sinus.
- b. Superior petrosal sinus.
- c. Inferior petrosal sinus.
- d. Transverse sinus.
- e. Straight sinus.

46. Emissary veins of cavernous sinus pass through all the following foramina Except:

- a. Foramen lacerum.
- b. Foramen oval.
- c. Carotid canal.
- d. Mastoid foramen.
- e. Emissary sphenoidal foramen.

1. All are contents of infratemporal fossa Except:

- a. Medial pterygoid muscle.
- b. Lateral pterygoid muscle.
- c. Pterygoid venous plexus.
- d. Middle meningeal artery.
- e. Third part of maxillary artery.

2. Regarding temporalis, All are true Except:

- a. It is attached to neck of mandible.
- b. It elevates the mandible.
- c. Its posterior fibers retract the mandible.
- d. It is a fan shaped muscle.
- e. It is covered by temporal fascia.

3. A muscle attached to the coronoid process of mandible:

- a. Buccinator.
- b. Lateral pterygoid.
- c. Medial pterygoid.
- d. Temporalis.
- e. Masseter.

4. The most horizontal fibers of temporalis muscles are:

- a. Deep.
- b. Superficial.
- c. Posterior.
- d. Middle.
- e. Anterior.

5. The pterygoid venous plexus receives tributaries from:

- a. Sphenopalatine vein.
- b. Inferior alveolar vein.
- c. Posterior superior alveolar vein.
- d. Middle meningeal veins.
- e. All of these veins.

6. All of the following nerves are derived from the posterior division of mandibular nerve Except:

- a. Auriculotemporal nerve.
- b. Nerve to medial pterygoid.
- c. Inferior alveolar.
- d. Lingual.

7. A muscle which elevates and retracts the mandible:

- a. Masseter.
- b. Temporalis.
- c. Medial pterygoid.
- d. Lateral pterygoid,
- e. Digastric.

8. Opening of the jaw is helped by all the following muscles Except:

- a. Lateral pterygoid.
- b. Masseter.
- c. Digastric.
- d. Geniohyoid.
- e. Mylohyoid.

9. Pterygoid venous plexus communicates with:

- a. Cavernous sinus.
- b. Facial vein.
- c. Inferior ophthalmic vein.
- d. All of the above.
- e. A and B only.

10. The following muscle can protract the mandible:

- a. Platysma.
- b. Buccinator.
- c. Temporalis.
- d. Masseter.
- e. Lateral pterygoid.

11. The buccal branch of the mandibular nerve supplies:

- a. Buccinator muscle.
- b. Skin and mucous membrane of cheek.
- c. Only skin of cheek.
- d. Only mucous membrane of cheek.
- e. Molar teeth.

12. The following are muscles of mastication Except:

- a. Medial pterygoid.
- b. Lateral pterygoid.
- c. Buccinator.
- d. Temporalis.
- e. Masseter.

13. Regarding medial pterygoid muscle, All are true Except:

- a. It has two heads of origin.
- b. It lies on lateral surface of ramus of mandible.
- c. It is supplied by trunk of mandibular nerve.
- d. It can elevate the mandible.
- e. Lingual nerve lies in the space between the muscle and the ramus of mandible.

14. Retraction of mandible is produced by:

- a. Upper fibers of lateral pterygoid muscle.
- b. Anterior fibers of Masseter muscle.
- c. Posterior fibers of temporalis muscle.
- d. Anterior belly of digastric muscle.
- e. Superficial head of medial pterygoid muscle.

15. Anterior division of mandibular nerve gives the following branches Except:

- a. Nerve to medial pterygoid.
- b. Nerve to lateral pterygoid.
- c. Buccal nerve.
- d. Deep temporal nerves.
- e. Nerve to Masseter.

16. Concerning the otic ganglion, select the false statement:

- a. It lies below foramen oval.
- b. It lies deep to the mandibular nerve.
- c. It is functionally connected to the glossopharyngeal nerve.
- d. It is closely related to the nerve to the medial pterygoid muscle.
- e. Its sympathetic root is derived from the plexus around internal carotid artery.

17. Regarding pterygoid venous plexus, All are true statements Except:

- a. It can transmit infection from face to cavernous sinus.
- b. Its tributaries correspond to branches of the maxillary artery.
- c. It drains into the maxillary vein.
- d. It is connected to inferior ophthalmic vein by a vein that passes through inferior orbital fissure.
- e. It surrounds Masseter muscle.

18. The inferior alveolar nerve supplies:

- a. Hyoglossus muscle.
- b. Stylohyoid.
- c. Genioglossus.
- d. Geniohyoid muscle.
- e. Mylohyoid.

19. Difficulty in retraction of mandible would indicate damage to the:

- a. Lateral pterygoid muscle.
- b. Temporalis muscle.
- c. Masseter muscle.
- d. Mylohyoid muscle.
- e. Medial pterygoid muscle.

20. All of the following nerves exit the middle cranial fossa EXCEPT:

- a. Abducent nerve.
- b. Facial nerve.
- c. Trochlear nerve.
- d. Oculomotor nerve.
- e. Trigeminal nerve.

21. A young girl complains of dryness of the nose and the palate. This would indicate a lesion of which of the following ganglia?

- a. Vagal ganglion.
- b. Pterygopalatine ganglion.
- c. Otic ganglion.
- d. Submandibular ganglion.
- e. Ciliary ganglion.

22. Which of the following arteries is a branch of maxillary artery?

- a. Tonsillar artery.
- b. Ascending pharyngeal artery.
- c. Lingual artery.
- d. Middle meningeal artery.
- e. Suprahyoid artery.

23. A muscle that can depress the mandible is the:

- a. Temporalis.
- b. Masseter.
- c. Lateral pterygoid.
- d. Stylohyoid.
- e. Medial pterygoid.

24. The nerve that carries taste sensation from anterior 2/3 of tongue is:

- a. Chorda tympani.
- b. Greater petrosal nerve.
- c. Lesser petrosal nerve.
- d. Greater palatine nerve.
- e. Lesser palatine nerve.

25. The nerve which may be injured on ligating submandibular duct is:

- a. Inferior alveolar nerve.
- b. Lingual nerve.
- c. Maxillary nerve.
- d. Buccal nerve.
- e. Masseteric nerve.

26. The maxillary nerve supplies:

- a. Upper eyelid.
- b. skin of forehead.
- c. Lower lip.
- d. Upper jaw teeth.
- e. Mucosa lining buccinator.

27. The structure attached to the lingula of the mandible is the:

- a. Stylomandibular ligament.
- b. Pterygomandibular ligament.
- c. Sphenomandibular ligament.
- d. Pterygomandibular ligament.
- e. Mylohyoid muscle.

28. The sympathetic root to the submandibular ganglion is derived from the plexus around the:

- a. Lingual Artery.
- b. Facial artery.
- c. Occipital artery.
- d. Maxillary artery.
- e. Ascending pharyngeal artery.

29. Affection of which of the following muscles would result in difficulty in protruding the jaw?

- a. Anterior belly of digastric.
- b. Lateral pterygoid.
- c. Medial pterygoid.
- d. Masseter.
- e. Temporalis.

30. Retraction of mandible is produced by:

- a. Upper fibers of lateral pterygoid muscle.
- b. Anterior fibers of masseter muscle.
- c. Posterior fibers of temporalis muscle.
- d. Anterior belly of digastric muscle.
- e. Superficial head of medial pterygoid muscle.

31. A muscle which elevates and retracts the mandible:

- a. Masseter.
- b. Temporalis.
- c. Lateral pterygoid.
- d. Medial pterygoid.
- e. Digastric.

32. The following muscle can protract the mandible:

- a. Platysma.
- b. Buccinator.
- c. Temporalis.
- d. Masseter.
- e. Lateral pterygoid.

33. Regarding temporalis, all are true Except:

- a. It is attached to neck of mandible.
- b. It elevates the mandible.
- c. Its posterior fibers retract the mandible.
- d. It is a fan shaped muscle.
- e. It is covered by temporal fascia.

34. Anterior division of mandibular nerve gives all the following Except:

- a. Nerve to medial pterygoid.
- b. Nerve to lateral pterygoid.
- c. Buccal nerve.
- d. Deep temporal nerves.
- e. Nerve to masseter.

35. All of the following nerves are derived from the posterior division of Mandibular nerve Except:

- a. Auriculotemporal nerve.
- b. Nerve to mylohyoid.
- c. Nerve to medial pterygoid.
- d. Inferior alveolar nerve.
- e. Lingual nerve.

36. Regarding pterygopalatine fossa, all are true Except:

- a. It is an inverted pyramidal space.
- b. It is bounded anteriorly by posterior surface of maxilla.
- c. It communicates with middle cranial fossa through foramen rotundum.
- d. It communicates with the orbit through superior orbital fissure.
- e. It communicates with nasal cavity through sphenopalatine (nasopalatine) foramen.

37. All the following statements regarding trigeminal ganglion are true except:

- a. It lies in middle cranial fossa.
- b. It lies within a fold of dura.
- c. It contains bipolar nerve cells.
- d. Central processes of its cells form sensory root of trigeminal.
- e. Motor root of trigeminal lies beneath it.

38. Maxillary nerve:

- a. Leaves the skull through foramen oval.
- b. Enters the orbit through superior orbital fissure.
- c. Gives the posterior superior alveolar (dental) branch.
- d. Has otic ganglion attached to it.
- e. Supplies skin of lower lip.

39. The maxillary nerve:

- a. Arises from the geniculate ganglion.
- b. Runs in the lateral wall of the cavernous sinus.
- c. Has the otic ganglion attached to it.
- d. Supplies the skin of the bridge of the nose.
- e. Has sensory and motor fibers.

40. All is true regarding maxillary nerve Except:

- a. It is entirely sensory.
- b. It supplies upper lip.
- c. It supplies lower eyelid.
- d. It supplies skin over zygomatic bone.
- e. It runs through foramen oval.

41. Regarding the mandibular nerve, all are true Except:

- a. It supplies all lower jaw teeth.
- b. It has motor and sensory fibers.
- c. It passes through foramen oval.
- d. It passes in lateral wall of cavernous sinus.
- e. It gives a meningeal branch.

42. The lingual nerve:

- a. Is a branch of the posterior division of maxillary nerve.
- b. Leaves skull through foramen rotundum.
- c. Carries fibers from first cervical spinal nerve.
- d. Carries taste from posterior third of tongue.
- e. Is closely related to the third molar tooth.

43. A lesion of mandibular nerve immediately after its exit from foramen oval may lead to:

- a. Loss of parotid gland secretion.
- b. Loss of taste sensation from anterior part of tongue.
- c. Loss of general sensations from posterior part of tongue.
- d. Paralysis of Buccinator.
- e. Loss of sensation of upper lip.

44. Injury of motor root of trigeminal nerve would cause paralysis of:

- a. Buccinator.
- b. Lateral pterygoid.
- c. Orbicularis oculi.
- d. Posterior belly of digastric.
- e. Platysma.

45. Regarding chorda tympani; all are correct statements Except:

- a. It is a branch of facial nerve.
- b. It carries taste fibers.
- c. It joins lingual nerve deep to lateral pterygoid.
- d. It is related to medial side of lower third molar.
- e. It carries parasympathetic fibers.

46. The following glands receive their parasympathetic innervations from the facial nerves Except:

- a. Lacrimal gland.
- b. Submandibular gland.
- c. Parotid gland.
- d. Nasal glands.
- e. Sublingual gland.

47. The facial nerve supplies all the following Except:

- a. Skin of face.
- b. Platysma.
- c. Secretomotor fibers to sublingual gland.
- d. Stapedius.
- e. Secretomotor fibers to lacrimal gland.

48. All the following muscles are supplied by facial nerve Except:

- a. Buccinator.
- b. Tensor tympani.
- c. Posterior belly of digastric.
- d. Auricular muscles.
- e. Stylohyoid.

49. The deep petrosal nerve is composed of nerve fibers whose cell bodies are located in which ganglion?

- a. Geniculate ganglion.
- b. Trigeminal ganglion.
- c. Pterygopalatine ganglion.
- d. Superior cervical sympathetic G.
- e. Inferior vagal ganglion.

1. The anterior triangle of the neck, include the following Except one:

- a. The muscular triangle.
- b. The carotid triangle.
- c. The digastric triangle.
- d. ½ the submental triangle.
- e. The occipital triangle.

2. Which of the following muscles subdivides the posterior triangle of the neck:

- a. Sternocleidomastoid.
- b. Superior belly of omohyoid.
- c. Inferior belly of omohyoid.
- d. Posterior belly of digastric.
- e. Subclavius.

3. The following structures form the floor of the posterior triangle of the neck Except:

- a. Levator scapulae.
- b. Splenius capitis.
- c. Scalenus Medius.
- d. Investing layer of deep cervical fascia.
- e. Prevertebral layer of deep cervical fascia.

4. Regarding Infrahyoid muscles, all are true Except:

- a. Include sternohyoid, sternothyroid, omohyoid and thyrohyoid muscles.
- b. Can depress hyoid bone.
- c. Are all supplied by the Ansa cervicalis.
- d. Intermediate tendon of omohyoid is attached to clavicle.
- e. Superior belly of omohyoid helps to bound carotid triangle.

5. The sternocleidomastoid, omohyoid and trapezius bound which of the following triangles?

- a. Carotid.
- b. Digastric.
- c. Occipital.
- d. Muscular.
- e. Submental.

6. Regarding digastric triangle. All are true Except:

- a. It is bounded anteriorly by anterior belly of digastric.
- b. It is bounded posteriorly by posterior belly of digastric and stylohyoid.
- c. It is bounded superiorly by hyoid bone.
- d. It contains submandibular gland.
- e. It contains submandibular lymph nodes.

7. The following muscle bounds the carotid triangle posteriorly:

- a. Digastric (posterior belly).
- b. Digastric (anterior belly).
- c. Omohyoid (superior belly).
- d. Omohyoid (inferior belly)
- e. Sternocleidomastoid.

8. The cutaneous nerve supply over the anterior triangle of the neck is largely from:

- a. Dorsal rami of cervical nerves.
- b. Great auricular nerve.
- c. Transverse cutaneous nerve.
- d. Suprascapular nerve.
- e. Supraclavicular nerve.

9. The following lie in the posterior triangle of the neck Except:

- a. Spinal part of accessory nerve.
- b. First part of subclavian artery.
- c. External jugular vein.
- d. Trunks of Brachial plexus.
- e. Occipital artery

10. Regarding anterior triangle of neck all are true Except:

- a. It is bounded anteriorly by midline of neck.
- b. It is bounded posteriorly by posterior border of sternocleidomastoid.
- c. Its roof is formed of skin, superficial fascia and investing layer of deep cervical fascia.
- d. Its roof contains transverse cutaneous nerve.
- e. It is subdivided into four smaller triangles.

11. The roots of the brachial plexus pass between:

- a. Scalenus anterior and sternomastoid.
- b. Scalenus anterior and clavicle.
- c. Scalenus anterior and scalenus medius.
- d. Scalenus medius and scalenus posterior.
- e. Scalenus posterior and levator scapulae.

12. The posterior triangle of the neck:

- a. Is bordered by sternocleidomastoid, mandible and trapezius.
- b. Has the accessory nerve beneath fascia of its floor.
- c. Is roofed by prevertebral layer of fascia.
- d. Contains the internal jugular vein.
- e. Contains trunks of the brachial plexus.

13. One of the following arteries is not a content of the carotid triangle:

- a. The internal carotid artery.
- b. The superior thyroid artery...
- c. The occipital artery.
- d. The superficial temporal artery.
- e. The facial artery.

14. Regarding carotid triangle, select the correct statement:

- a. It is bounded by inferior belly of omohyoid.
- b. Hyoglossus forms its floor posteriorly.
- c. Internal jugular vein lies medial to carotid arteries.
- d. Sympathetic chain is enclosed in carotid sheath.
- e. It contains the carotid sinus.

15. Which of the following is correct regarding posterior triangle of the neck?

- a. Contains the spinal accessory nerve within its fascial roof.
- b. Contains the third part of subclavian artery beneath its fascial floor.
- c. Branches from thyrocervical trunk lie between its roof and its floor.
- d. All of the above.
- e. None of the above.

16. Regarding supraclavicular triangle, All are true statements Except:

- a. It is the lower smaller division of the posterior triangle.
- b. It is bounded superiorly by superior belly of omohyoid.
- c. It contains part of subclavian artery.
- d. Trunks of brachial plexus can be felt in the triangle.
- e. Suprascapular vessels run transversely in the triangle.

17. One of the following structures pierces the roof of the posterior triangle of neck:

- a. Phrenic nerve.
- b. Suprascapular artery.
- c. Transverse cervical artery.
- d. External jugular vein.
- e. Occipital artery.

18. The following muscles are in the muscular triangle Except:

- a. Sternohyoid.
- b. Thyrohyoid.
- c. Omohyoid
- d. Hyoglossus.
- e. Sternothyroid.

19. The following are contents of the digastric triangle Except:

- a. Submandibular salivary gland.
- b. Facial artery and vein.
- c. Hypoglossal nerve.
- d. Submental lymph nodes.
- e. Nerve to mylohyoid.

20. All the following are branches of facial nerve Except:

- a. Chorda tympani.
- b. Greater superficial petrosal.
- c. Deep petrosal.
- d. Nerve to stapedius.
- e. Nerve to posterior belly of digastric.

21. The ophthalmic nerve:

- a. Enters the orbit through the optic foramen.
- b. Carries corneal sensation.
- c. Is motor to Levator palpebrae superioris.
- d. Is sensory to the upper teeth.
- e. When injured the light reflex is abolished on the same side.

22. A lesion of the mandibular nerve after its exit from foramen ovale may result in:

- a. Loss of parotid gland secretion.
- b. Loss of taste sensation from anterior part of tongue.
- c. Loss of general sensations from posterior part of tongue.
- d. Paralysis of Buccinator.
- e. Loss of sensation of lower lip.

23. Maxillary Nerve:

- a. Leaves the skull through foramen oval.
- b. Enters the orbit through superior orbital fissure.
- c. Gives the posterior superior alveolar (dental) branch.
- d. Has otic ganglion attached to it.
- e. Supplies skin of lower lip.

24. One of the following is not given from the superior cervical sympathetic ganglia:

- a. The laryngopharyngeal branch.
- b. The internal carotid nerve.
- c. The cardiac branches.
- d. Communicating branches to the last four cranial nerves.
- e. The Ansa subclavian.

25. Choose one correct statement regarding the lingual nerve:

- a. It is a branch of the posterior division of maxillary nerve.
- b. It leaves skull through foramen rotundum.
- c. It carries fibers from first cervical spinal nerve.
- d. It carries taste from posterior third of tongue.
- e. It is closely related to the third molar tooth.

26. Concerning the maxillary nerve, one statement is false:

- a. It enters the pterygopalatine fossa through the pterygomaxillary fissure.
- b. Gives two sensory roots to the sphenopalatine ganglion.
- c. Enters the orbit through the infra-orbital fissure.
- d. Gives off a zygomatic branch.
- e. Contains only sensory fibers.

27. All is true regarding maxillary nerve Except:

- a. It is entirely sensory.
- b. Supplies upper lip.
- c. Supplies lower eyelid.
- d. Supplies skin over zygomatic bone.
- e. Runs through foramen oval.

28. The lingual nerve:

- a. Leaves the skull through the pterygomaxillary fissure.
- b. Carries fibers from the first cervical spinal nerve.
- c. Is closely related to the first molar tooth.
- d. Joins chorda tympani on hyoglossus muscle.
- e. Carries sensory fibers from the anterior part of tongue.

29. Regarding the mandibular nerve, All are true Except:

- a. Supplies all lower jaw teeth.
- b. Has motor and sensory fibers.
- c. Passes through foramen oval.
- d. Passes in lateral wall of cavernous sinus.
- e. Gives a meningeal branch

30. Injury to the motor root of the trigeminal nerve would cause paralysis of:

- a. Buccinator.
- b. Lateral pterygoid.
- c. Orbicularis oculi.
- d. Posterior belly of digastric.
- e. Platysma.

31. All the following statements regarding trigeminal ganglion are true Except:

- a. It lies in middle cranial fossa.
- b. It lies within a fold of dura.
- c. It contains bipolar nerve cells.
- d. Central processes of its cells form sensory root of trigeminal.
- e. Motor root of trigeminal lies beneath it.

32. Regarding chorda tympani; all are correct statements Except:

- a. It is a branch of facial nerve.
- b. It crosses the tympanic membrane.
- c. It joins lingual nerve beneath lateral pterygoid.
- d. It is related to medial side of lower third molar.
- e. It carries taste and parasympathetic fibers.

33. The Ansa cervicalis supplies one of the following muscles:

- a. Mylohyoid.
- b. Stylohyoid.
- c. Geniohyoid.
- d. Omohyoid.
- e. Thyrohyoid.

34. The following glands receive their parasympathetic innervation from the facial nerves Except:

- a. Lacrimal.
- b. Submandibular.
- c. Parotid.
- d. Nasal.
- e. Sublingual.

35. The maxillary nerve:

- a. Arises from the geniculate ganglion.
- b. Runs in the lateral wall of the cavernous sinus.
- c. Has the otic ganglion attached to it.
- d. Supplies the skin of the bridge of the nose.
- e. Has sensory and motor fibers.

36. The cervical branch of facial nerve supplies:

- a. Sternomastoid.
- b. Masseter.
- c. Zygomaticus major.
- d. Platysma.
- e. Buccinator.

37. The trochlear nerve is peculiar (different) from other cranial nerves because:

- a. It emerges from dorsal aspect of brain stem.
- b. It decussates with that of the opposite side.
- c. Its nucleus lies in the midbrain.
- d. It receives ipsilateral and contralateral cortico nuclear fibers.
- e. A & B only.

38. The maxillary nerve supplies all of the following Except:

- a. All upper jaw teeth.
- b. Sensory fibers to the palate.
- c. Meninges.
- d. The upper lip:
- e. The upper eyelid.

39. The facial nerve supplies all the following Except:

- a. Skin of face.
- b. Platysma.
- c. Secretomotor fibers to sublingual gland.
- d. Stapedius.
- e. Secretomotor fibers to lacrimal gland.

40. The glossopharyngeal nerve supplies the following:

- a. The otic ganglion.
- b. The stylopharyngeus muscle.
- c. The mucous membrane covering the tonsil.
- d. All the above.
- e. A & B only.

41. If you ask a patient to protrude his tongue and it deviates to the left, this indicates injury to which nerve?

- a. Left hypoglossal.
- b. Right hypoglossal.
- c. Lingual.
- d. Left glossopharyngeal.
- e. Right glossopharyngeal.

42. The glossopharyngeal nerve:

- a. Supplies all pharyngeal muscles.
- b. Carries taste sensation from circumvallate papillae.
- c. Passes superficial to hyoglossus muscle.
- d. All the above.
- e. None of the above.

43. The facial nerve:

- a. Leaves the skull through foramen spinosum.
- b. Supplies sensory fibers to the cornea.
- c. Is sensory to skin covering buccinator.
- d. Carries secreto-motor fibers for the parotid gland.
- e. Provides one limb of the corneal reflex arc.

44. The following are cutaneous branches of the cervical plexus Except

- a. Transverse cutaneous.
- b. Great auricular.
- c. Lesser occipital.
- d. Dorsal scapular.
- e. Supraclavicular.

45. The following are branches of the vagus nerve in the neck Except:

- a. Branches to the carotid body.
- b. Left recurrent laryngeal nerve.
- c. Pharyngeal branch.
- d. Superior laryngeal nerve.
- e. Cardiac branches.

46. Which branch of the following nerves synapses in the ciliary ganglion?

- a. Abducent nerve.
- b. Nerve to inferior oblique.
- c. Nerve to superior oblique.
- d. Ophthalmic.
- e. Superior division of oculomotor.

47. Which of the following nerves is a dorsal ramus of the second cervical nerve?

- a. Great auricular.
- b. Greater occipital.
- c. Lesser occipital.
- d. Transverse cervical.
- e. Supraclavicular.

48. Regarding vagus nerve; all are true Except:

- a. It is the longest cranial nerve.
- b. It is joined by the spinal root of the accessory nerve.
- c. It has two ganglia.
- d. It descends within the carotid sheath.
- e. Its injury leads to palpitation and tachycardia (increase in heart rate).

49. Which of the following fibers is contained in nerve of pterygoid canal:

- a. Preganglionic sympathetic.
- b. Postganglionic sympathetic.
- c. Postganglionic parasympathetic.
- d. Taste from anterior part of tongue.
- e. Taste from posterior part of tongue.

50. Regarding pharyngeal branch of vagus, all are true statements Except:

- a. It carries fibers that originate from cranial part of accessory nerve.
- b. It shares in forming the pharyngeal plexus of nerves.
- c. It supplies all pharyngeal constrictors.
- d. It supplies all soft palate muscles Except tensor palati.
- e. It supplies all muscles attached to posterior border of thyroid cartilage.

51. Anterior division of mandibular nerve gives the following branches Except:

- a. Nerve to medial pterygoid.
- b. Nerve to lateral pterygoid.
- c. Buccal nerve.
- d. Deep temporal nerves.
- e. Nerve to masseter.

52. The hypoglossal nerve supplies the following muscles Except:

- a. Hyoglossus.
- b. Inferior longitudinal muscle of the tongue.
- c. Genioglossus.
- d. Styloglossus.
- e. Mylohyoid.

53. Concerning the chorda tympani nerve; all are true Except:

- a. It is a branch of the facial nerve.
- b. It leaves the skull through stylomastoid foramen.
- c. It carries taste sensation from the anterior 2/3 of tongue.
- d. It supplies the submandibular gland.
- e. It joins the lingual nerve in the infratemporal fossa.

54. Which is the correct statement regarding cervical sympathetic trunk?

- a. It lies anterior to carotid sheath.
- b. It contains ascending postganglionic axons.
- c. It receives white rami communicates from cervical nerves.
- d. Supplies branches to the heart.
- e. Its middle ganglion is the largest.

55. If you ask a patient to say "Ah" and you notice deviation of the uvula to the right side, you suspect injury to which nerve?

- a. Trigeminal.
- b. Accessory.
- c. Glossopharyngeal.
- d. Hypoglossal.
- e. None of the above.

56. The parasympathetic root of otic ganglion is a branch of which nerve:

- a. Oculomotor.
- b. Trochlear.
- c. Facial.
- d. Glossopharyngeal.
- e. Vagus.

57. The hypoglossal nerve:

- a. Emerges from medulla between olive and inferior cerebellar peduncle.
- b. Leaves the skull through jugular foramen.
- c. Lies within the upper part of carotid sheath.
- d. Is joined by a branch from C2.
- e. It sensory to the anterior 2/3 of the tongue.

58. All are branches of nasociliary nerve Except:

- a. Long ciliary.
- b. Lacrimal.
- c. Anterior ethmoidal.
- d. Posterior ethmoidal.
- e. Infra trochlear.

59. All the following muscles are supplied by the hypoglossal nerve Except:

- a. Anterior belly of digastric.
- b. Hyoglossus.
- c. Genioglossus.
- d. Styloglossus.
- e. Intrinsic muscles of tongue.

60. The deep petrosal nerve is composed of nerve fibers whose cell bodies are located in which ganglion?

- a. Geniculate.
- b. Trigeminal.
- c. Pterygopalatine.
- d. Superior cervical sympathetic.
- e. Inferior vagal ganglion.

61. Injury of abducent nerve leads to:

- a. Internal squint.
- b. Inability to move eye upwards.
- c. Inability to move eye downwards.
- d. A dilated pupil.
- e. Loss of accommodation.

62. Inability to whistle or suck on a straw may indicate a lesion of which nerve:

- a. Maxillary.
- b. Mandibular.
- c. Facial.
- d. Glossopharyngeal.
- e. Hypoglossal.

63. All The following muscles are supplied by facial nerve Except:

- a. Buccinator.
- b. Posterior belly of digastric.
- c. Tensor tympani.
- d. Auricular muscles.
- e. Stylohyoid.

64. All the following muscles receive motor fibers from the cervical plexus Except:

- a. Thyrohyoid.
- b. Omohyoid.
- c. Sternohyoid.
- d. Mylohyoid.
- e. Geniohyoid.

65. Regarding the accessory nerve; All the following statements are true Except:

- a. It is a purely motor nerve.
- b. Its cranial part takes origin from the nucleus ambiguus.
- c. It emerges at the groove between pyramid and olive.
- d. Its spinal part runs between investing and prevertebral layers of cervical fascia.
- e. Its cranial part joins the vagus.

66. The following are branches of the cervical plexus Except:

- a. Phrenic.
- b. Great auricular.
- c. Suprascapular.
- d. Supraclavicular.
- e. Lesser occipital.

67. The Ansa cervicalis:

- a. Is a nerve loop on the internal jugular vein.
- b. Is formed of fibers from the upper three cervical nerves.
- c. Supplies most of the infrahyoid muscles.
- d. All of the above.
- e. A & B only.

68. Regarding cervical spinal nerves, All are true statements Except:

- a. Their dorsal rami supply skin and muscles of back.
- b. Upper four ventral rami form cervical plexus.
- c. Lower four ventral rami form brachial plexus.
- d. They receive grey rami communicates from the cervical sympathetic ganglia.
- e. They give white rami communicates to the cervical sympathetic ganglia.

69. Regarding the superior cervical sympathetic ganglion, all are true Except:

- a. It is connected by white rami communicates to the upper four cervical nerves.
- b. It forms plexus around internal carotid artery.
- c. It forms plexus around external carotid artery.
- d. It sends a pharyngeal branch.
- e. It gives cardiac branch.

70. All the following are branches of the middle cervical sympathetic ganglion Except

- a. Pharyngeal.
- b. Cardiac.
- c. Gray rami communicates to the fifth and sixth cervical nerves.
- d. Connections with the cervicothoracic ganglion.
- e. Branches to thyroid gland.

71. Preganglionic parasympathetic fibers to the lacrimal gland run in which nerve?

- a. Chorda tympani.
- b. Zygomaticotemporal.
- c. Greater petrosal.
- d. Lesser petrosal.
- e. Ophthalmic nerve.

72. One of the following cranial nerves contains parasympathetic fibers:

- a. The oculomotor nerve.
- b. The abducent nerve.
- c. The trochlear nerve.
- d. The vestibulocochlear.
- e. The olfactory nerves.

73. The glossopharyngeal nerve supplies the mucous membrane of:

- a. Nasopharynx.
- b. Oropharynx.
- c. Laryngopharynx.
- d. Larynx.
- e. All of the above.

74. One of the following cranial nerves does not contain preganglionic parasympathetic fibers:

- a. The 3rd.
- b. The 5th.
- c. The 7th.
- d. The 9th.
- e. The 10th.

75. Regarding the spinal accessory nerve, all are true Except:

- a. Arises entirely from the spinal cord.
- b. Enters the skull via the foramen magnum.
- c. Leaves the skull via the jugular foramen.
- d. Contributes to the pharyngeal plexus of nerves.
- e. Its injury in the posterior triangle results in wasting of the trapezius on the same side.

1. The medial surface of thalamus:

- a. Forms lateral wall of 3rd ventricle
- b. Is limited inferiorly by hypothalamic sulcus
- c. Is connected to the opposite surface by interthalamic adhesions
- d. All of the above
- e. A & B only

2. Which of the following is or are included in the meta thalamus:

- a. Geniculate bodies
- b. Pineal body
- c. Habenular nuclei
- d. Posterior commissure
- e. None of the above

3. Which is the thalamic surface related to tela-choroidae of two ventricles?

- a. Superior
- b. Inferior
- c. Medial
- d. Lateral
- e. None of the above

4. The thalamic nucleus in which spinal & medial lemnisci terminate is:

- a. Ventralis posterior medialis (VPM)
- b. Ventralis anterior
- c. Ventralis intermedius
- d. Ventralis posterior lateralis (VPL)
- e. None of the above

5. Which is the thalamic surface related to posterior limb of internal capsule?

- a. Superior
- b. Inferior
- c. Medial
- d. Lateral
- e. None of the above

6. Astereognosis & receptive aphasia characterize lesions in:

- a. Dominant parietal lobe
- b. Frontal lobe
- c. Temporal lobe
- d. Occipital lobe.
- e. Non dominant parietal lobe

7. A lesion in one cortical auditory area results in:

- a. Unilateral slight hearing loss only
- b. Bilateral impaired hearing
- c. Inability to understand the meanings of words.
- d. All of the above
- e. None of the above

8. One of the following structures does not contribute to the floor of 3rd ventricle.

- a. Lamina terminalis
- b. Optic chiasma
- c. Mamillary bodies
- d. Tuber cinereum
- e. Posterior perforated substance

9. Which is the thalamic surface that shares in the formation of the floor of body of lateral ventricle?

- a. Medial part of superior surface
- b. Lateral part of superior surface
- c. Inferior surface
- d. Medial surface
- e. Lateral surface

10. The lateral wall of the third ventricle is formed by:

- a. Thalamus
- b. Hypothalamus
- c. Lamina terminalis
- d. All of the above
- e. A & B only

11. Sensations received by the thalamus are Except:

- a. Exteroceptive
- b. Proprioceptive
- c. Gustatory
- d. Visual & auditory
- e. Smell

12. Nucleus Ventralis posterior of thalamus receives the following Except:

- a. Medial lemniscus
- b. Lateral lemniscus
- c. Trigeminal lemniscus
- d. Spinal lemniscus
- e. Fibers from nucleus solitarius

13. The following are efferent hypothalamic fibers Except:

- a. Corticohypothalamic.
- b. Supraoptic hypophyseal
- c. Tuberoinfundibular
- d. Mamillo tegmental
- e. Para ventriculo hypophyseal

14. The anterolateral boundary of interpeduncular fossa is formed by:

- a. Optic chiasma
- b. Optic tract
- c. Crus cerebri
- d. Pons
- e. Posterior perforated substance

15. Regarding Broca's area, select the correct answer:

- a. It is a sensory speech center.
- b. It is located in middle frontal gyrus.
- c. It is present in the dominant hemisphere.
- d. It is supplied by anterior cerebral artery.
- e. None of the above is correct.

16. Frontal lobe in the dominant hemisphere is concerned with the following functions Except:

- a. Production of speech
- b. Performance of skilled movements
- c. Initiation of conjugate lateral eye movements
- d. Understanding of received auditory information.
- e. Control of behavior

17. Frontal lobe in the non-dominant hemisphere contains the following areas Except:

- a. Primary motor area (4)
- b. Frontal eye field (I)
- c. Supplementary motor area (MS II)
- d. Premotor area (6)
- e. Werneck's speech area

18. Regarding the first visual area (17):

- a. Lies above and below the post calcarine sulcus.
- b. Extends below the pericalcarine sulcus.
- c. Is supplied by posterior cerebral artery.
- d. Its lesion produces contralateral homonymous hemianopia.
- e. All of the above is true.

19. Regarding the motor speech center, all the statements are true Except:

- a. Found in the inferior frontal gyrus.
- b. Present in dominant hemisphere only.
- c. Its lesion leads to expressive aphasia.
- d. Is connected to Werneck's area.
- e. Is supplied by anterior cerebral artery.

20. Parietal lobe in the dominant hemisphere is concerned with the following Except:

- a. Facial recognition
- b. Stereognosis
- c. Perception of cutaneous stimuli
- d. Perception of proprioceptive impulses
- e. Comprehension of received auditory & visual information.

21. Cortical areas in the parietal lobe of the non-dominant hemisphere include:

- a. supplementary motor area
- b. Werneck's area
- c. First & second somatosensory area
- d. Facial recognition area
- e. Secondary acoustic area

22. Lesion of frontal cortex in the dominant hemisphere results in all but one of the following:

- a. Expressive aphasia
- b. Receptive aphasia
- c. Contralateral paralysis
- d. Increased tendon jerks
- e. Behavioral & mood changes

23. A pituitary tumor pressing the central part of optic chiasma will lead to:

- a. Total blindness
- b. Bitemporal hemianopia
- c. Binasal hemianopia
- d. Homonymous hemianopia
- e. None of the above

24. Posterior limb of internal capsule contains all of the following Except:

- a. Cortico-spinal fibers
- b. Superior thalamic radiation
- c. Cortico-rubral fibers
- d. Fasciculus lenticularis
- e. Anterior thalamic radiation

25. Regarding the optic tract, which is an untrue statement?

- a. Its fibers synapse in medial geniculate body
- b. Emerges from optic chiasma.
- c. Its lesion produces contralateral homonymous hemianopia.
- d. Contains fibers for light reflex.
- e. Is connected to pretectal nucleus.

26. Bulb of posterior horn of lateral ventricle is formed of:

- a. Tapetum
- b. Optic radiations
- c. Calcarine sulcus
- d. Forceps major
- e. Collateral sulcus

27. Projection fibers include:

- a. Forceps major
- b. Cingulum
- c. Fasciculus uncinatus
- d. Fornix
- e. Optic radiations

28. Select one of the cortical structures forming the limbic system:

- a. Hypothalamus
- b. Hippocampus
- c. Anterior thalamic nuclei
- d. Amygdaloid complex
- e. Anterior commissure

29. All is true about the occipital eye field Except:

- a. Is independent of visual stimuli.
- b. Is responsible for reflex conjugate lateral movements of the eyes.
- c. Is confined to areas 18 & 17.
- d. Is connected to frontal eye field.
- e. Is concerned with following moving objects.

30. All statements about optic nerve are correct Except:

- a. It originates from ganglion cell layer of retina.
- b. Its fibers are myelinated.
- c. Its fibers are ensheathed with Schwann cells.
- d. Its fibers pierce sclera at lamina cribrosa
- e. Its average length is 40 mm.

31. In horizontal section through the brain including frontal & occipital poles the following basal nuclei appear clearly Except:

- a. Head of caudate
- b. Claustrum
- c. Globus pallidus
- d. Amygdaloid complex
- e. Putamen

32. The external capsule lies between:

- a. Insula & claustrum
- b. Thalamus & lentiform
- c. Claustrum & lateral surface of lentiform
- d. Caudate & lentiform
- e. None of the above

33. Cortico nuclear fibers in the internal capsule are located in:

- a. Anterior limb
- b. Genu
- c. Posterior limb
- d. Retro lentiform part
- e. Sub lentiform part.

34. Regarding arterial supply of posterior limb of internal capsule:

- a. Upper half by anterior cerebral
- b. Anterior 1/3 of inferior 1/2 by posterior communicating
- c. Posterior 2/3 of inferior 1/2 by posterior choroidal
- d. All of the above is true.
- e. None of the above is true.

35. Tapetum of corpus callosum is formed by:

- a. Forceps minor
- b. Anterior fibers of splenium
- c. Posterior fibers of trunk
- d. All of the above
- e. B & C only

36. Corpus callosum is composed of:

- a. Rostrum
- b. Splenium
- c. Genu
- d. Trunk
- e. All of the above

37. Arterial supply of anterior ends of caudate & lentiform nuclei is derived from:

- a. Central branches of middle cerebral
- b. Central branches of posterior cerebral
- c. Central branches of anterior cerebral
- d. All of the above
- e. None of the above

38. Arterial supply of upper 1/2 of internal capsule is derived from:

- a. Central branches of middle cerebral
- b. Central branches of posterior cerebral
- c. Central branches of anterior cerebral
- d. All of the above
- e. None of the above

39. The following are commissural fibers Except:

- a. Forceps major
- b. Fasciculus uncinatus
- c. Tapetum
- d. Fornix
- e. Forceps minor

40. Acoustic radiation fibers in the internal capsule are located in:

- a. Anterior limb
- b. Genu
- c. Posterior limb
- d. Retro lentiform part
- e. Sub lentiform part.

41. Frontal lobes are connected by:

- a. Forceps major
- b. Fibers in the rostrum
- c. Forceps minor
- d. All of the above
- e. B & C only

42. The following include descending projection fibers:

- a. Cortico nuclear
- b. Auditory radiation
- c. Anterior thalamic radiations
- d. Fornix
- e. Forceps minor

43. Regarding the fourth ventricle:

- a. Is the cavity of hind brain.
- b. Communicates with subarachnoid space.
- c. Has a large median aperture
- d. All of the above
- e. None of the above

44. Regarding the lateral ventricle select an untrue statement:

- a. Is the cavity of diencephalon.
- b. Is lined with ependyma.
- c. Has a body & 3 horns.
- d. Communicates with third ventricle.
- e. Contains C.S.F.

45. Regarding the lateral ventricle, the head of caudate nucleus is related to which part?

- a. Anterior horn
- b. Posterior horn
- c. Inferior horn
- d. Body of the ventricle
- e. None of the above

46. Association fibers include the following Except:

- a. Corona radiata
- b. Cingulum
- c. Short arcuate fibers
- d. Uncinate fasciculus
- e. Fronto-occipital fasciculus

47. Regarding cerebrospinal fluid (C.S.F.) all statements are true Except:

- a. Is formed by choroid plexuses of ventricles.
- b. Helps in removal of metabolites.
- c. Finally reaches the blood stream.
- d. Reaches cisterna Magna through the aqueduct of Sylvius.
- e. Is filtered through arachnoid granulations.

48. In the dominant hemisphere cortical branches of middle cerebral artery supply the following areas Except:

- a. Auditory area
- b. Broca's area
- c. Frontal eye field
- d. Wernicke's area
- e. The whole primary motor area

49. Tail of caudate nucleus & hippocampus are present in which part of lateral ventricle?

- a. Anterior horn
- b. Posterior horn
- c. Inferior horn
- d. Body of the ventricle
- e. None of the above

50. One of the following arteries does not share in formation of circle of Willis:

- a. Middle cerebral
- b. Anterior cerebral
- c. Posterior cerebral
- d. Internal carotid
- e. Posterior communicating

51. Which of the following veins end in cavernous sinus:

- a. Superior cerebral veins
- b. Inferior cerebral veins
- c. Deep middle cerebral vein
- d. Superficial middle cerebral vein
- e. None of the above

52. Vertebrobasilar system supplies the following Except:

- a. Cerebellum
- b. Inner ear
- c. Visual cortex
- d. Temporal pole
- e. Pons

53. Anterior cerebral artery supplies:

- a. Paracentral lobule
- b. Auditory cortex
- c. Inferior parietal lobule
- d. Temporal pole
- e. Visual cortex

54. Posterior cerebral artery supplies:

- a. Lateral surface of occipital lobe
- b. Visual cortex
- c. Lower border of temporal lobe
- d. Thalamus
- e. All of the above

55. The vein that opens in the straight sinus is:

- a. Great cerebral
- b. Internal cerebral
- c. Deep middle cerebral
- d. All of the above
- e. None of the above



1. A muscle derived from the third pharyngeal arch:

- a. Stapedius.
- b. Superior parathyroid.
- c. Stylopharyngeus.
- d. Cricothyroid.
- e. Platysma.

2. The hyoid bone is derived from:

- a. Migrating paraxial mesoderm of 1st & 2nd arches.
- b. Migrating paraxial mesoderm of 2nd & 3rd arches.
- c. Migrating paraxial mesoderm of 3rd & 4th arches.
- d. Migrating neural crest cells of 1st & 2nd arches.
- e. Migrating neural crest cells of 2nd & 3rd arches.

3. The two bones that develop from the 1st pharyngeal arch are:

- a. Malleus & incus.
- b. Malleus & stapes.
- c. Incus & stapes.
- d. Incus & styloid process.
- e. Styloid process & malleus.

4. The body of the hyoid bone is derived from the:

- a. 1st & 2nd arches.
- b. 2nd & 3rd arches.
- c. 3rd & 4th arches.
- d. 4th & 5th arches.
- e. 5th & 6th arches.

5. The muscle that develops from the 4th branchial arch is the:

- a. Platysma.
- b. Masseter.
- c. Stapedius.
- d. Stylopharyngeus.
- e. Cricothyroid.

6. The nerve that supplies muscles derived from 2nd branchial arch is:

- a. Hypoglossal nerve.
- b. Facial nerve.
- c. Glossopharyngeal nerve.
- d. Trigeminal nerve.
- e. Vagus nerve.

7. All the following muscles arise from second pharyngeal arch, Except:

- a. Buccinator.
- b. Posterior belly of digastric.
- c. Temporalis.
- d. Platysma.
- e. Orbicularis oculi.

8. The intermaxillary segment is formed by the fusion of:

- a. Maxillary prominences.
- b. Mandibular prominences.
- c. Palatine shelves.
- d. Lateral nasal processes.
- e. Medial nasal processes.

9. The median part of the upper lip is derived from the:

- a. Maxillary prominence.
- b. Mandibular prominences.
- c. Frontal prominence.
- d. Lateral nasal prominence.
- e. Intermaxillary segment.

10. What type of facial anomaly that starts at lip & extends to orbit?

- a. Oblique facial cleft.
- b. Median cleft.
- c. Cleft lip.
- d. Anterior cleft.
- e. Posterior cleft.

11. The major portion of the palate is derived from:

- a. Maxillary swellings (prominences).
- b. Medial nasal process (prominences).
- c. Mandibular swellings (prominences).
- d. Nasal septum.
- e. Frontal prominence.

12. Oblique facial cleft is due to:

- a. Failure of fusion of lingual swellings.
- b. Failure of fusion of nasal septum and palate.
- c. Failure of fusion of medial and lateral nasal folds.
- d. Failure of fusion of lateral nasal fold and maxillary swelling.
- e. Failure of fusion of medial nasal fold and maxillary swelling.

13. Failure of closure of anterior neuropore of the neural tube leads to:

- a. Anencephaly.
- b. Spina bifida.
- c. Microcephaly.
- d. Hydrocephalus.
- e. Scaphocephaly.

14. The structure(s) derived from spongioblasts of neural crest is (are):

- a. Dura mater.
- b. Oligodendrocytes.
- c. Microglia.
- d. Pia mater.
- e. Chromaffin cells.

15. Neuroepithelial cells give rise to:

- a. Neuroblasts.
- b. Glioblasts.
- c. Ependymal cells.
- d. All of the above.
- e. None of the above.

16. Alar plates form:

- a. Lateral horns.
- b. Dorsal horns.
- c. Ventral horns.
- d. All of the above.
- e. None of the above.

17. Neural crest gives all the following Except:

- a. Spinal ganglia.
- b. Red nucleus.
- c. Arachnoid mater.
- d. Suprarenal medulla.
- e. Sympathetic ganglia.

18. The spinal cord in newly born infant reaches the level of:

- a. 1st lumbar vertebra.
- b. 2nd lumbar vertebra.
- c. 3rd lumbar vertebra.
- d. 5th lumbar vertebra.
- e. 1st sacral vertebra.

19. The commonest site of spino bifida is:

- a. Cervical.
- b. Upper thoracic.
- c. Mid-thoracic.
- d. Thoraco-lumbar.
- e. Lumbo-sacral.

2

Section



Physiology

Chapter 1 ?

Overview of CNS

1. Concerning synapses:

- a. Chemical synapses are not widely distributed in CNS.
- b. Electric synapses are characterized by delayed transmission.
- c. Electric synapse is a unidirectional synapse.
- d. Chemical synapses are affected by many drugs & chemicals.
- e. In chemical synapses, gap junction conduct impulses between presynaptic & postsynaptic structure.

2. Regarding neurotransmitters:

- a. ACh is an inhibitory transmitter.
- b. Dopamine function is related to mood.
- c. Catecholamines are slow potent transmitters.
- d. Glutamate is an inhibitory transmitter.
- e. GABA is an excitatory transmitter.

3. The mechanism of synaptic transmission includes:

- a. Ca^{++} ions are released from the presynaptic knob.
- b. Presynaptic knobs release only an excitatory transmitter.
- c. Chemical transmission causes development of only excitatory postsynaptic state.
- d. Excitatory transmitters hyperpolarize the postsynaptic membrane.
- e. The released transmitter combines with specific receptors on postsynaptic membrane.

4. Regarding neurotransmitters:

- a. Neuropeptides are rapidly acting transmitters.
- b. Rapidly acting transmitters are slowly inactivated.
- c. The action of a neurotransmitter whether excitatory or inhibitory depends on its receptors.
- d. Dopamine is an important excitatory transmitter.
- e. Substance P function to decrease pain transmission.

5. Regarding neurotransmitters:

- a. Neurohormones are secreted at synapses to affect neurons activity.
- b. Neuromodulator is secreted by neuron into blood.
- c. The same neurotransmitter may excite a target organ and inhibit another one.
- d. A neurotransmitter has a specific one type of receptor.
- e. Neuropeptides mediate acute responses of the nervous system.

6. Regarding Acetylcholine:

- a. Is involved in arousal and motor activities.
- b. Its receptor sensitivity is increased in myasthenia gravis.
- c. Is always excitatory to its target cells.
- d. Is involved in regulation of feeding.
- e. Its action is potentiated by benzodiazepines.

7. Regarding neurotransmitters:

- a. Glycine causes depolarization of postsynaptic membrane.
- b. GABA is antagonized by strychnine.
- c. GABA causes increase Na^{+} influx to postsynaptic neuron.
- d. Glycine is a facilitatory transmitter.
- e. Benzodiazepines potentiate the actions of GABA.

1. Which of the following does not contain channels that are activated by mechanical distortion:

- a. Muscle spindle.
- b. Pacinian corpuscles.
- c. Hair cells in cochlea.
- d. Hair end organs.
- e. Cones.

2. Regarding sensory receptors:

- a. Receptors are non-specific to stimuli according to Muller's law.
- b. Stimulation of certain receptors can produces different modalities of sensations
- c. Receptors are stimulated by adequate stimuli at a low threshold.
- d. Receptor potential is inversely related to the intensity of stimulus according to Weber-Fechner law.
- e. The same receptor may be stimulated by different stimuli to produce different sensations.

3. Regarding sensory receptors:

- a. The magnitude of receptor potential and the magnitude of action potentials discharged in afferent nerve are directly related to logarithm of stimulus intensity.
- b. Receptors that decrease their receptor potential rapidly in response to constant stimulation are tonic receptors.
- c. Phasic receptors continue to discharge impulses even with a constant stimulus.
- d. Slowly adapting receptors help in body protection.
- e. Teleceptors detect cutaneous changes in environment.

4. Compared to the back of the body, tip of finger is characterized by having:

- a. Less number of receptors.
- b. Less number of afferent neurons.
- c. Less convergence of afferents.
- d. Less number of sensory units.
- e. Less size of cortical representation area.

5. Locality of the stimulus is discriminated by:

- a. Muller's law.
- b. Weber Fichner law.
- c. Law of specific nerve energies.
- d. Projection Theory.
- e. Labelled-line theory.

6. Regarding sensations, all of the following are true EXCEPT:

- a. Cutaneous sensations include touch, pain & proprioception.
- b. Deep sensations include proprioception & pain.
- c. Mixed sensations include stereognosis & vibration.
- d. Tactile sensations include touch, pressure & vibration.
- e. Kinesthetic sensations inciude sense of position and movement.

7. Which of the following receptor is specific for the rate of movement:

- a. Muscle spindle.
- b. Golgi tendon organ.
- c. Ruffini end organ.
- d. Paccinian corpuscle.
- e. Receptors in capsules and ligaments of joints.

8. Receptive field concept:

- a. The larger the receptive field, the less the receptor number.
- b. Receptive field is small in high sensory value areas.
- c. The less the receptive field, the less the threshold distance.
- d. There is overlap of adjacent receptive fields
- e. All of the above.

9. Adaptation of a sensory receptors to a stimulus, includes:

- a. A more intense sensation.
- b. A sensation sharply localized to an area in the body.
- c. Diminished sensation in the presence of applied stimulus.
- d. Increase frequency of receptors action potential.
- e. Increase amplitude of receptor potential.

10. Discrimination of stimulus type depends upon:

- a. Muller's law.
- b. Number of receptors activated by the stimulus.
- c. Phantom limb phenomenon.
- d. Weber Fichner law.
- e. Frequency of action potentials in the afferent nerve.

11. Concerning Coding of sensory information, All are true EXCEPT:

- a. Modality discrimination depends on specificity of receptors and Muller's law.
- b. Locality discrimination depends on law of projection
- c. Intensity discrimination depends on the number of activated receptors (recruitment of receptors) and the frequency of action potentials according to Weber. Fechner law.
- d. All of the above
- e. None of the above.

12. Tonic receptors include:

- a. Touch receptors.
- b. Nociceptors.
- c. Thermal receptors.
- d. Taste receptors.
- e. Electromagnetic receptors.

13. Receptors:

- a. Are detectors of only adequate stimulus.
- b. Act as transducers.
- c. Are located centrally in relation to afferent nerve.
- d. Are only modified organs.
- e. Generate action potential in response to their stimulation.

14. The relationship between the intensity of stimulus applied to receptors and the frequency of action potential discharged in the afferent nerve is:

- a. A logarithmic one.
- b. A plateau one.
- c. A directly simple proportional one.
- d. An inverse one.
- e. Is the "Muller's law".

15. Phasic receptors include:

- a. Pain receptors.
- b. Proprioceptors.
- c. Touch receptors.
- d. Pressure receptors.
- e. Thermoreceptors.

16. Regarding sensory receptors:

- a. They can-only detect changes in external environment.
- b. They respond to stimuli by local non propagated potential changes in receptor membrane.
- c. All receptors adapt to constant stimulation.
- d. They can produce action potential in receptor membrane.
- e. All receptors release transmitters to conduct impulses to afferent neurons.

17. Mechanoreceptors include:

- a. Rods and cones.
- b. Krause end bulb.
- c. Smell receptors.
- d. Nociceptors.
- e. Hair end organ.

18. Pain receptors:

- a. Adapt rapidly to a constant stimulus.
- b. Are densely present in wall of viscera.
- c. Their mechanism of stimulation is mechanical displacement.
- d. Are stimulated by a rise in the local K^+ concentration.

19. Discrimination of stimulus intensity depends upon:

- a. Number of receptors activated by the stimulus.
- b. Muller's law.
- c. Specific pathway for each stimulation.
- d. Adequate stimulus detected by the receptor.
- e. Law of projection.

20. Which of the following could NOT be an exteroceptor:

- a. Muscle spindle.
- b. Thermoreceptor.
- c. Nociceptor.
- d. Pacinian corpuscle.
- e. Rods.



Chapter 3 ? Anterolateral system & spinothalamic tract

1. Regarding Crude Touch sensation, choose the most correct statement:

- a. Is perceived at the level of the sensory cortex.
- b. Is transmitted through dorsal column pathway.
- c. Its receptors are Meissner's corpuscles.
- d. Is better felt in non-hairy skin areas.
- e. It is one of the protopathic sensations.

2. Impulses serving pain sensation in the right leg are carried:

- a. By fibers in the right spinothalamic tract.
- b. By left posterior root ganglion.
- c. By the same spinal cord tract which serves proprioception.
- d. To the thalamus on the left side.
- e. To the left area 6.

3. In Thermal sensations:

- a. Receptors for warm sensations are Paccinian corpuscles.
- b. Receptors for cold are Krause end bulb and Meissner's corpuscles
- c. Thermoreceptors are stimulated by mechanical displacement of receptors.
- d. Warm receptors send impulses through C fibers, while cold receptors send impulses through As and C fibers.
- e. Thermal hot and cold spots of receptors are equally distributed all over the body

4. Regarding warm receptors, the following is correct:

- a. Warm receptors are rapidly adapting phasic receptors.
- b. There are 3-10 times more warm receptors than cold receptors.
- c. Signals from warm receptors are transmitted along C fibers.
- d. Are sensitive in the range of 10-43 °C.
- e. Transmit impulses in the lemniscal system.

5. Crude touch sensation:

- a. Is carried along Aδ fibers.
- b. Includes two-point discrimination.
- c. Includes stereognosis.
- d. Is perceived by Meissner's corpuscles.
- e. Has tonic receptors.

6. The antero-lateral system of the spinal cord:

- a. Carries crude touch sensation.
- b. Is a fast-conducting system.
- c. Transmit signals faster than dorsal column.
- d. Carries vibration sense.
- e. Carries only epicritic senses.

7. Paleospinothalamic pathway:

- a. Has Aδ type of afferents.
- b. Carries impulses that arrives to lamina marginalis.
- c. Relays most of its fibers in the postcentral gyrus.
- d. Relay small percentages of its impulses to reticular formation.
- e. SGR is where its 1st order neuron terminates.

8. In Thermal sensations, all of the following are true EXCEPT:

- a. Cold receptors are more numerous than Hot receptors, but Hot receptors adapt more rapidly than cold receptors.
- b. Are transmitted through lateral division of anterolateral system.
- c. Temperature higher than 45°C produces hot pain and temperature lower than 10°C produces cold pain.
- d. The maximal discharge of hot receptors occurs at 45°C and of cold receptors occurs at 25°C.
- e. They are carried by A_β fibers.

9. The spinothalamic system transmits the following sensation:

- a. Kinesthetic sensations.
- b. Vibration sensation.
- c. Pain and temperature.
- d. two-point discrimination
- e. Deep pressure

10. A person immersed in a pool with a temperature of the water at 20°C, will in the first few minutes, feels:

- a. Pain.
- b. Equally as cold as after 5 minutes of immersion.
- c. Less cold than after 5 minutes of immersion.
- d. Much colder than after 5 minutes of immersion.
- e. A warm sensation at first, then he feels cold.

11. Action potentials recorded after stimulation of thermal receptors show that:

- a. The maximal discharge in the cold nerve fibers is at 25°C.
- b. Action potentials are recorded in warm nerve fibers between 10-35°C.
- c. Action potentials are recorded in cold pain fibers at 0°C.
- d. The maximum rate of discharge from warm pain and cold pain fibers occurs at 40°C.
- e. Decrease frequency of action potential in cold receptor afferents rapidly than that of warm receptor afferents.

12. Concerning temperature sensation:

- a. Temperature receptors are stimulated by mechanical displacement of receptors.
- b. Cold receptors continue to be activated even when the skin temperature is below zero°
- c. Cold and warm receptors have very specific non-overlapping ranges of temperature sensitivity.
- d. Warm but not cold receptors exhibit adaptation.
- e. The thalamus detects the crude degrees of temperature.

13. The fast pain sensation that accompanies a painful cutaneous trauma is carried to C.N.S. by fibers that belong to:

- a. A α fibers.
- b. A δ fibers.
- c. A β fibers.
- d. C fibers.
- e. B fibers.

14. Regarding itch and tickling sensations, all of the following are true EXCEPT:

- a. Tickle is felt by local repeated mechanical stimulation.
- b. Itch is caused by chemicals such as histamine.
- c. They are protopathic sensations.
- d. The afferent fiber for these sensations is a slowly conducting un-myelinated fiber.
- e. They are transmitted through medial lemniscal system.

15. Which of the followings is NOT a stimulus for nociceptors:

- a. Ischemia.
- b. Spasm.
- c. Overdistention.
- d. Accumulation of metabolites.
- e. A temperature of 40°C.



1. Pain produced by tissue damage is due to release of:

- a. Excess norepinephrine.
- b. ATP.
- c. Kinins.
- d. Heparin.
- e. Dopamine.

2. Deep pain:

- a. Is due to stimulation of cutaneous pain receptors.
- b. Is pricking in character.
- c. Is poorly localized.
- d. Is carried along A delta fibers.
- e. Is associated with sympathetic overactivity.

3. Which of the following is NOT likely to stimulate pain receptors in the intestine:

- a. Inflammation of the wall.
- b. Sharp cut through their wall.
- c. Distention.
- d. Spasmodic contraction behind an obstruction.
- e. Ischemia.

4. Cutaneous pain:

- a. Occurs as fast pain and then slow pain.
- b. May be referred to other structures.
- c. Is associated with depressor autonomic effects.
- d. Carried by C fibers only.
- e. Is rapidly adapting than deep pain.

5. Compared to fast pain, slow pain is characterized by:

- a. Occurs before fast pain.
- b. Perceived at the sensory cortex.
- c. Sharply localized.
- d. Occurs in skin, deep structures and viscera.
- e. Glutamate is the neurotransmitter involved in its conduction.

6. Slow pain, all of the following are true EXCEPT:

- a. May occur without tissue damage.
- b. Occurs due to ischemia and spasm.
- c. Occurs due to over distension of viscera.
- d. Transmitted through paleo-lateral spinothalamic tract.
- e. Carried by Aδ and C fibers.

7. Body reactions to pain include all of the following, EXCEPT:

- a. Deep and visceral pain induce somatic rigidity of muscles.
- b. Fast pain induces sympathetic autonomic effects.
- c. Reticular formation and intralaminar thalamic nuclei receive pain sensation & project impulses to the whole cortex to produce arousal reaction.
- d. Fast cutaneous pain induces emotional depression.
- e. Deep and visceral pains are poorly localized than cutaneous fast pain.

8. Concerning visceral pain, which of the followings statements is NOT true

- a. May cause reflex vomiting.
- b. May cause contraction of overlying skeletal muscle.
- c. May refer to other somatic structures.
- d. Is well localized.
- e. May cause change in arterial blood pressure.

9. In hyperalgesia:

- a. Hyperalgesia is a physiological exaggeration of pain sensation.
- b. In 1^{ry} hyperalgesia, pain threshold is normal while it is decreased by kinins in 2^{ry} hyperalgesia.
- c. 1^{ry} hyperalgesia occurs in healthy skin, while 2^{ry} hyperalgesia occurs in inflamed skin.
- d. 1^{ry} hyperalgesia is due to facilitation in the spinal cord, while 2^{ry} hyperalgesia is due to kinins.
- e. In 1^{ry} hyperalgesia, a non-painful stimulus becomes painful, while in 2^{ry} hyperalgesia a painful stimulus becomes more painful.

10. Tic douloureux, all of the following are true EXCEPT:

- a. Is due to inflammation in the sensory root of trigeminal.
- b. It is characterized by spontaneous attacks of pain along the distribution of the 9th cranial nerve or along the distribution of one of the branches of the 5th cranial nerve.
- c. The pain may be initiated by a non-noxious stimulus e.g. smiling or swallowing.
- d. Is treated by cutting the sensory root of the involved nerve.

11. Pain strictly limited to a dermatome should suggest the diagnosis of:

- a. Syphilis.
- b. Herpes zoster.
- c. Tuberculosis.
- d. B12 deficiency
- e. Syringomyelia

12. Concerning primary hyperalgesia:

- a. Pain threshold is high.
- b. May be due to facilitation of receptors by chemicals released from injury.
- c. Skin is normal.
- d. Explained by convergence-facilitation in the spinal cord.
- e. A painful stimulus is more painful.

13. Regarding secondary hyperalgesia:

- a. Pain threshold is low.
- b. Occurs in the healthy skin around injured skin.
- c. Is due to facilitation of receptors.
- d. A non-painful stimulus becomes painful.
- e. Explained by Weber-Fechner law.

14. Pain pathways:

- a. Neo-spinothalamic tract carries pain to the thalamus.
- b. As fibers carry pain to the cortex.
- c. 90% of neo-spinothalamic fibers end in reticular formation.
- d. Deep and visceral pains are carried by neo-lateral spinothalamic pathway.

15. Cutaneous pain:

- a. Receptors adapt to stimulation more rapidly than touch receptors
- b. Can be caused by excitation of receptors by K^+ ions released in injured tissues.
- c. Can be caused by over-stimulation of touch receptors.
- d. Is carried into the spinal cord by a long A beta fiber.
- e. Transmitted to the higher centers through the lemniscal system.

16. Visceral pain can NOT result from which of the following:

- a. Ischemia.
- b. Escape of HCL through rupture duodenal ulcer.
- c. Overdistension of hollow organ.
- d. Visceral spasm.
- e. Cutting through the viscera.

17. Slow pain is associated with:

- a. Depressor autonomic effects.
- b. Type A delta fibers.
- c. Skin temperature between 35°C and 45°C.
- d. Inactivation of warmth receptors
- e. Transmission of signals in neospinothalamic tract.

18. Slow pain, all of the following are true EXCEPT:

- a. Substance P is the transmitter involved in its transmission.
- b. Perceived mainly at reticular formation, midbrain and intralaminar thalamic nuclei.
- c. Is poorly localized due to low perception at the sensory cortex.
- d. Transmitted through neo-spinothalamic pathway.
- e. May be accompanied by parasympathetic effects.



1. Referred pain, all of the following are true EXCEPT:

- a. Is common with visceral and deep pain.
- b. Is due to facilitation of afferent pain pathways of skin at SGR.
- c. Is due to irritation of skin receptors by the visceral inflammation.
- d. Pain is felt at skin due to convergence of afferents from skin and viscera.
- e. May be due to projection of pain sensation to the skin when the viscera are inflamed.

2. Headache is NOT commonly produced by which of the following:

- a. Irritation of meninges.
- b. Dilatation of intracranial blood vessels.
- c. Otitis media.
- d. Loss of cerebrospinal fluid
- e. Hemorrhage.

3. Which sentence is NOT true regarding Intracranial headache:

- a. Is a type of referred pain.
- b. May be caused by error of refraction.
- c. Can be caused by lumbar puncture.
- d. Can be produced by hypertension.
- e. Is a frontal headache if the supratentorial region is irritated.

4. Extracranial headache may NOT be produced by:

- a. Sinusitis.
- b. Errors of refraction.
- c. Sustained contraction of the neck muscles.
- d. Otitis media
- e. Hypertension.

5. Gallbladder pain is referred to the:

- a. Umbilicus.
- b. Tip of the right scapula.
- c. Left shoulder.
- d. Left lumbar region.
- e. Inguinal region and testis.

6. Prodromal symptoms (hallucinations) experienced by patients with migraine headache is caused by:

- a. An increased blood flow to the brain tissue.
- b. Vasospasm of intracranial vessels.
- c. Decreased GABA secretion in the sensory areas of the cortex.
- d. Excessive sleep and inactivity.
- e. Severe dilatation of intracranial blood vessels.

7. In Headache:

- a. Spasm of head muscles produces intracranial headache.
- b. Is due to inflammation in brain tissue.
- c. May be intracranial due to meningeal irritation.
- d. May be extracranial due to hypertension.
- e. May be due to spasm of extracranial blood vessels.

8. Pain can be blocked by all of the following EXCEPT:

- a. Cut of spinothalamic tract.
- b. Cut of posterior root.
- c. Electrical stimulation of selected skin areas.
- d. Electrical stimulation of PAG or non-specific thalamic nuclei.
- e. Cut of the dorsal column pathway.

**1. Substantia Gelatinosa of Rolandi can NOT be inhibited by:**

- a. Stimulation of A β fibres.
- b. Release of endorphins from the hypothalamus.
- c. Counter irritant.
- d. Aquipuncture
- e. Herpes Zoster.

2. Pain is NOT inhibited by which of the following:

- a. Impulses transmitted along A β fibres.
- b. Injection of morphine.
- c. Antero lateral cordotomy.
- d. Endogenous opioids.
- e. Substance P

3. Which sentence is NOT true about Opioid receptors in the brain:

- a. Can be stimulated by endorphins.
- b. Are blocked by naloxone.
- c. Are activated by endogenous peptides.
- d. Their activation produces hyperalgesia.
- e. They are involved in stress analgesia.

4. Regarding pain analgesic system, all of the following are true EXCEPT:

- a. Periventricular area of hypothalamus releases β -endorphin.
- b. Periaqueductal grey area of brain stem releases enkephalin.
- c. Raphe nucleus of medulla secretes serotonin.
- d. Pain inhibitory complex in spinal cord dorsal horn secretes endorphin.
- e. Enkephalin inhibits the release of substance P from pain conducting C fibers.

5. Pain sensation can be inhibited by:

- a. Adaptation of pain receptors.
- b. Kinin.
- c. Injection of naloxone.
- d. Stimulation of opioid receptors.
- e. Stimulation of lateral spinothalamic tract.

6. The analgesia system includes:

- a. Raphe magnus nucleus in the post horn of the spinal cord.
- b. The periaqueductal area in the mid-brain.
- c. The sensory area 1.
- d. Supra-optic nuclei of the hypothalamus.
- e. Dorsal root ganglion.

7. Serotonin is used as neurotransmitter in pain suppression by neurons in the following region:

- a. Pain inhibitory complex in dorsal horn.
- b. Periaqueductal grey matter.
- c. Post-central gyrus.
- d. Nucleus cuneatus.
- e. Raphe magnus nucleus.

8. Regarding pain analgesic system, all of the following are true EXCEPT

- a. Produces presynaptic inhibition of pain transmission at the spinal cord.
- b. Occurs in stress conditions.
- c. Depends on the presence of opioid peptides and opioid receptors in CNS.
- d. It also inhibits the reticular formation and the non-specific thalamic nuclei.
- e. Its main chemical transmitters are GABA & glycine.

9. Regarding Gate theory of analgesia, all of the following are true EXCEPT

- a. It is the block of pain transmission at the important gates like; SGR in layers I & II of dorsal horn, thalamus, reticular formation and limbic cortex.
- b. Occurs in stress situations.
- c. Aquipuncture acts by gate theory and also by psychogenic effect through enkephalin & endorphin to produce analgesia.
- d. Rubbing the skin and counter-irritant are some of its applications.
- e. GABA, enkephalin and serotonin released from large sensory nerves produce presynaptic inhibition of pain transmission at the spinal cord.



1. Stereognosis sensation needs:

- a. Intact all sensation.
- b. Previous cortical experience.
- c. Intact sensory cortex specially area 5,7.
- d. High degree of cortical processing.
- e. All of the above.

2. Vibration sense:

- a. Is sensed by free nerve endings,
- b. Depends on past cortical experience.
- c. Is a poorly localized sensation.
- d. Is carried on spinothalamic tract.
- e. Is carried on gracile and cuneate tracts.

3. The distance by which two touch stimuli must be separated to be perceived as two separate stimuli is greatest on:

- a. The lips.
- b. The palm of the hand.
- c. The skin over the scapula.
- d. The tips of the finger.
- e. The tip of tongue.

4. Fine touch sensations:

- a. Are not sharply localized.
- b. Need a relatively high threshold stimulus.
- c. Are transmitted via dorsal column.
- d. Are detected by free nerve endings.
- e. Are thalamic sensations.

5. Kinesthetic sensations:

- a. Have their receptors in the joint capsule.
- b. Allow weight discrimination.
- c. Are conducted via ventral spinothalamic tract.
- d. Are the perception of rapid repetitive stimuli.
- e. Are carried by Aδ fibers.

6. The main final destination of signals that carry conscious kinesthetic sensations is the:

- a. Non-specific thalamic nuclei.
- b. Ventrolateral nuclei of the thalamus.
- c. Cerebellar cortex.
- d. Caudate nucleus.
- e. Post-central gyrus of the cerebral cortex.

7. Threshold distance:

- a. Is the minimal distance between 2 points to be perceived as 2 separate points.
- b. Is constant all over the body.
- c. Areas with low threshold distance have low receptor density.
- d. Areas with high threshold distance have large number of afferents.
- e. Is examined for by using cotton wool test.

8. Vibration sensation:

- a. Is felt only over bony prominences
- b. Is transmitted through spinothalamic tract.
- c. Its receptors are Pacinian and Meissner's corpuscles.
- d. It is produced due to stimulation of cutaneous receptors only.
- e. It is a slowly adapting sensation.

9. Free nerve endings are receptors for all of the following sensations, EXCEPT

- a. Pain.
- b. Crude touch.
- c. Crude pressure.
- d. Vibration.
- e. Thermal sensations.

10. Which of the following is an epicritic sensation, choose the most correct:

- a. Cold sensation.
- b. Crude pressure.
- c. Itch and tickle
- d. Slow pain.
- e. Tactile localization.

11. Proprioceptive sensations:

- a. Detect change of position.
- b. Detect touch of body parts.
- c. Have their receptors in the skin.
- d. Are transmitted in the spinothalamic tract.
- e. Are protopathic sensations.

12. Regarding tactile sensations:

- a. Merkel's discs detect fine touch.
- b. Fine touch relayed in non-specific thalamic nuclei.
- c. Vibration is transmitted via spinothalamic tract.
- d. Proprioception is a phasic sensation.

13. Paccinian corpuscles are receptors for all of the following sensations, EXCEPT:

- a. Fine touch.
- b. Vibration.
- c. Static kinesthetic sensation.
- d. Sense of movement.
- e. Deep pressure.

14. Which of the following is NOT a protopathic sensation:

- a. Crude touch.
- b. Slow pain.
- c. Crude temperature.
- d. Itch and tickle.
- e. Proprioception.

15. Two-point discrimination is more accurate:

- a. At the back.
- b. The more the convergence of afferents.
- c. The more the number of receptors.
- d. The less the cortical area of representation.
- e. The larger the receptive field.

16. In conscious Kinesthetic sensations, all of the following are true EXCEPT:

- a. It occurs due to stimulation of deep receptors.
- b. Is transmitted through spinocerebellar tracts.
- c. Its receptors are tonic receptors.
- d. Is carried by A β fibers and perceived at the level of sensory cortex.



1. A patient who has had destruction of somatic sensory area I on the left side will show:

- a. Total lack of pain sensation on the same side:
- b. Total lack of temperature sensation of the same side.
- c. Inability to identify objects put in the left hand.
- d. Astereognosis in the right hand.

2. The 1st Somatic sensory cortex:

- a. Is the major site for perception of slow pain and crude touch.
- b. In it, body representation is proportionate to the size of body organs.
- c. It receives sensory impulses from the same side of the body.
- d. Face receptors are represented in its lower part.

3. Lesion in 1st somatic sensory cortex produces, choose the most correct:

- a. Sensory loss on the same side of the body.
- b. Loss of pain sensation.
- c. Loss of fine localization and orientation of different body parts, accompanied by loss of judgement of critical degrees of weights and pressure contralaterally.
- d. Complete loss of thermal sensations on the opposite side of the body.
- e. Intact stereognosis on the opposite side of the body.

4. Sensory association cortex, all of the following are true EXCEPT:

- a. Is the center of stereognosis.
- b. Occupies Brodman's area 5 & 7 in frontal lobe.
- c. Its lesion causes forgetting the opposite body side (amorphosynthesis) and neglect of the opposite side of the world.
- d. It collects sensory information to understand the meaning.
- e. Its lesion results in astereognosis & autotopognosia.

5. Thalamus performs all of the following functions, EXCEPT:

- a. Transmission of olfactory senses to the cortex.
- b. Perception of slow pain and other crude sensations.
- c. Motor coordination by connecting basal ganglia and cerebellum to the motor cortex.
- d. Consciousness and arousal by receiving impulses from ARAS and project it to the whole cortex.
- e. Memory and emotions through its connections to the limbic cortex
- f. Higher intellectual functions through its connections to the association cortex.
- g. A role in personality and behavior through its connections to the prefrontal cortex.

6. Regarding the somatosensory cortex, which sentence is NOT true:

- a. S-II is characterized by poor localization compared to S-I.
- b. S-II is important in tactile learning and memory.
- c. Lesion in sensory association area results in partial neglect of stimuli of the contralateral side of lesion.
- d. Lesion in sensory association area results in inability to feel both sides of an object.
- e. Sensory association area lesion results in loss of learning by manipulation.

7. The size of representation of various body parts in the primary somatic sensory cortex is dependent on:

- a. Relative size of the body parts.
- b. Density of the receptors in that part.
- c. Size of the muscles in that body part.
- d. Conduction velocity of the afferent fibers.
- e. Frequency of impulses discharged from receptors.

8. Regarding the thalamus, which sentence is **NOT** true:

- a. Its non-specific nuclei transmit epicritic sensations to the somatosensory cortex.
- b. It is the site of perception of crude sensations.
- c. Is involved in motor activities by connecting the basal ganglia and cerebellum to the motor cortex.
- d. Plays a role in consciousness by receiving impulses from reticular formation to activate the whole cortex.
- e. Connected to the limbic cortex to help in emotional responses.



Chapter 1 ?

Stretch reflex & other reflexes

1. Regarding alpha motor neurons, which sentence is **NOT** true:

- a. They are larger and more numerous than gamma neurons.
- b. They supply extrafusal skeletal muscle fibers.
- c. Alpha in the lateral part of anterior horn is concerned with fine movements.
- d. Alpha of flexors is dorsally present to alpha of extensors
- e. All alpha cells that supply a specific muscle represent the motor unit.

2. Regarding alpha motor neurons, which sentence is **NOT** true:

- a. A single alpha cell + its axon + fibers innervated represent the motor unit.
- b. Alpha is the final common pathway for muscle contraction.
- c. The motor unit of distal limb muscle involves 200 fibers.
- d. Its discharge is affected by projections from higher centers.
- e. Its discharge is affected by spinal cord interneurons.

3. Interruption of γ -motor neuron discharge to a skeletal muscle produces:

- a. contraction of the muscle
- b. contraction of the spindle fibers
- c. increased sensory discharge from the spindle fibers.
- d. relaxation of the spindle fibers

4. When the γ -motor neuron discharge to a muscle decreases, it causes:

- a. increased muscle spindle discharge
- b. shortening of the spindle fibers
- c. stronger contraction of the extrafusal muscle fibers
- d. decreased spindle sensitivity to stretch.

5. Compared to alpha motor neurons, gamma motor neurons are:

- a. More numerous
- b. Supply ordinary contractile skeletal muscle fibers.
- c. One type
- d. Innervates the muscle spindle.
- e. Transmit impulses from the central part of intrafusal fibers.

6. Regarding the stretch reflex, which sentence is **NOT** true:

- a. Can be elicited by passive stretch.
- b. Depends on the muscle spindle which is in parallel with the extrafusal muscle fibers.
- c. It is a polysynaptic reflex.
- d. Stretch of a muscle is followed by its contraction.
- e. Is associated with simultaneous inhibition of antagonist muscle.

7. Which statement is **NOT** true about Renshaw cells:

- a. Involved in lateral inhibition for sharpening of signals
- b. Involved in recurrent inhibition to prevent overexcitation in spinal cord
- c. Activates nearby alpha motor neurons
- d. A glycinergic neuron mediate its action
- e. Excited by a cholinergic neuron

8. The stretch reflex:

- a. Is a polysynaptic reflex.
- b. Is a monosynaptic reflex.
- c. Receptor is the Pacinian corpuscle.
- d. Has a long central delay.
- e. Results in relaxation of the stretched muscle.

9. Regarding the muscle spindle, which sentence is **NOT** true:

- a. The primary afferent fibers responds to the static length of muscle as well as to its rate of change in length.
- b. Stretching the extrafusal fibers causes the intrafusal muscle fibers to be stretched.
- c. They are excited during overstretch of the muscle.
- d. They are parallel to the extrafusal muscle fibers.
- e. Consists of two types of intrafusal fibers

10. The gamma efferent system:

- a. Controls spindle sensitivity during voluntary movements.
- b. Supplies the central zone of intrafusal muscle fibers.
- c. Is inhibited by the vestibulo-spinal tract.
- d. When stimulated, causes less discharge of muscle spindles.
- e. Produces direct contraction of extrafusal fibers.

11. The muscle spindles when stimulated transmit information about:

- a. The degree of stretch only.
- b. The rate of stretch only.
- c. Both the degree and rate of stretch.
- d. The muscle tension.
- e. The muscle tone.

12. The primary endings innervating the muscle Spindle:

- a. Only innervates the nuclear chain fibers.
- b. Innervate both nuclear chain & nuclear bag fibers.
- c. Are a slowly conducting nerves
- d. Discharge during the static response only.
- e. End on interneurons in the spinal cord.

13. The nuclear bag intrafusal fibers that signal The rate of changes in muscle length are innervated By:

- a. Corticospinal neurons.
- b. Alpha motor neurons.
- c. Dynamic motor neurons.
- d. Static gamma motor neurons.
- e. Secondary afferents.

14. Events of reflex actions include all the following, except

- a. stimulation of sensory receptors
- b. stimulation of central synapses
- c. inhibition of sensory receptors
- d. inhibition of central synapses

15. A typical example of a monosynaptic reflex is:

- a. stretch reflex
- b. superficial abdominal reflex
- c. withdrawal reflex
- d. light reflex

16. Functions of stretch reflex does **NOT** include Which of the following:

- a. Signal averaging of impulses reaching the spinal cord.
- b. Make muscle contraction smooth.
- c. Prevents oscillations in muscle contraction.
- d. Maintains upright posture.
- e. Prevent overexcitation in the spinal cord.

17. Gamma motor fibers:

- a. Transmit signals from intrafusal fibers to the spinal cord.
- b. Cause contraction of the middle portions of the muscle Spindle.
- c. Cause relaxation of the intrafusal fibers.
- d. Cause direct contraction of extrafusal fibers.
- e. Cause contraction of the peripheral part of the intrafusal Fibers.

18. Functions of stretch reflex does NOT include Which of the following:

- a. Servoassist voluntary movements.
- b. Allows muscle contraction to increase with minimal brain Energy
- c. Compensates for muscle weakness.
- d. Maintains body position inspite of application of loads.
- e. Prevent rupture of tendon in cases of severe muscle stretch.

19. Compared to static stretch reflex, the dynamic Stretch reflex characterized by:

- a. Its stimulus is maintained muscle stretch.
- b. Its receptor is nuclear bag
- c. Its afferents are 1^{ry} & 2^{ry} afferents.
- d. Its center is the cerebellum.
- e. Its response is maintained contraction to support the muscle.

20. Compared to nuclear chain fibers, nuclear bag Intafusal fibers are characterized by:

- a. Monitoring the degree of stretch.
- b. Being a tonic receptor.
- c. Transmit impulses by 1^{ry} & 2^{ry} afferents.
- d. Acts dynamically.
- e. Is the main receptor responsible for muscle tone.

21. Compared to secondary afferent of muscle Spindle, the primary afferent is characterized by:

- a. Type-II nerve fiber.
- b. Slowly conducting.
- c. Transmit impulses from nuclear bag and nuclear chain Fibers.
- d. Acts dynamically, but not statically.
- e. Synapses with many spinal cord interneurons.

22. Which of the following is NOT a property of stretch reflex:

- a. Reciprocal innervation.
- b. Can by controlled by higher centers.
- c. Monosynaptic and fast
- d. After discharge.
- e. Is inhibited by severe muscle stretch.

23. Functions of gamma efferents do NOT include:

- a. Prevents shortening of intrafusal fibers during shortening of extrafusal fibers.
- b. Prevents unloading and maintain receptor sensitivity.
- c. Its stimulation produces reflex contraction.
- d. Is one of the sites subjected to supraspinal control on stretch reflex.
- e. Monitor muscle length.

24. Stretch reflex activity is facilitated by:

- a. Cortical area 4.
- b. Paleocerebellum.
- c. Medullary reticular formation.
- d. Red nucleus.
- e. Putamen nucleus of basal ganglia.

25. Muscle spindles:

- a. Are the only receptors present in the skeletal muscle.
- b. Are supplied by the gamma as well as the alpha efferents.
- c. Are length detectors.
- d. Are in series with extrafusal fibers.
- e. Are phasic receptors.

26. Concerning the role of stretch reflex for the control of voluntary motor activity:

- a. Its damping function maximize energy utilization by the nervous system.
- b. It protects muscle tendons from high tension.
- c. Through its servo assist function it co-activates the extrafusal muscle fibers.
- d. Among its disadvantages, the unloading of receptor during muscle contraction.
- e. Among its disadvantages, the decreased receptor sensitivity.

27. Regarding muscle tone, which sentence is NOT True:

- a. Is defined as resistance offered by muscle to passive Stretch.
- b. Is maintained by gamma efferent discharges.
- c. Maintains body posture against effect of gravity.
- d. Is decreased in UMNL.
- e. Is the static type of stretch reflex.

28. Muscle relaxation evoked by the inverse stretch reflex:

- a. Is operated monosynaptically,
- b. Is initiated by the muscle spindle.
- c. Involves inhibition of LMN supplying the muscle.
- d. Can be demonstrated in the right leg when a sharp painful Stimulus is given to the opposite limb.
- e. Allows muscle contraction with minimal brain energy

29. Which sentence is NOT true regarding the autogenic inhibition reaction (inverse stretch reflex):

- a. Involves a receptor innervated by group Ib fibers.
- b. Involves excitation of alpha motor neurons.
- c. Can be elicited when the Golgi tendon apparatus is stimulated.
- d. Equalizes the contraction force between muscle fibers during muscle contraction.

30. Golgi tendon organ:

- a. Has a threshold of stimulation equal to the muscle spindle.
- b. Lies in parallel with the extrafusal muscle fibers,
- c. Is stimulated in a monosynaptic muscle reflex.
- d. When stimulated, reflexly relaxes the muscle.
- e. When stimulated, damp the voluntary movements and prevent its oscillation

31. The inverse stretch reflex:

- a. Depends on a receptor which is situated in the fleshy part of the muscle.
- b. Is elicited when the muscle is subjected to excessive stretch.
- c. Leads to increased force of muscle contraction.
- d. Is a monosynaptic reflex.
- e. Prevents unloading

32. Impulses from Golgi tendon organ pass to inhibitory interneurons in the spinal cord via:

- a. Ia
- b. Ib
- c. II a
- d. C fibers

33. The flexion withdrawal reflex is initiated by Simulation of the following receptors:

- a. Pacinian corpuscles.
- b. Golgi tendon organ.
- c. Free nerve endings.
- d. Muscle spindles.
- e. Proprioceptors.

34. The withdrawal reflex:

- a. Is monosynaptic.
- b. Characterized by after discharge.
- c. Is more rapid than the stretch reflex.
- d. Has type Ia afferents.

35. A reflex action:

- a. is a fast brief response to a sensory stimulus
- b. is involved in regulation of voluntary movements
- c. includes at least three sequential sets of neurons in its pathway.
- d. is always initiated whenever its specific receptors are effectively stimulated by specific stimulus.

36. Concerning the flexor withdrawal reflex:

- a. It involves reciprocal innervation.
- b. Its central delay is less than 0.05 msec.
- c. It is a monosynaptic reflex.
- d. It provides an example of inverse stretch reflex.
- e. It servoassists voluntary movements.

37. Regarding the plantar reflex, which sentence is NOT true:

- a. Is a superficial spinal reflex.
- b. Its center is S1 & S2.
- c. Is lost in UMNL.
- d. The normal response is plantar flexion of all toes.
- e. Is a modification performed by the pyramidal tract to an Infantile reflex.

38. The crossed extensor reflex:

- a. Provides a typical example of servoassist function.
- b. Is monosynaptic.
- c. Does not involve reciprocal inhibition.
- d. Supports the body from falling.
- e. Is due to stimulation of nuclear bag fibers.

39. Regarding the crossed extensor reflex, which Sentence is NOT true:

- a. Is a supporting reflex.
- b. Is a polysynaptic spinal reflex.
- c. Has a longer latency than the stretch reflex.
- d. Is elicited by ipsilateral painful stimulus.
- e. An intersegmental interneuron is involved in this reflex.

40. +ve Babinski sign is NOT present in:

- a. LMNL.
- b. Chronic UMNL.
- c. Deep sleep.
- d. Coma.
- e. Before the age of 1 year.

41. Which of the following reflexes is correctly paired with its receptor?

- a. Autogenic inhibition - muscle spindle
- b. Reciprocal inhibition: Golgi tendon organ
- c. Reciprocal inhibition - Pacinian corpuscle
- d. Stretch reflex - muscle spindle

42. Sensory impulses from spindles of a stretched muscle could antagonistic muscles by:

- a. Directly inhibiting γ -motor neurons of the antagonistic muscles
- b. Directly inhibiting the α -motor neurons of the antagonistic muscle
- c. Inhibiting the transmitter release from the central terminals of afferents from the spindles of the antagonistic muscles
- d. Activation of inhibitory interneurons

43. The receptor for the stretch reflex is the:

- a. muscle spindle
- b. Golgi tendon organ
- c. Meissner's corpuscle
- d. Free nerve endings

44. Muscle spindles:

- a. Are found in all skeletal muscles
- b. Are found only in large skeletal muscles
- c. Consist of small numbers of extrafusal muscle fibers
- d. Consist of a large number of extrafusal muscle fibers

45. Babinski sign is NOT present in:

- a. Pyramidal lesion.
- b. Extrapyramidal lesion.
- c. Cerebellar lesion.
- d. Deep sleep.
- e. Comma.

46. Physiologically, an intact innervated muscle can normally be made to contract by stimulation of which of the following neurons innervating that muscle?

- a. α -motor neurons
- b. γ -motor neurons
- c. Either one of the above
- d. None of the above

47. The muscle spindle remains capable of responding to stretch even the muscle is contracting. This is because of:

- a. presence of stretch receptors in the tendon
- b. co-activation of α - and γ -motor neurons
- c. reciprocal innervation
- d. autogenic inhibition

48. A reflex arc includes:

- a. At least two sets of sequential neurons
- b. At least two sequential sets of central synapses
- c. At least two types of sensory receptors
- d. At least two types of efferent neurons

49. The receptor for the inverse stretch reflex is:

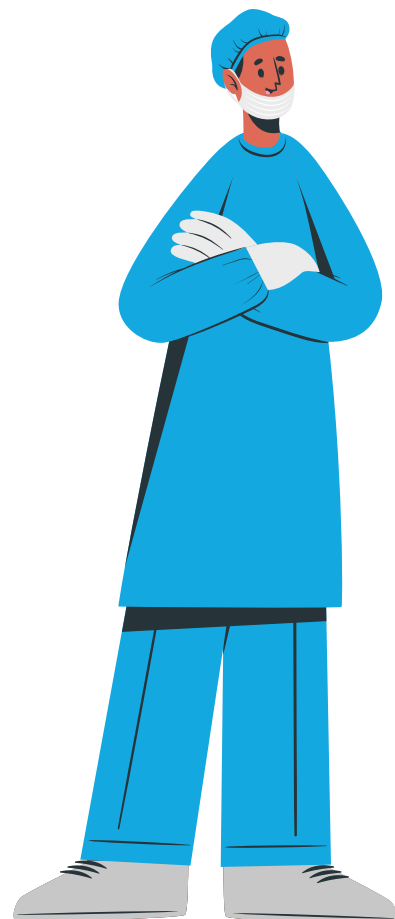
- a. muscle spindle
- b. Golgi tendon organ
- c. C fiber
- d. delta fibers

50. Muscle contraction in response to maintained stretch is initiated by sensory inputs from:

- a. nuclear bag fibers)
- b. nuclear chain fibers
- c. Golgi tendon organ
- d. γ -motor neurons

51. Increased γ -motor neuron discharge stimulates muscle spindles

- a. Produces stretch of the extrafusal muscle fibers
- b. Causes stretch of the peripheral regions of the intrafusal fibers
- c. Causes stretch of the central region of the intrafusal fibers
- d. Stimulates directly the sensory fibers innervating muscle spindles



1. Regarding cortical motor areas.

Which Sentence is **NOT** true:

- a. Body is represented in the motor cortex 4 crossed, inverted And disproportionate to the real size of muscles.
- b. Upper part of the face has bilateral representation.
- c. Supplementary motor area is involved in planning of Complex motor sequences.
- d. Premotor area initiates fine movements.
- e. Supplementary motor area is involved in planning of Internal directed movements.

2. The larger area of representation in the motor Cortex includes muscles:

- a. That extend the shoulder.
- b. That flex the knee.
- c. Of the trunk.
- d. Of vocalization.
- e. Of the back.

3. Partial +ve Babinski sign due to area 4 lesion is:

- a. Plantar flexion of all toes.
- b. Dorsiflexion of all toes.
- c. Dorsiflexion of big toe.
- d. Plantar flexion of big toe.
- e. Fanning of lateral toes.

4. Specialized functions of the pre-motor cortex do **NOT** include:

- a. Voluntary eye movement.
- b. Head rotation.
- c. Speech.
- d. Increase muscle tone.
- e. Skilled hand movement

5. Regarding the primary motor cortex (area 4), Which sentence is **NOT** true:

- a. Is responsible for fine delicate movements.
- b. Parts of the body are represented in an area which is Directly proportional in size to the complexity of movement Carried by that part.
- c. Sends facilitatory impulses to gamma motor neurons.
- d. Is stimulatory to stretch reflex.
- e. Is essential for +ve Babinski sign.

6. The cortical motor area 4 on the left side of the Body:

- a. There is a map representing body parts on the right side.
- b. The body is represented in an erect (upright) pattern.
- c. Controls motor activity of the left side of the body.
- d. The area representing a particular body muscle is directly Proportional in size to the size of that muscle.
- e. Its lesion produces contralateral hemiplegia.

7. Which sentence is **NOT** true regarding the Primary motor area:

- a. Is present in the precentral gyrus.
- b. Is highly excitable.
- c. Sends impulses to AHCs along the cortico-spinal tract.
- d. Controls activity of the hand muscles on the contralateral Side.
- e. Its lesion produces loss of memory of learned motor acts.

8. Which sentence is NOT true regarding the Premotor area:

- a. Broca's area program sequences of contraction of muscles Of vocalization.
- b. Voluntary eye movements controls conjugate eye Movement.
- c. It plans for internal directed complex movements.
- d. Hand skill area helps coordination of hand movement.
- e. Is less excitable than area 4

9. Which sentence is NOT true regarding a lesion In premotor area:

- a. It does not produce paralysis.
- b. It produces paresis of gross movements.
- c. It produces hypertonia and hyperreflexia.
- d. It produces motor apraxia.
- e. It produces loss of plantar and grasp reflexes.

10. Which area is NOT involved in planning of Movements

- a. Premotor area.
- b. Supplemental motor area.
- c. 1^{ry} motor area.
- d. Basal ganglia.
- e. Cerebellum.

Chapter 3 ?**Descending motor tracts****1. The pyramidal tract:**

- a. Arises only from subcortical areas.
- b. Carries its functions immediately after birth.
- c. Most of its fibers do not cross.
- d. Responsible for fine movements.
- e. Some fibers are facilitatory and others are inhibitory to Stretch reflex.

2. In controlling the fine movement of the hands, The corticospinal tract terminates directly on:

- a. Renshaw cells.
- b. Spinal cord interneurons.
- c. Spinal lateral alpha motor neurons.
- d. Posterior horn cells.
- e. Lateral horn cells.

3. The lateral motor system includes:

- a. The lateral Reticulospinal tract
- b. The lateral corticospinal tract
- c. The lateral vestibulospinal tract
- d. The tectospinal tracts
- e. All the above tracts.



Chapter 4 ? Sensory & motor system lesions

1. Hyporeflexia is present in:

- a. Area 6 lesion.
- b. Tabes dorsalis.
- c. Parkinsonism
- d. UMNL.
- e. Paleocerebellar lesion.

2. Tendon jerks are exaggerated in:

- a. Anterior horn cell disease.
- b. Dorsal root lesions
- c. Chorea.
- d. Internal capsule lesion.
- e. Parkinsonism.

3. Muscle tone is NOT reduced in the following Condition:

- a. Lower motor neuron lesion.
- b. Upper motor neuron lesion.
- c. Cerebellar lesions.
- d. Caudate lesion.
- e. Immediately following complete transection of the spinal Cord

4. The ankle jerk is exaggerated:

- a. Immediately after complete transection of spinal cord at mid-thoracic region.
- b. In Parkinsonism.
- c. In neocerebellar disease.
- d. On the right side after damage in the left internal capsule.
- e. In LMNL

5. Damage which interrupt upper motor neuron Pathways in the internal capsule would NOT lead to:

- a. Paralysis and loss of fine voluntary movements especially Of the hand & fingers.
- b. Wasting of muscles.
- c. Loss of abdominal reflex.
- d. Positive Babinski sign.
- e. Hyperreflexia on the contralateral side.

6. Clonus:

- a. Is a flexion withdrawal reflex:
- b. Occurs on the same side of the body after unilateral Damage to the internal capsule.
- c. Can be elicited by sudden maintained stretch of a Hypertonic muscle.
- d. Is present in muscles with lower motor neuron lesion.

7. UMN lesion is manifested by:

- a. Spastic paralysis.
- b. Fibrillations.
- c. Muscle wasting.
- d. Normal planter response.
- e. Ipsilateral localized paralysis.

8. The signs of hemiplegia do NOT include:

- a. Clonus.
- b. Inability to move the eye.
- c. Spastic tone.
- d. Hyperactive deep reflexes.
- e. Circumduction gait.

9. Which sentence is NOT true about clonus:

- a. Can be elicited by doing sudden maintained stretch of a Hypertonic muscle.
- b. Occurs on the opposite side of the body after damage to Internal capsule.
- c. Is rhythmic alternating cycles of stretch-inverse stretch Reflexes.
- d. Is present in muscle with lower motor neuron lesion.
- e. Is a sure sign of UMNL.

10. In lower motor neuron disease, the subject may Show:

- a. Wasting of muscles.
- b. Rigidity of muscles.
- c. Hyperactive reflexes.
- d. Clonus.
- e. Contralateral hemiplegia

11. Babinski sign is present in:

- a. UMNL
- b. Cerebellar lesion,
- c. Basal ganglia lesion.
- d. Complete section of a peripheral nerve.
- e. Syringomyelia

12. Which sentence is NOT true about LMN lesion:

- a. Loss of superficial reflexes but exaggerated deep reflexes.
- b. Loss of tone of affected muscle.
- c. Fasciculation of the affected muscle.
- d. Wasting of the affected muscle
- e. Ipsilateral localized paralysis.
- f. Is associated with marked muscle atrophy.

13. The lower motor neuron lesion is NOT Characterized by:

- a. Loss of voluntary movements.
- b. Lead-pipe rigidity.
- c. Fasciculation of the affected muscle.
- d. Marked atrophy of the affected muscles.
- e. Loss of all reflexes of the affected muscles.

14. A person suffering from UMNL lesion shows:

- a. Loss of muscle tone.
- b. Extensive paralysis of one side of the body.
- c. Muscle wasting in the early stages of the disease.
- d. Presence of shuffling gait.
- e. Fibrillations

15. A LMN lesion:

- a. Can result from poliomyelitis.
- b. Is characterized by exaggerated deep reflexes.
- c. Leads to generalized effects on contralateral muscles.
- d. Is accompanied with clonus.
- e. Is associated with circumduction gait.

16. Lesion of the internal capsule results in:

- a. Spastic paralysis of the contralateral side of the body.
- b. Flaccid paralysis of both sides of the body.
- c. Clonus in muscles of the same side.
- d. Exaggerated tendon jerk on the same side of the lesion.
- e. Loss of muscle tone in the opposite side.

17. Which of the following observation would Suggest that an upper motor neuron lesion rather Than a lower motor neuron lesion is present?

- a. Marked muscle atrophy.
- b. Hyporeflexia.
- c. Spasticity.
- d. Fasciculations.
- e. Localized paralysis.

18. In Hemiplegia affecting the right side of the Body, which sentence is NOT true:

- a. Is characterized by spasticity of muscle on the right side of The body.
- b. Is usually accompanied by marked muscle wasting on the Right side of the body.
- c. Is associated with increased tendon reflexes on the right Side of the body.
- d. Can be due to bleeding in the left internal capsule.
- e. Is associated with +ve Babinski sign in the right side.

19. Hypertonia of UMN lesions is characterized by:

- a. Increased inhibitory discharge from the premotor area
- b. Inhibition of pontine reticular formation
- c. Increased γ -motor neuron discharge
- d. Decreased muscle spindle discharge.
- e. Cog-wheel rigidity.

20. The stage of spinal shock is NOT characterized By:

- a. Failure of spinal reflexes below the level of the lesion
- b. Loss of sensations from the body below the level of the Lesion
- c. Loss of voluntary movement from the body below the level Of the lesion
- d. Exaggerated tendon jerks below the level of the lesion

21. In humans the usual duration of the stage of Spinal shock is :

- a. From 2-6 hours
- b. From 2-6 days
- c. From 2-6 weeks
- d. From 2-6 months
- e. From 2- 6 minutes

22. Spinal shock is due to:

- a. Severe pain felt at the site of the lesion
- b. Severe hypotensive shock
- c. Interruption of the ascending sensory pathways
- d. Interruption of the descending facilitatory tracts
- e. Internal capsule lesion.

23. Complete transection of the spinal cord does NOT produce:

- a. Permanent loss of all sensations mediated by the cord Below level of lesion
- b. Permanent loss of voluntary movements by muscles Innervated by the cord below level of lesion
- c. Permanent loss of reflexes mediated by the cord below Level of lesion
- d. Temporary loss of micturition reflexes
- e. Hypotension.

24. Complete transection of the spinal cord did not Affect arterial blood pressure when the lesion occurs At level of :

- a. Mid-cervical segments
- b. Upper thoracic segments
- c. Lower thoracic segments
- d. Mid-lumbar segments
- e. Upper cervical

25. The earliest spinal reflex that recovers after the Stage of spinal; shock is:

- a. The micturition reflex
- b. The scratch reflex
- c. The stretch reflex
- d. The flexor reflex
- e. Babinski sign

26. Recovery of micturition reflexes following the Stage of shock :

- a. Is due to recovery of supraspinal facilitation to the Micturition center in the sacral segments
- b. Is due to recovery of activity of the micturition center in The sacral segments
- c. Causes retention with overflow
- d. Causes normal micturition
- e. Causes emptying of bladder after filling 400 ml volume of Urine

27. Brown-Sequard syndrome is NOT characterized By:

- a. Loss of vibration sense on the opposite side below level of the lesion
- b. Loss of voluntary movements on the same side below the Level of the lesion
- c. Loss of reflex movements on the same side at the level of The lesion
- d. Loss of pain sensation on the opposite side below the level Of the lesion
- e. Loss of all sensations at the level and on the same side of Lesion.

28. Lesion of the dorsal column may impair:

- a. Pain sensation.
- b. Vibration sense.
- c. Itch sensation.
- d. Thermal sensations.
- e. Voluntary movements

29. Tabes dorsalis is:

- a. Due to virus infection of the posterior root ganglia:
- b. Accompanied with shuffling gait.
- c. Accompanied with loss of vestibular sensations.
- d. Is a central spinal cord lesion
- e. Accompanied with incoordination of voluntary movements.

30. Which sentence is NOT true regarding Syringomyelia:

- a. Fine touch is preserved.
- b. Is due to dilatation of the central canal
- c. There is dissociated sensory loss.
- d. Dorsal column sensations are not lost.
- e. There is unilateral loss of pain.

31. Tabes dorsalis may NOT lead to:

- a. Loss of sense of position.
- b. High steppage gait.
- c. Sensory ataxia
- d. Deafferentation of all reflexes of lower limb muscles.
- e. Early loss of pain sensation.

32. Which sentence is NOT true regarding Syringomyelia:

- a. May lead to damage of anterior horn of spinal cord.
- b. May lead to horner's syndrome.
- c. May lead to syringobulbia.
- d. Has a jacket distribution of pain and temperature loss.
- e. Associated with sensory ataxia.

33. A subject suffers from right hemisection of the Spinal cord shows:

- a. Upper motor neuron paralysis at the level of the Hemisection.
- b. Loss of pain and temperature on the right side below the Level of the section.
- c. Loss of fine touch and vibration sense on the left side Below the level of the section.
- d. Loss of all sensations on the same side at the level of the Section
- e. Loss of voluntary movements in both lower limbs

34. Tabes dorsalis may early impair the following, EXCEPT:

- a. The ability to stand steadily with the eyes closed.
- b. Pain sensation.
- c. Vibration sensation.
- d. Kinesthetic sensations.
- e. Fine touch

35. Which sentence is NOT true regarding thalamic Syndrome:

- a. Fine sensations are lost on the opposite side of lesion.
- b. Crude sensations return after sometime.
- c. There is 2nd hyperalgesia.
- d. Associated with ataxia.
- e. Proprioception is not lost.

36. In tabes dorsalis:

- a. Ataxia is present.
- b. Vibration is lost early.
- c. Pain is the last to be lost.
- d. Associated with damage of afferents of lower limb Reflexes.
- e. The patient has a shuffling gait.

37. In peripheral nerve lesions:

- a. Mononeuropathy results in loss of sensations in a dermatome.
- b. Peripheral neuritis manifestations start proximally and then spread distally.
- c. Dorsal root lesion results in intact reflexes.
- d. Polyneuritis sensory effects are paresthesia then hypoesthesia, and its motor effects are paresis then paralysis appear first in glove and stocks pattern.
- e. Polyneuritis is accompanied by ankle and wrist drop, but the reflexes are intact.

38. In Syringomyelia, all of the following are true EXCEPT:

- a. Is due to increased gliosis and cavitation of central canal in spinal cord.
- b. Affects mainly the A β dorsal column fibers.
- c. Leads to jacket distribution of sensory loss in upper limbs.
- d. Results in dissociated loss of pain and thermal sensations only.
- e. Accompanied by intact all dorsal column sensations of the upper limb.

39. Syringomyelia:

- a. Never results in autonomic or motor effects.
- b. Causes loss of thermal and pain sensation mainly in the lower limbs.
- c. Affects all body sensations.
- d. Fine touch, proprioception and vibration are preserved (intact).
- e. Crude touch is lost in the upper limbs.

40. Tabes dorsalis:

- a. Is due to increased cavitation in spinal cord.
- b. Characterized by severe attacks of pain in lower limbs.
- c. Results in incoordination of movements (ataxia) due to paralysis of lower limb muscles.
- d. Slow pain is the first sensation lost.
- e. C fibers are affected early in the diseases.

41. In Tabes dorsalis, all of the following are true EXCEPT:

- a. Is due to inflammation at dorsal root.
- b. Most common in Lumbo-Sacral spinal segments.
- c. Vibration is the last sensation lost in the disease.
- d. Romberg's sign test is positive.
- e. May lead to loss of pupillary light reflex "Argyll Robertson pupil" when the effect involves the pretectal area in midbrain.

42. In Hemisection of the spinal cord, all of the following are true EXCEPT:

- a. Hyperesthesia may present above level of lesion on the same side due to irritation of dorsal root.
- b. Flaccid paralysis and lost muscle reflexes is found ipsilaterally at the level of lesion.
- c. Spastic paralysis and exaggerated muscle reflexes is found contralaterally below level of lesion.
- d. Anesthesia is found at the level of lesion ipsilaterally.
- e. Dorsal column sensation arc lost ipsilaterally while spinothalamic sensations are lost contralaterally below the level of lesion.

43. Thalamic syndrome:

- a. Results in ipsilateral (on the same side) sensory loss.
- b. Affects only sensations of spinothalamic pathway.
- c. The anterior thalamic nuclei are the most commonly affected nuclei.
- d. Characterized by return of fine touch and vibration after sometime.
- e. Leads to ataxia due to loss of proprioception and due to loss of connections between motor cortex and basal ganglia & cerebellum.

44. In Thalamic syndrome, all of the following are true EXCEPT:

- a. Results from vascular lesion in thalamogeniculate artery.
- b. Accompanied by loss of all sensations, then return of crude sensations.
- c. Epicritic sensations are not permanently lost.
- d. Pain returns severe, prolonged and spontaneous (thalamic pain) due to facilitation of non-specific thalamic nuclei.
- e. Accompanied by incoordination of movements in absence of paralysis on the opposite side of lesion.

45. Brown sequard syndrome:

- a. Is due to a complete damage of a segment of spinal cord.
- b. Results in loss of pain and thermal sensations only at the level of lesion.
- c. Results in loss of all sensations below the level of lesion.
- d. Fine touch and proprioception are intact below the level of lesion on the opposite side.
- e. At level of lesion on the same side, muscles become spastic with exaggerated reflexes.



1. Motor apraxia:

- a. Is due to lesion in primary motor area 4.
- b. Is due to lesion in sensory area.
- c. Is accompanied with muscle paralysis.
- d. Is characterized by loss of memory of learned voluntary Movements.
- e. Is associated with stamping gait.

2. In the nigrostriated lesion there is:

- a. Kinetic tremors.
- b. Lead pipe rigidity.
- c. Positive Babinski sign.
- d. Rebound phenomenon.
- e. Spasticity

3. The basal ganglia are NOT primarily concerned With:

- a. Control of the extra-pyramidal activities.
- b. Planning and programming of movements.
- c. Tonically increase muscle tone activity.
- d. The associated gross subconscious movements.
- e. Inhibition of excess cortical activity

4. A patient suffering from damage of basal Ganglia may NOT show:

- a. Static tremors.
- b. Hyperreflexia and clonus.
- c. Shuffling gait.
- d. Degeneration of dopaminergic neurons.
- e. Hypokinesia in all movements

5. Basal ganglia lesion may Not cause:

- a. Akinesia disorder.
- b. Mask face
- c. Cogwheel rigidity.
- d. Monotonous speech.
- e. Flaccidity

6. Parkinson's disease is characterized by:

- a. Tremors that worsen by movements.
- b. Paralysis of one side of the body.
- c. Increase of dopamine in the substantia nigra.
- d. Lead-pipe rigidity.
- e. Clonus.

7. Parkinson's disease:

- a. Is caused by a lesion in the substantia nigra.
- b. Is caused by lesion in the posteroventral nucleus of the Thalamus.
- c. Is accompanied by kinetic tremors.
- d. Accompanied by clasp-knife rigidity.
- e. Accompanied by pendular jerks

8. Which of the following is NOT a feature of Parkinsonism:

- a. A tremor at rest.
- b. A deficiency of dopamine in the substantia nigra.
- c. A clasp-knife type of spastic rigidity.
- d. Difficulty in initiating the voluntary movements.
- e. Normoreflexia

9. Chorea:

- a. Is due to lesion in the substantia nigra.
- b. May be due to lesion in the internal capsule.
- c. Is associated with hypotonia.
- d. Is associated with exaggerated tendon jerk.
- e. Is associated with increased gamma activity

10. Athetosis is associated with movement which is NOT:

- a. Spontaneous.
- b. Spasmodic.
- c. Twisting.
- d. Present at rest.
- e. Decreased by emotions

11. Concerning the basal ganglia function, which is NOT true:

- a. Caudate nucleus plan the sequence of patterns of Movement.
- b. Caudate nucleus modifies the timing of movements.
- c. Putamen helps execution of subconscious learned pattern of Movement.
- d. Basal ganglia mainly increases the muscle tone.
- e. Basal ganglia prevent involuntary movements

12. The neurotransmitter used by the axons of Substantia nigra neurons which project to the Caudate and putamen is :

- a. Acetylcholine.
- b. Serotonin.
- c. Dopamine.
- d. Norepinephrine
- e. Enkephalin.





1. Higher motor commands originate in all the following centers, except:
 - a. Cerebral cortex
 - b. Thalamus
 - c. Caudate nucleus
 - d. Cerebellum
2. Regarding the cerebellum, which is NOT true:
 - a. Cerebellum helps in planning and timing of voluntary Movements.
 - b. The cerebellum is stimulated by the motor activity which Originates in the cerebral cortex.
 - c. The receptor activity of the spindles and Golgi tendons Organs reaches the cerebellum through the spinocerebellar Tract.
 - d. The cerebellum has direct stimulatory effect on alpha Motor neurons.
 - e. The cerebellum predicts and plans for the successive Movements
3. The cerebellum is NOT involved in:
 - a. The maintenance of posture.
 - b. Continuous feedback of information even during action.
 - c. The performance of rapid alternating movements.
 - d. Coordination of movements.
 - e. Neocerebellum decreases muscle tone.
4. Which one of the following is NOT true Regarding the cerebellum:
 - a. Each cerebellar hemisphere controls movements of the Ipsilateral muscles.
 - b. Cerebellar ataxia is a sensory ataxia.
 - c. Cerebellar lesions cause pendular knee jerks.
 - d. The intention tremor associated with cerebellar disease is Worsened by movement.
 - e. The neocerebellar syndrome gait is zig-zag like.
5. Neocerebellar syndrome in humans is NOT Characterized by:
 - a. Hypotonia.
 - b. Pendular knee jerk.
 - c. Disturbance of balance
 - d. Dysarthria.
 - e. Static tremors.
6. The following is NOT a neurological features of Cerebellar lesion:
 - a. Abnormal coordination tests.
 - b. Rebound phenomenon.
 - c. Dysdiadochokinesia.
 - d. Shuffling gate.
 - e. Kinetic tremors.
7. Adiadochokinesia:
 - a. Is failure of the patient to perform rapid coordinate Movements.
 - b. Is due to lesion in caudate nucleus.
 - c. Is due to lesion in subthalamus.
 - d. Is due to lesion in vestibulocerebellum.
 - e. Is associated with hypertonia
8. Which sentence is NOT true about kinetic Tremors:
 - a. Occurs during rest.
 - b. Is due to neocerebellar lesion.
 - c. Is accompanied with hypotonia.
 - d. Disappear during sleep.
 - e. Associated with motor ataxia
9. Neocerebellum syndrome is NOT associated With:
 - a. Ataxia.
 - b. Hypertonia.
 - c. Inability to do complex movements.
 - d. Rebound phenomenon.
 - e. Decomposition of movements

10. Neocerebellum disease is associated with:

- a. Shuffling gait.
- b. Zigzag gait.
- c. Stamping gait.
- d. Circumduction gate.
- e. Drunken gait

11. The following is NOT a neurological features of Cerebellar lesion:

- a. Dysmetria.
- b. Lead-pipe rigidity.
- c. Dysdiadochokinesia.
- d. Rebound phenomenon.
- e. Incoordination of voluntary movements

12. Sensory ataxia is characterized by:

- a. Stamping gait.
- b. Staccato speech.
- c. Kinetic tremors.
- d. Failure of coordination even with opened eyes
- e. Cerebellar lesion.

13. Motor ataxia is NOT characterized by:

- a. Zig zag gait.
- b. Staccato speech.
- c. Kinetic tremors.
- d. Failure of coordination on closing eyes
- e. Cerebellar lesion.

14. The following conditions may produce ataxia, EXCEPT:

- a. Peripheral neuritis.
- b. Neurosyphilis affecting lumbosacral spinal segments.
- c. Vascular block of thalamic vessels.
- d. Transection of spinal cord.
- e. Hyperalgesia.

Chapter 7



Disorders of muscle tone

1. Spasticity is characterized by:

- a. Affecting all body muscles.
- b. Due to increase alpha and gamma cells activities.
- c. Occurs with basal ganglia lesion.
- d. Associated with static tremors.
- e. Is of clasp-knife nature.

2. Rigidity is characterized by:

- a. Affecting antigravity muscles.
- b. Due to increase gamma cells activities.
- c. Occurs with UMNL lesion.
- d. Associated with static tremors.
- e. Associated with circumduction gait.

3. Spasticity is characterized by:

- a. Associated with shuffling gait.
- b. Associated with normoreflexia.
- c. Is of Cog-wheel or lead pipe nature.
- d. Associated with clonus.
- e. Occur in parkinsonism.

Section 3

Higher Brain Function

Chapter 1 ?

Reticular formation

- 1. RAS activity is enhanced by:**
 - a. Prolonged activity of its neurons.
 - b. Activation of sleep-inducing centers.
 - c. General anesthesia.
 - d. Withdrawal of sensory and motor impulses affecting reticular formation.
 - e. Thyroid hormones.
- 2. Pontine reticular formation is NOT characterized by:**
 - a. Has intrinsic activity.
 - b. Has descending and ascending fibers.
 - c. Has a sleep center in its upper part.
 - d. Non-specifically activates the whole cortex, mainly directly.
 - e. Its activity is inhibited by the medullary reticular formation.
- 3. Medullary reticular formation is stimulated by:**
 - a. Primary motor cortex.
 - b. Cerebrocerebellum.
 - c. Basal ganglia.
 - d. Vestibular nucleus.
 - e. Gamma motor neurons of spinal cord anterior horn.
- 4. Which sentence is NOT true about the reticular formation:**
 - a. Its excitatory component is located in the pons.
 - b. Its medullary component is inhibitory to all cortical areas.
 - c. Its excitatory components constitutes the ARAS.
 - d. Its pontine part is excitatory to spinal cord motor neurons.
 - e. It is stimulated by both of Corticofugal and sensory fibers.
- 5. Which of the following is NOT involved in decreasing the activity of the pontine reticular formation?**
 - a. Cortical inhibitory areas.
 - b. Red nucleus.
 - c. Medullary reticular formation.
 - d. Basal ganglia.
 - e. Corticofugal fibers.
- 6. Which of the following is NOT a character of ARAS:**
 - a. Present in pons and midbrain.
 - b. Indirectly and directly excites the whole cortex.
 - c. Receives excitatory impulses from cortical areas.
 - d. Facilitates spinal cord motor neurons of antigravity muscles.
 - e. Responsible for arousal.
- 7. Concerning reticular formation, all are true EXCEPT:**
 - a. It receives impulses from various somatic and special sense pathways.
 - b. Its inhibitory motor part inhibits the spinal cord motor neurons of antigravity muscles.
 - c. Its facilitatory motor part excites both the cortex and the spinal cord.
 - d. Its intrinsic activities characterized by diffuse electric activity of all its components.
 - e. Red nucleus and paleocerebellum inhibit its pontine component.
- 8. Concerning reticular formation, all are true EXCEPT:**
 - a. It excites all parts of cortex through ARAS.
 - b. It is a link between higher and lower CNS centers.
 - c. It is responsible for perception and localization of fast pain.
 - d. It is responsible for brain electrical activity.
 - e. Its descending motor facilitatory component plays a role in body support.

1. Secondary evoked potential is characterized by:

- a. Localized to a specific cortical area.
- b. Consists of a small +ve followed by a small -ve wave.
- c. Similar to beta waves of EEG.
- d. Due to stimulation of non-specific thalamic nuclei.
- e. Recorded when the person shows mental activity.

2. Electroencephalogram:

- a. Recorded in an anaesthetized person.
- b. Represents summation of EPSP & IPSP in brain neurons.
- c. Theta waves dominate in adult during awake state with mental rest.
- d. Beta waves are recorded in light sleep.
- e. A diffuse generalized wave is recorded after an initial localized small wave.

3. EEG:

- a. Is recorded to map the brain.
- b. Is used for identification of sleep stages.
- c. Desynchronized waves are replaced by synchronized waves in response to opening of eyes.
- d. Show low frequency- high voltage waves in case of brain death.
- e. Is recorded in an anesthetized person after receptor stimulation.

4. Brain waves:

- a. Regular synchronized alpha waves are recorded in paradoxical sleep.
- b. Desynchronized, low voltage, high frequency waves are recorded due to activation of RAS.
- c. Highest voltage low frequency waves are recorded in adults during awake relaxed state.
- d. Intrinsic activity of reticular formation produces beta waves.
- e. Gamma waves are recorded in deep sleep.

5. The EEG wave recorded in high mental activity, hyperaroused state is:

- a. Alpha (8-13 HZ, 50 μ v, regular, synchronized).
- b. Beta (13-30 HZ, 25 μ v, irregular, desynchronized).
- c. Theta (4-7 HZ, 100 μ v, regular, synchronized).
- d. Delta (0.5-3.5 HZ, 100 μ v, regular, synchronized).
- e. Gamma (30-80 HZ, 0.5-2 μ v, irregular, desynchronized).

6. The EEG wave recorded in children is:

- a. Alpha (8-13 HZ, 50 μ v, regular, synchronized).
- b. Beta (13-30 HZ, 25 μ v, irregular, desynchronized).
- c. Theta (4-7 HZ, 100 μ v, regular, synchronized).
- d. Delta (0.5-3.5 HZ, 100 μ v, regular, synchronized).
- e. Gamma (30-80 HZ, 0.5-2 μ v, irregular, desynchronized).

7. β waves of EEG are:

- a. Character of relaxed awake state.
- b. Faster than α waves, but slower than δ waves.
- c. Recorded in children.
- d. Disappear when the person becomes alert.
- e. Recorded in paradoxical sleep.

8. Regarding Evoked brain potentials:

- a. It is used for mapping of the brain.
- b. It is recorded in a conscious person.
- c. It consists of a primary diffuse wave followed by a secondary localized wave.
- d. It is due to activity of RAS.
- e. It represents summation of excitatory and inhibitory potentials in brain neurons

1. Which of the following sentences about sleep and arousal is true:

- a. Posterior hypothalamus is a Non-REM sleep center.
- b. Cholinergic neurons of upper brain stem induce arousal.
- c. Serotonergic brain stem neurons induce REM sleep.
- d. Suprachiasmatic nucleus is a REM sleep center.
- e. Orexin hormone of the hypothalamus regulates brain stem nuclei activity and switches the activity between sleep and arousal.

2. REM sleep differs from Non-REM sleep in:

- a. Has longer duration in the sleep cycles.
- b. Low voltage desynchronized brain waves are recorded during it.
- c. Presence of sleep walking & talking.
- d. Low threshold of arousal.
- e. Increased level of growth hormone.

3. Sleep, all are true EXCEPT:

- a. It helps brain rest and brain plasticity.
- b. It may be due to synaptic fatigue of RAS neurons.
- c. It may be due to active inhibition of RAS by sleep centers.
- d. It may be due to increased activity of raphe magnus and locus coeruleus brain stem nuclei.
- e. It may be due to accumulation of sleep-inducing lipids or peptide in the CSF.

4. Compared to REM sleep, Non-REM sleep is characterized by:

- a. Dreams and penile erection.
- b. Irregular HR and ABP.
- c. Marked decrease in muscle tone.
- d. Increase in growth hormone secretion.
- e. Recording of high frequency, low voltage, desynchronized brain waves.

5. Paradoxical sleep is characterized by:

- a. Occurs at the start of sleep cycle.
- b. Includes 4 sleep stages.
- c. Movement of eye, finger and toe muscles.
- d. Marked decrease in blood pressure & heart rate.
- e. Important for physical rest.

6. Which of the following is a REM sleep center:

- a. Raphe nucleus.
- b. Locus coeruleus.
- c. Reticular formation.
- d. Anterior hypothalamus.
- e. Suprachiasmatic nucleus of hypothalamus.

7. Which of the following is a neurotransmitter for REM sleep:

- a. Acetylcholine.
- b. Norepinephrine.
- c. Serotonin.
- d. Histamine.
- e. GABA.

**1. Which of the following is NOT a function of the prefrontal association area:**

- a. Operation of working memory.
- b. Storage of coordinated vocalization programs.
- c. Decision making and personality traits.
- d. Thoughts.
- e. Language comprehension.

2. Concerning speech, all are true EXCEPT:

- a. Sensory aphasia occurs due to defect in visual association area, auditory association area or Wernicke's area.
- b. Expressive aphasia occurs due to lesion in Broca's area or Exner's area.
- c. Anomic aphasia is due to lesion in and around auditory cortex.
- d. In anomic aphasia, the patient cannot give name for seen pictures and words.
- e. Aphasia occurs only in case of lesion in the categorical hemisphere.

3. Aphasia:

- a. Is an abnormal speech due to defect in articulation.
- b. Is of fluent type due to lesion in Wernicke's area.
- c. May be due to lower motor neuron lesion of vocalization muscles.
- d. Is expressive in type due to lesion in auditory association cortex.
- e. Broca's aphasia characterized by excessive meaningless speech.

4. Lesion in frontal lobe speech center produces:

- a. Speech muscle paralysis.
- b. Inability to select appropriate words.
- c. Inability to process visual languages.
- d. Sensory aphasia.
- e. Expressive defect in speech.

5. Speech centers do NOT include:

- a. Wernicke's area.
- b. Broca's area.
- c. Somatic sensory association area.
- d. Auditory association area.
- e. Visual association area.

6. Sensory aphasia occurs with lesion in:

- a. Visual pathway.
- b. Broca's area.
- c. Exner's area.
- d. Auditory association area.
- e. Tongue muscles.

7. Concerning learning:

- a. Habituation is due to increase intracellular calcium leading to augmented response.
- b. Sensitization occurs by presynaptic facilitation.
- c. Operant conditioning occurs by inducing a response to a neutral stimulus.
- d. Classical conditioning occurs by stimulation of reward and punishment centers.
- e. -ve operant conditioning means adding of something to induce learning of a specific task or behavior

8. Motor aphasia occurs with lesion in:

- a. Primary motor area.
- b. Auditory association area.
- c. Wernicke's area.
- d. Angular gyrus.
- e. Exner's area.

9. Concerning Learning:

- a. In habituation, a repeated benign stimulus produces gradual loss of response.
- b. Associative learning is present when a noxious stimulus occurs with a benign one.
- c. Pairing of stimuli is a type of simple learning.
- d. Adding a reward to decrease a certain behavior is a +ve reinforcement.
- e. In classical conditioning, an unconditioned stimulus can produce a certain response when paired with a neutral conditioned stimulus.

10. Regarding dysarthria, all of the following are true EXCEPT:

- a. It is a speech abnormality due to defect in articulation .
- b. It occurs in UMNL & LMNL affecting speech muscles.
- c. Is of slow monotonous type in Parkinsonism.
- d. Is of spastic type due to lesion in Wernicke's area.
- e. Is of ataxic staccato type in neocerebellar syndrome



1. Which sentence is **NOT** correct for long-term potentiation:

- a. A rapid repeated brief stimulation leads to a persistent enhancement of response.
- b. Glutamate transmitter and its NMDA receptors may play a role.
- c. Increase Ca^{++} in pre and post-synaptic neurons may play a role.
- d. Protein synthesis, growth and increased neuronal connections are possible mechanisms.
- e. Involved in cerebellar learning.

2. In memory:

- a. Implicit memory is the conscious recall of information.
- b. Explicit memory is stored in basal ganglia and cerebellum.
- c. Explicit memory types include priming and procedural memory.
- d. Episodic memory is stored in the hippocampus and neocortex.
- e. Semantic memory is the recall of events and experiences.

3. Short-term memory is:

- a. Stored after 2 hours from getting the information.
- b. Brain capacity for it is unlimited.
- c. Encoded and stored in the neocortex.
- d. Due to transient synaptic modifications such as increased release of neurotransmitters.
- e. Involves gene activation and protein synthesis

4. Long-term memory is:

- a. Stored immediately after getting the information.
- b. Brain capacity for it is limited.
- c. Encoded in hippocampus and stored in the frontal lobe.
- d. Due to transient synaptic modifications such as increased release of neurotransmitters.

- e. Involves gene activation and protein synthesis.

5. Which sentence is **NOT** true regarding consolidation of memory:

- a. It is the conversion of short-term memory to a long-term one.
- b. Completed in 5 minutes.
- c. Represents the time needed for the structural and functional synaptic changes to take place.
- d. Occurs in neocortex.
- e. Increases and fastens by focusing and rehearsal.

6. Mechanisms of long-term memory do **NOT** include:

- a. Transient synaptic modifications.
- b. Increase presynaptic terminals and vesicles.
- c. Structural and functional synaptic changes.
- d. Formation of new synapses.
- e. Gene activation and protein synthesis

7. The primary role of amygdala in memory is:

- a. Storage of semantic memory.
- b. Consolidation of memory.
- c. Operation of short-term memory.
- d. Processing and remembering the emotional component of memories.
- e. Initiation of synaptic changes needed for memory storage.

8. Consolidation of memory increases by:

- a. Anesthesia.
- b. Sleep.
- c. Concussion.
- d. Electroshock therapy.
- e. Mental fatigue.

9. Consolidation of memory decreases by:

- a. Interesting information.
- b. Lack of REM sleep.
- c. Rehearsal.
- d. Mental rest.
- e. Quiet sleep.

10. Habituation:

- a. Is the gradual increase in response in presence of a repeated stimulus.
- b. Is involved in operant conditioning.
- c. Explained by gradual inactivation of Ca^{++} channels in the postsynaptic neuron.
- d. Leads to the formation of new synapses.
- e. Is due to gradual decrease in the neurotransmitter release in the involved pathway.

11. Which sentence is NOT true about sensitization:

- a. A prolonged augmented response occurs when a noxious stimulus paired to a benign stimulus.
- b. It is due to serotonin-induced facilitation.
- c. Is involved in short-term memory.
- d. Is involved in long-term memory.
- e. It continues up to 60 seconds after removal of the stimulus.

12. Post-tetanic facilitation:

- a. Is a gradual loss of response to a repeated benign stimulus.
- b. The postsynaptic response continues as long as the presynaptic stimulation is maintained.
- c. Explained by accumulation of Ca^{++} in the presynaptic neuron upon rapid short-term stimulation.
- d. Is associated with decrease postsynaptic response after removal of presynaptic stimulation.
- e. Is the basis of classical conditioning.

13. In long-term depression:

- a. A rapid repeated brief stimulation leads to a persistent enhancement of response.
- b. A slow, prolonged, low frequency stimulation leads to decrease strength of synaptic transmission.
- c. A rapid tetanizing presynaptic stimulation leads to continuation of postsynaptic response after removal of the stimulus.
- d. A noxious stimulus leads to presynaptic facilitation of a habituated pathway.
- e. A continuous benign stimulus leads to gradual increase in response.





1. Lateral nuclei of hypothalamus are NOT involved in:

- a. Feeding and thirst.
- b. Release of orexin hormone.
- c. Rage.
- d. Approach motivation.
- e. Heat gain.

2. Functions of the limbic system do NOT include:

- a. Perception and discrimination of olfaction.
- b. Control of feeding, maternal & sexual behaviors.
- c. Control of emotions.
- d. Control of water balance.
- e. Control of motivations.

3. Which sentence is NOT true about the Amygdala:

- a. Is involved in processing of the emotional components of memories.
- b. Is involved in processing of the emotional responses to odors.
- c. Its lesion leads to hyperphagia and omniphagia.
- d. Its stimulation leads to rage and aggression.
- e. Is the area for consolidation of memory.

4. The hypothalamus performs all of the following functions EXCEPT:

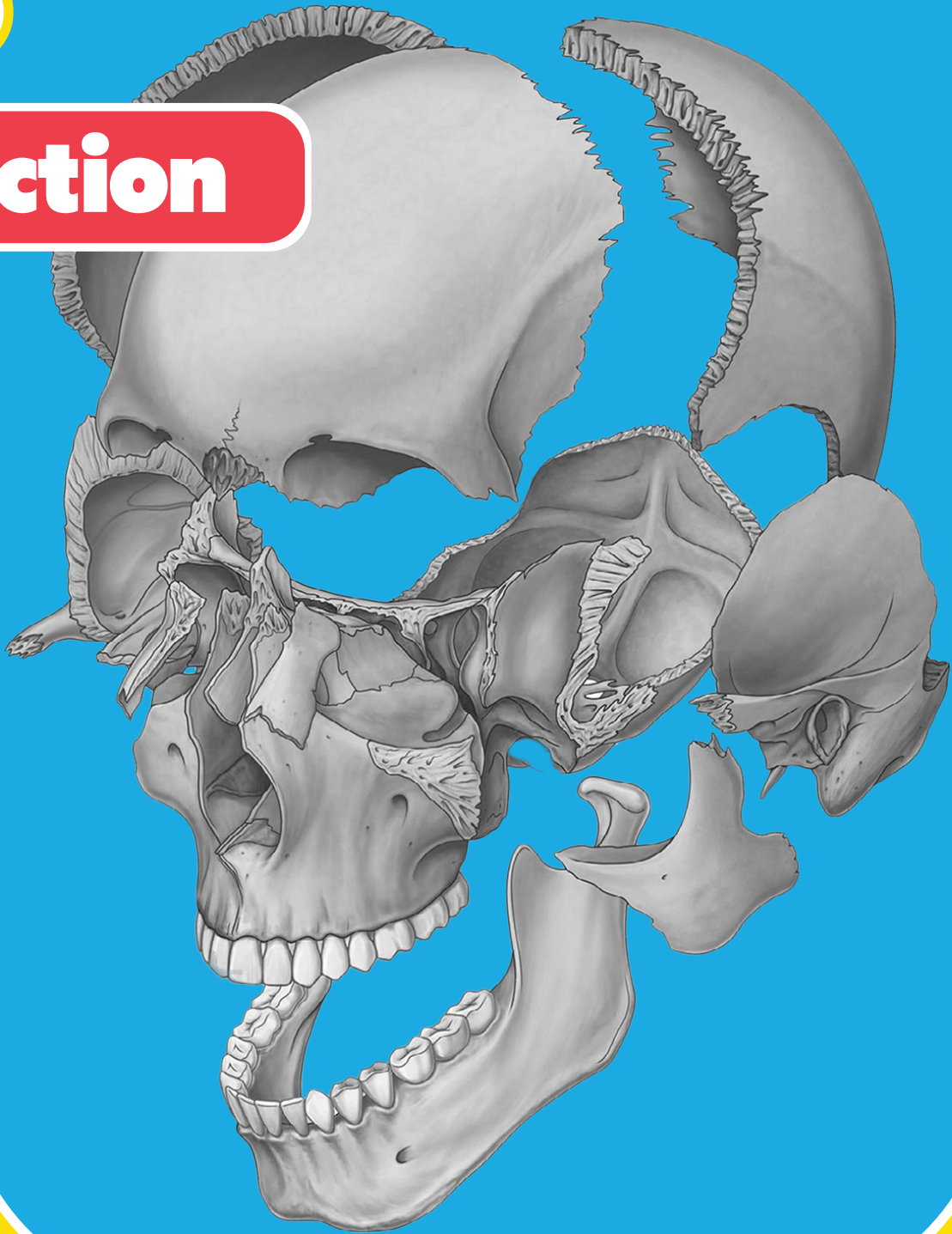
- a. Control of autonomic functions.
- b. Regulation of anterior and posterior pituitary hormonal secretion.
- c. Control of voluntary movements.
- d. Regulation of body temperature.
- e. Control of appetite and food intake.

5. Regarding the hypothalamus:

- a. Satiety center has intrinsic activity.
- b. Heat gain center is present in the anterior hypothalamus.
- c. Ventromedial nuclei synchronize body activities according to dark-light cycles.
- d. Posterior hypothalamus function as a sleep center.
- e. Anterior hypothalamic nuclei play a role in all of sleep, heat loss, parasympathetic activity and sexual drive.

3

Section



Histology



1. Receptor/s for proprioception is/are:

I. Pacinian corpuscle.

II. Meissner's corpuscle.

III. Ruffini's corpuscle.

a. I only.

c. III only.

b. II only.

d. I, II and III.

2. Which of the following is a receptor for vibration?

a. Meissner's corpuscle.

b. Ruffini's corpuscle.

c. Merkel's disc.

d. Pacinian corpuscle.

3. One of the following is NOT a character of Meissner's corpuscle:

a. It is an encapsulated nerve ending.

b. It is present in dermal papillae.

c. The Schwann cells are arranged in two cores.

d. The nerve fibers run spirally between the Schwann cells.

4. All about muscle spindle true except:

a. Present in parallel with sk. Muscle fibers.

b. Contain collagen fibers.

c. Has gamma afferent fibers.

d. Capsulated receptors

5. Krause end bulb, all true except:

a. Has gelatinous core

b. Is a mechanoreceptor.

c. Present in conjunctiva & mouth cavity.

d. Has flattened Teloglia cells.

6. Which of the following is a character of muscle spindle:

a. Present In series with muscle fibers.

b. Innervated by afferent & efferent nerve fiber.

c. Contain intrafusal collagen fibers.

d. Responds to tension on muscle.

e. Non capsulated receptor

7. Pacinian corpuscle:

a. Lies in dermal papillae of skin.

b. Noncapsulated receptor

c. Has transversely arranged schwann cells.

d. Nerve fibers run in spiral coarse.

e. Contain concentric layers of Teloglia.

8. Which of the following receptors are mechanoreceptors?

a. Plexus of Bounet

b. Merkel's disc

c. Meissner's corpuscle

d. All of the above

9. The sensory nerve enters the Meissner's corpuscle from:

a. lower part

d. Any site of the corpuscle

b. Side

c. Central core

10. Free nerve endings are widely distributed in

a. Epidermis of the skin

b. Cornea

c. Mucous membrane of visceral organs

d. Serous membranes

e. All of the above

11. What do free nerve endings respond to?

a. Noxious(painful) stimuli

c. Thermal stimuli

d. All of the above

b. Crude touch

12. What is the site of Merkel's tactile disc?

a. Deep layers of the epidermis of the palmer surfaces of hands and planter surfaces of feet

b. Transitional sites between hairy & non-hairy skin

c. All of the above

d. None of the above

13. Pacinian corpuscles are found in

a. Dermis & hypodermis

c. Urinary bladder

d. All of the above

b. Pancreas

Chapter 2 ? Spinal cord "Intro" & "Grey matter"

1. **One of the following is a character of lumbar level of spinal cord.**
 - a. It never has lateral horns.
 - b. Small in size.
 - c. Smallest whiter matter /grey matter ratio.
 - d. Ventral central canal.
2. **Which of the following is a character of thoracic level of spinal cord?**
 - a. Oval in shape.
 - b. Large in size.
 - c. It has 4 horns.
 - d. It has 2 lateral horns.
3. **Which of the following is a character of cervical level of spinal cord?**
 - a. Rounded in shape.
 - b. Has largest white matter/grey matter ratio
 - c. Small in size.
 - d. Central canal is central.
4. **About spinal nerve all true except:**
 - a. Lateral division of dorsal root carry pain, temp
 - b. Medial division carry proprioception & touch.
 - c. Dorsal root ganglia contain cells of same size.
 - d. Thick myelinated fibers carry proprioception.
5. **Which of the following is true about the dorsal root ganglion:**
 - a. It is a collection of nerve cells inside the central nervous system.
 - b. It is located at the viscera.
 - c. It has nerve cells of variable size.
 - d. It has no connective tissue capsule.
6. **Nerve cells of dorsal root ganglia are characterized by:**
 - a. Of the same size
 - b. Have multiple processes.
 - c. Have eccentric nucleus.
 - d. Associated with myelinated & unmyelinated nerve fibers.
 - e. Have shwann cells around its cell bodies.
7. **Nucleus dorsalis Clark's nucleus, all true except:**
 - a. In the medial aspect of the base of the posterior horn, In level from C8-L3
 - b. Involved in lamina 7 of gray matter.
 - c. large cells with eccentric nucleus.
 - d. Give rise to ventral spinocerebellar tract.
8. **One of the following nuclei is not present in all segments of the spinal cord:**
 - a. Postero-marginal nucleus.
 - b. Nucleus dorsalis (Clark's nucleus).
 - c. Substantia gelatinosa.
 - d. Main sensory nucleus (nucleus proprius dorsalis).
9. **Which of the following statements concerning the nucleus dorsalis of Clark is TRUE?**
 - a. It is found in the ventral horn.
 - b. It is present at all spinal levels.
 - c. It is homologues in function to the main sensory nucleus.
 - d. It projects to the cerebellum.
10. **Which of the following statements concerning the lateral horn of the spinal cord is TRUE?**
 - a. Gives origin to pre-ganglionic sympathetic fibers.
 - b. It is most prominent at sacral levels.
 - c. It is present in all spinal cord levels.
 - d. It contains pre-ganglionic parasympathetic neurons.

11. Destruction of the ventral horn results in one of the following:

- a. Loss of crude touch
- b. Loss of fine touch
- c. Flaccid paralysis
- d. Loss of pain sensation

12. Which of the following is true about the grey matter:

- a. It is located outside the white matter in spinal cord.
- b. It is formed of nerve cells mainly.
- c. It has satellite cells sheath around its cells.
- d. It has no neuroglia.

13. Anterior horn of spinal cord contains the following nuclei EXCEPT

- a. anteromedial
- b. anterolateral
- c. middle
- d. Posterolateral
- e. Posteromedial

14. Preganglionic sympathetic neurons are present in lamina of spinal cord gray matter:

- a. II
- b. III
- c. VII
- d. VI
- e. X

15. The main sensory nucleus of spinal cord is called:

- a. substantia gelatinosa
- b. nucleus proprius
- c. Clark nucleus
- d. posteromarginal nucleus
- e. none of the above

16. Anterior, posterior and lateral horns are present in....spinal cord segment

- a. Cervical
- b. Thoracic
- c. Sacral
- d. All of the above

17. Which spinal cord segment has a thin posterior horn and a thick anterior horn?

- a. Cervical
- b. Thoracic
- c. Lumber
- d. Sacral

18. Which spinal cord segment has a thick posterior horn and a thick anterior horn?

- a. Cervical
- b. Thoracic
- c. Lumber
- d. Sacral

19. White matter is formed of all of the following except

- a. Nerve fibers
- b. Neuroglia
- c. Blood vessels
- d. None of the above

20. Intermediomedial nucleus is found in... horn, but intermediolateral nucleus is found in.. horn

- a. Posterior , anterior
- b. Anterior , posterior
- c. Lateral , anterior
- d. Anterior , lateral
- e. Posterior , lateral

21. Grey matter is formed by all of the following except

- a. Nerve cell
- b. Neuroglia
- c. Blood capillaries
- d. None of the above

22. Posteromarginal nucleus is found in

- a. Lamina I
- b. Lamina II
- c. Lamina III
- d. Lamina V

23. Substantia Gelatinosa is found in

- a. Lamina I
- b. Lamina II
- c. Lamina III
- d. Lamina V

24. Main sensory nucleus is found in

- a. Lamina I
- b. Lamina II
- c. Lamina III & IV
- d. Lamina V

25. Clarke's nucleus is found in

- a. Lamina I
- b. Lamina II
- c. Lamina VI
- d. Lamina VII

26. Intermediomedial nucleus is found in

- a. Lamina I
- b. Lamina II
- c. Lamina VI
- d. Lamina VII

27. Intermediolateral nucleus is found in

- a. Lamina I
- b. Lamina II
- c. Lamina VI
- d. Lamina VII

28. Medial and lateral motor nuclear group is found in

- a. Lamina I
- b. Lamina II
- c. Lamina VI
- d. Lamina IX

29. Which lamina represents the central gray substance surrounding the central canal?

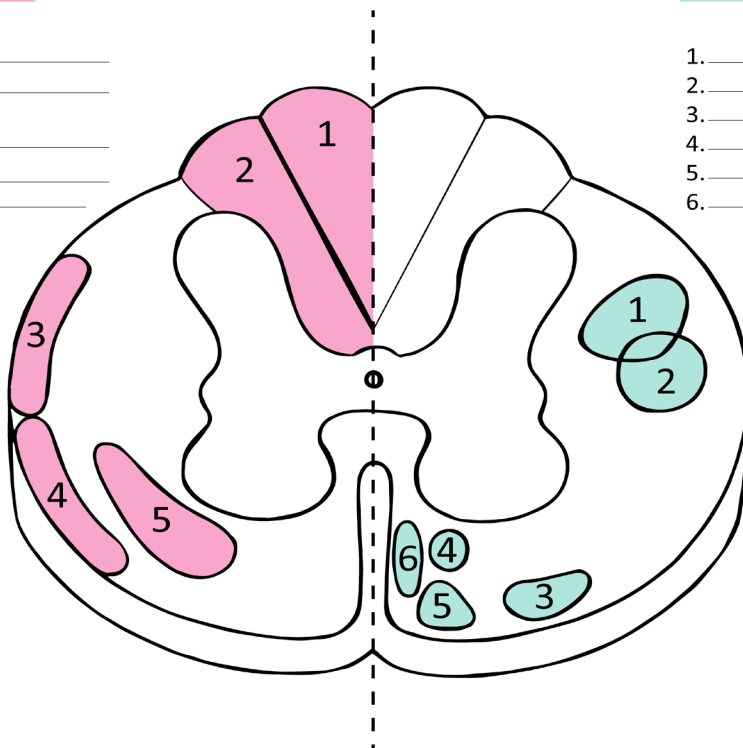
- a. Lamina I
- b. Lamina II
- c. Lamina IX
- d. Lamina X

ASCENDING TRACTS

1. _____
2. _____
3. _____
4. _____
5. _____

DESCENDING TRACTS

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____





1. Which of the following tracts lie in the dorsal funiculus:

- a. Cuneate
- b. Rubrospinal
- c. Vestibulospinal
- d. MLB
- e. Spinocerebellar

2. Which of the following tracts is present in all segments of spinal cord:

- a. Gracile
- b. MLB
- c. Cuneate
- d. Spinothalamic

3. One of the following tracts carries sensations to subcortical level:

- a. Dorsal spinocerebellar tract.
- b. Gracile tract.
- c. Lateral spinothalamic.
- d. Ventral spinothalamic.

4. All the following nuclei are present in all segments of the spinal canal EXCEPT

- a. Postero- marginal nucleus
- b. nucleus dorsalis (Clarke nucleus)
- c. substantia gelatinosa
- d. main sensory nucleus (nucleus proprius dorsalis)

5. The first order neuron in proprioception from the lower half of the body is in:

- a. Gracile nucleus.
- b. Cuneate nucleus.
- c. Spinal ganglia.
- d. Main sensory nucleus.

6. The second order neuron in proprioception from lower half of the body is in:

- a. Gracile nucleus on the same side.
- b. Cuneate nucleus on the same side.
- c. Gracile nucleus on opposite side.
- d. Cuneate nucleus on opposite side.

7. One of the following is true about gracile tract:

- a. Presents only in cervical&thoracic levels of spinal cord.
- b. Carries proprioception from the upper half of the body.
- c. Terminates at gracile nucleus in medulla.
- d. Arises from spinal ganglia in upper half of body.

8. One of the following is true about cuneate tract:

- a. It arises from spinal ganglia in upper half of body.
- b. It arises from cuneate nucleus on same side.
- c. It arises from cuneate nucleus on opposite side.
- d. It is present in lateral column of white matter in spinal cord.

9. 38 years old male complains of unsteadiness in gait especially in the Dark, his complaint may be due to lesion in:

- a. Lateral spinothalamic tract
- b. Ventral spinothalamic tract
- c. Dorsal spinocerebellar tract
- d. Dorsal column tracts

10. One of the following statements is true about syringomyelia:

- a. There is congenital dilatation of central canal in lumbar region.
- b. There is pressure on dorsal column.
- c. There is pressure on lateral spinothalamic tracts.

**1. The central nervous system is soft because:**

- a. It consists of neurons joined by connective tissue.
- b. It is rich in blood vessels.
- c. It does not contain connective tissue.
- d. It contains fat cells.

2. Which of the following is/are true for central nervous system?

- I. It is rich in connective tissue cells.
- II. It is soft jelly like.
- III. It has grey matter and white matter.

- a. I only
- b. II only
- c. II and III only.
- d. I, II and III.

3. Dura matter characterized by being:

- a. Fibrous nonvascular layer
- b. Have many trabeculae.
- c. Vascular cellular layer
- d. Fibrous vascular layer

4. Concerning the pia matter, which of the following statements is true:

- a. It is highly vascular.
- b. It contains spaces for CSF.
- c. It contains venous sinuses.
- d. It is most outer layer.

5. The processes of protoplasmic astrocytes characterized by:

- a. Few long thin
- b. Few short spiny
- c. Numerous short
- d. Few short

6. Tanycytes line which of the following:

- a. All cavities in brain
- b. Choroid plexus
- c. Choroid villi
- d. 3rd ventricle

7. The apical surface of ependymal Cells has:

- a. Microvilli
- b. Cilia
- c. Sterocilia
- d. Microfolds

8. All of the followings have choroid Plexus EXCEPT:

- a. 3rd ventricles
- b. 4th ventricles
- c. Lateral ventricle
- d. Cerebral aqueduct

9. The central canal is lined by:

- a. Simple squamous epithelium
- b. Stratified squamous epithelium.
- c. Simple columnar to cuboidal epithelium partially ciliated.
- d. Simple columnar non ciliated epithelium.

10. The oligodendrocyte is derived from ... while microglia is derived from ...:

- a. Ectoderm; mesoderm
- b. Ectoderm
- c. Mesoderm, ectoderm
- d. Mesoderm, mesoderm

11. Which of the following cells Produce myelin sheath?

- a. Oligodendrocytes.
- b. Astrocytes
- c. Satellite cells.
- d. Ependymal

12. In the CNS, which of the following are phagocytic cells:

- a. Astrocytes.
- b. Microglia
- c. Oligodendrocytes.
- d. Ependymal cells

13. In the CNS Injury, which of the following cells are abundant:

- a. Astrocytes & microglia.
- b. Microglia & oligodendrocytes
- c. Oligodendrocytes & astrocytes
- d. Ependymal cells

14. Which of the following is involved in the blood brain barrier?

- a. Astrocytes.
- b. Ependymal cells.
- c. Oligodendrocytes.
- d. Schwann cells.

15. The distribution of grey and white matter depends on the distribution of:

- a. Blood vessels.
- b. Myelin.
- c. Pigments.
- d. Astrocytes.

16. One of the following a component of Blood CSF barrier:

- a. Border cells of arachnoid matter
- b. tight junctions between ependymal cells.
- c. astrocytes.
- d. Trabecular cells of arachnoid matter

17. One of the followings is not present in grey matter:

- a. Nerve cells.
- b. Myelinated nerve fibers.
- c. Neuroglial cells.
- d. Unmyelinated nerve fibers.

18. Martinotti cells of cerebral cortex are characterized by:

- a. Their dendrites are directed laterally.
- b. Their axons are directed upwards.
- c. Their cell bodies are rounded.
- d. Their neurotransmitter is acetyl choline.
- e. their cytoplasm is basophilic.

19. Inner Pyramidal cells are present in layer of cerebral cortex:

- a. I
- b. II
- c. III
- d. IV
- e. None of the above

20. Giant cells of Betz are present in which layer of cerebral cortex:

- a. Outer pyramidal
- b. Inner pyramidal
- c. Outer granular
- d. Molecular

21. Cerebral cortex is characterized by:

- a. Has smooth surface.
- b. Neurons of the same shapes
- c. Layers of the same thickness all over the brain
- d. Has both laminar & columnar arrangement
- e. It lacks plasticity.

22. Cells of Cajal are present in

- a. molecular layer of cerebrum
- b. molecular layer of cerebellum
- c. granular layer of cerebrum
- d. granular layer of cerebellum
- e. none of the above

23. All of the following are layers of cerebral cortex EXCEPT

- a. molecular layer
- b. purkinje layer
- c. external granular layer
- d. internal granular layer
- e. multiform layer

24. Pia matter is formed of:

- a. simple squamous epithelium.
- b. simple columnar epithelium.
- c. simple columnar ciliated epithelium.
- d. modified fibroblasts.

25. The inner surface of dura mater is lined by.

- a. Pseudostratified columnar epith.
- b. Simple columnar epith.
- c. Simple squamous epith.
- d. stratified columnar epith.

26. The brain CSF barrier is not formed of:

- a. Fenestrated endothelial cells
- b. Foot processes of astrocytes
- c. Basement membrane of ependymal cells
- d. Ependymal cells connected by tight junction.

27. The blood CSF barrier is not formed of:

- a. Endothelial cells of fenestrated blood capillaries
- b. Foot processes of astrocytes
- c. Basement membrane of blood capillaries
- d. Ependymal cells connected by tight junction.

28. tight junction between ependymal cells forms:

- a. blood brain barrier
- b. blood CSF barrier
- c. outer glia limitans
- d. inner glia limitans
- e. arachnoid barrier

29. Glial fibrillar acidic protein (GFAP) is marker of:

- a. Astrocyte
- b. Oligodendroglia
- c. Ependymal cells
- d. Microglia

30. Which of the following share in the cerebellar islands:

- a. Dendrites of stellate cells
- b. Dendrites of Golgi II cells
- c. Dendrites of granule cells
- d. Climbing fibers

31. Cerebellar glomeruli contain the following structures EXCEPT:

- a. Rosettes of mossy fibers
- b. Dendrites of granule cells
- c. Axons of Golgi cells
- d. Dendrites of basket cells

32. The main efferent fiber of the cerebellar cortex is formed by axons of:

- a. Purkinje cells
- b. Golgi cells
- c. Basket cells
- d. Granule cell

33. Cerebellar cortex:

- a. Contain martinotti cells in the molecular layer.
- b. Is of uniform structure all over cerebellum
- c. Formed of 6 layers
- d. Show sulci.

34. Climbing fibers:

- a. Axons of Golgi cells
- b. Olivocerebellar fibers
- c. End in granular layer
- d. Part of cerebellar island

35. Which of the following tracts carries the climbing fibers of cerebellum?

- a. Ventral spinothalamic tract
- b. Dorsal spinocerebellar tract
- c. Pontocerebellar tract
- d. Olivocerebellar tract

36. The molecular layer of cerebellum contains which of the following structures?

- a. Purkinje cell bodies
- b. Basket cell bodies
- c. Golgi cell bodies
- d. Cerebellar glomeruli

37. The deepest layer of the cerebellar cortex is

- a. Molecular layer
- b. Purkinje layer
- c. Granular layer

38. All of the following statements are correctly describing the pyramidal cells except:

- a. The apex is directed to the cortical surface, a thick branching dendrite arises from the base of the cell and passes into the underlying white matter.
- b. Form Bitz Cells(giant cells).
- c. They are well developed in motor centers.

39. All of the following are histological layers of the cerebral cortex except:

- a. Pyramidal
- b. Molecular
- c. Granular
- d. Purkinje

40. All of the following are cells of the cerebral cortex except:

- a. Pyramidal cells
- b. Granular cells
- c. Fusiform cells
- d. Cells of Martinotti
- e. None of the above

41. The outer band of baillarger is present in which layer

- a. Molecular layer
- b. External granular layer
- c. Inner granular layer
- d. Inner pyramidal layer

42. The inner band of baillarger is present in which layer

- a. Molecular layer
- b. External granular layer
- c. Inner granular layer
- d. Inner pyramidal layer

43. The cytoplasm of astrocytes contains

- a. Microtubules
- b. Glial Fibrillary acid protein(GFAP)
- c. Thread-like mitochondria
- d. All of the above

44. Which of the following is true about microglia?

- a. They are present in the retina and optic nerve
- b. They are small spindle shaped cells with branching poles
- c. The cell body and processes have minute spines on their surfaces
- d. All of the above



1. **Fibers of vagus nerve exit**
 - a. Between pyramids & olive
 - b. Between olive & inferior cerebellar peduncle
 - c. At pontomedullary junction
 - d. At spinomedullary junction
2. **Which of the following nuclei receive the GSA fibers of facial nerve:**
 - a. Spinal nucleus of trigeminal nerve
 - b. Mesencephalic nucleus
 - c. Main sensory nucleus
 - d. Solitary nucleus
3. **The cranial nerve whose component is Only GSE:**
 - a. Hypoglossal nerve
 - b. Vagus nerve
 - c. Glossopharyngeal N.
 - d. Accessory nerve.
4. **Spinal n. of trigeminal, equivalent to:**
 - a. Substantia gelatinosa
 - b. Clark's n
 - c. Main sensory n
 - d. Commissural nuclei
5. **Nuclei of the vagus nerve include the following except:**
 - a. Nucleus ambiguus
 - b. Dorsal motor nucleus of vagus
 - c. Inferior salivatory nucleus
 - d. Solitary nucleus
6. **Cranial nerve nucleus in floor of 4th ventricle close to midline in lower pons is:**
 - a. Hypoglossal nerve
 - b. Abducent nucleus
 - c. Motor facial nucleus
 - d. Motor vagus nucleus

7. **The only cranial nerve that decussate before its emergence is:**
 - a. Oculomotor
 - b. Trochlear
 - c. Abducent
 - d. Facial





1. Internal arcuate fibers are present in:

- a. Superior midbrain.
- b. Inferior midbrain.
- c. Open medulla.
- d. Closed medulla.

2. Each of the following statements concerning closed medulla is true **EXCEPT**

- a. in the lower part there are motor decussation
- b. at a higher level there are sensory decussation
- c. contains the main inferior olivary nucleus.
- d. contains gracile and cuneate nucleus.
- e. contains gracile and cuneate tracts.

3. Cranial nerves in medulla oblongata are the following **EXCEPT**

- a. hypoglossal
- b. accessory
- c. vagus
- d. facial
- e. glossopharyngeal

4. Gracile & Cuneate N. are present in:

- a. Dorsal aspect of closed medulla
- b. Ventral aspect of closed medulla
- c. Dorsal aspect of open medulla
- d. Ventral aspect of open medulla

5. Crossing fibers of gracile & cuneate tracts are called:

- a. Internal arcuate
- b. Dorsal external arcuate
- c. Ventral external arcuate
- d. Horizontal

6. Lowermost level of medulla shows:

- a. Internal arcuate fibers
- b. Decussating corticospinal fibers
- c. Inferior olive nucleus
- d. Superior olive nucleus

7. Arcuate nucleus is found in:

- a. Ventral to pyramids
- b. Ventral to cuneate nucleus
- c. Medial to olive
- d. Dorsal to Inferior cerebellar Peduncle

8. Ventral external arcuate fibers arise from:

- a. Accessory cuneate nucleus
- b. Cuneate nucleus
- c. Arcuate nucleus
- d. Pontine nucleus

9. Accessory cuneate nucleus give rise to:

- a. Uncrossed fibers to cerebellum
- b. Crossed fibers to cerebellum.
- c. Crossed fibers to medial lemniscus.
- d. Uncrossed fibers in MLB

10. The central gray in upper closed medulla contains:

- a. Spinal nucleus of V nerve
- b. Olive nucleus
- c. Nucleus solitaires
- d. Arcuate nucleus

11. Internal arcuate fibers are seen in:

- a. Lower level of closed medulla
- b. Upper level of closed medulla
- c. Open medulla
- d. Lower pans

12. The reticular formation controls the endocrine system due to its connections with:

- a. hypothalamus
- b. Spinal cord
- c. Thalamus
- d. Cerebral cortex

13. Inferior olivary nucleus:

- a. Receive rubro-olivary fibers.
- b. Present in the ventral aspect of closed medulla.
- c. Its efferent fibers reach the ipsilateral cerebellar cortex through the superior cerebellar peduncle.
- d. Its efferent fibers are the mousy fibers reaching the cerebellar cortex.

1. One of the relays nuclei of auditory pathway is:

- a. Inferior olive nucleus
- b. Superior olivary nucleus
- c. Vestibular nucleus
- d. Superior colliculus

2. Sections in lower level of pons show.

- a. Middle cerebellar peduncle:
- b. Superior cerebellar peduncle
- c. Mesencephalic nucleus of V
- d. Medial & Inferior vestibular nuclei

3. Basis pontis contains the following except:

- a. Corticospinal fibers
- b. Corticonuclear fibers
- c. Pontine reticulospinal
- d. Transverse pontine fibers

4. Auditory pathways contain the following except:

- a. Cochlear nuclei
- b. Trapezoid body
- c. Medial lemniscus
- d. Inferior colliculus
- e. Medial geniculate body

5. About substantia nigra, all true except:

- a. Formed of st. compactum & reticularis
- b. Connected to basal ganglia.
- c. Contains dopamine.
- d. Present dorsal to basis pontis

6. The superior colliculus of midbrain is formed of the following EXCEPT.

- a. Stratum basale
- b. Stratum zonale
- c. Stratum opticum
- d. Stratum lemnisci
- e. Stratum cinereum

7. The tract in MLB responsible for head righting reflex is:

- a. Tectospinal
- b. Vestibule spinal
- c. Reticulospinal
- d. Interstitiospinal

8. The specific feature of upper mid brain is presence of:

- a. Crus cerebri
- b. Substantia nigra
- c. Red nucleus
- d. Superior cerebellar peduncle decussation

9. Central tegmental tract contains fibers connect between:

- a. Red nucleus & olivary nucleus
- b. Superior colliculus & Inferior Colliculus
- c. Red nucleus & pontine nucleus
- d. Olivary nucleus & pontine nucleus

10. Medial longitudinal bundle, all true except:

- a. receive ascending & descending fibers from vestibular nuclei.
- b. Function for movement of the head & neck
- c. extending from brainstem till cervical segments of spinal cord.
- d. Extend the whole length of spinal cord.

11. Dorsal tegmental decussation:

- a. Formed of rubrospinal tract
- b. Its fibers descend in all segments of spinal cord.
- c. Present in inferior level of midbrain.
- d. Formed by tectospinal tract.

12. Superior cerebellar decussation:

- a. Present in inferior level of midbrain.
- b. Formed by tectospinal tract.
- c. Fibers pass through ICP.
- d. Its fibers descend in all segments of spinal cord.

13. Superior colliculus, all true except:

- a. Composed of small & large nuclei
- b. Connected with retina.
- c. Connected to occipital cortex.
- d. Give tectospinal tract & dorsal tegmental decussation.
- e. Present in tectum of the upper midbrain.

14. Which of the following nuclei extend in the midbrain:

- a. Vestibular nuclei
- b. Mesencephalic nuclei
- c. Motor n of facial
- d. Superior olive

15. Fibers derived from which nucleus run transversally to form the trapezoid body?

- a. Ventral cochlear nucleus
- b. Dorsal cochlear nucleus
- c. All of the above
- d. None of the above

16. Which of the following tracts are present in the tegmentum pontis?

- a. Medial longitudinal fasciculus
- b. Spinothalamic tract
- c. Central tegmental tract
- d. All of the above

17. What is the lateral lemniscus formed of?

- a. Three acoustic stria
- b. Fibers from superior olive
- c. Fibers from the nuclei of the trapezoid body
- d. All of the above

18. The dorsal tegmental decussation is formed by

- a. Tectospinal
- b. Rubrospinal
- c. Anterior spinocerebellar
- d. Posterior spinocerebellar

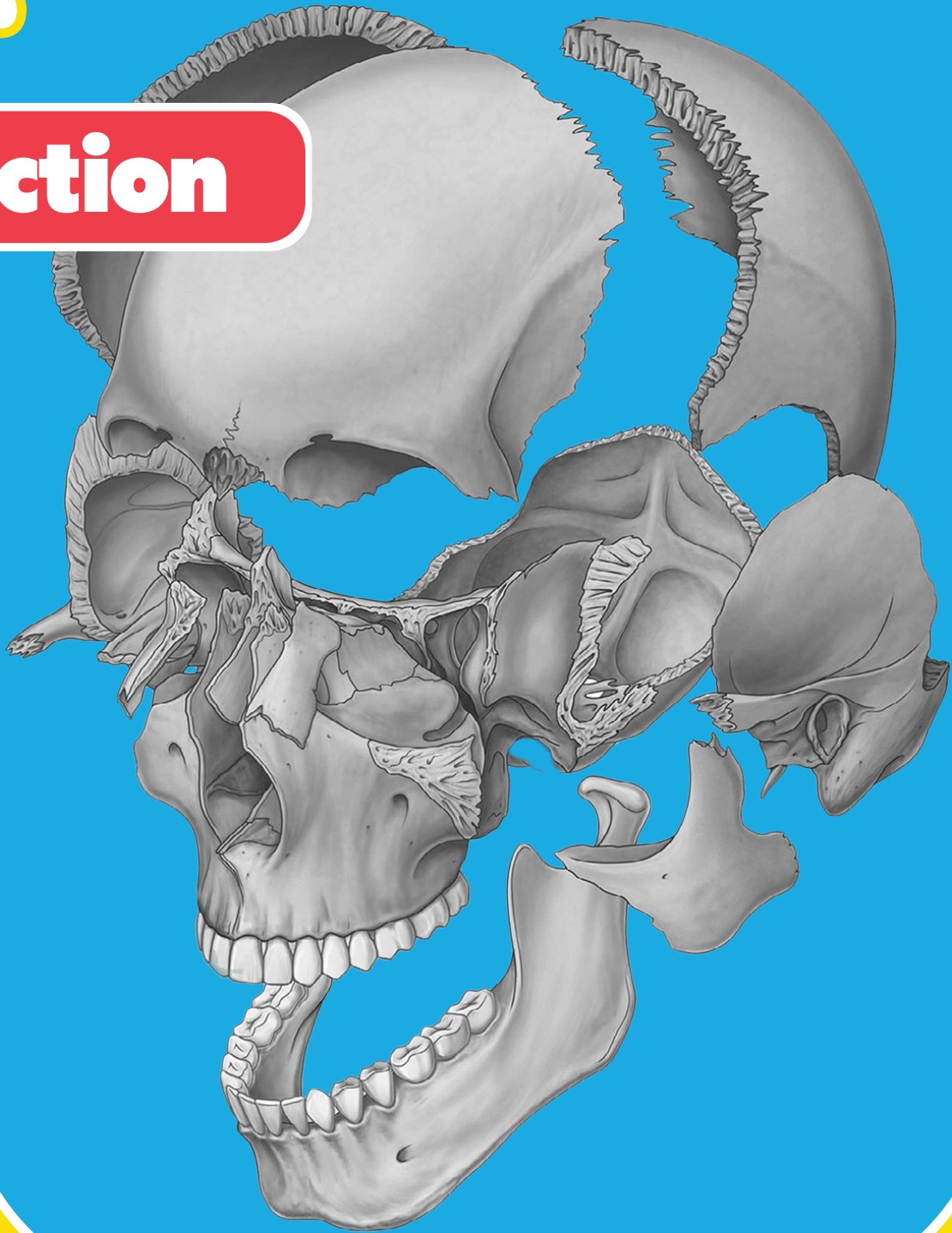
19. The ventral tegmental decussation is formed by

- a. Tectospinal
- b. Rubrospinal
- c. Anterior spinocerebellar
- d. Posterior spinocerebellar



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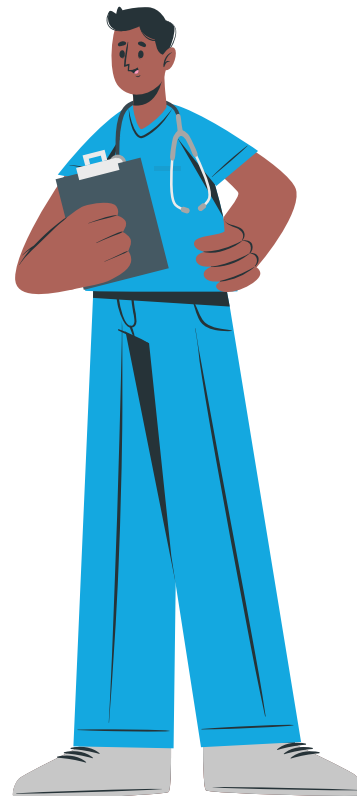
Section



Biochemistry



1. **Six hours after a meal, which of the following is (are) MOST important as a source of fuel for the brain?**
 - a. Glycogen from the muscle.
 - b. Ketone bodies from the liver.
 - c. Protein from the muscle.
 - d. Fatty acids from the adipose tissue.
 - e. Glycogen from the liver.
2. **Which of the following is an absolute requirement for the brain at all times?**
 - a. Glucose and oxygen.
 - b. Glucose and ketone bodies.
 - c. Ketone bodies and oxygen.
 - d. Glucose and fatty acids.
3. **In fasting conditions, glucose is provided for the brain by.**
 - a. Absorption from the intestine.
 - b. glycogenolysis in the liver.
 - c. gluconeogenesis in the liver
 - d. (b) and (c).
4. **Brain cells uptake glucose by**
 - a. GLUT1 and GLUT2.
 - b. GLUT1 and GLUT3.
 - c. GLUT4.
 - d. SGLT2.
5. **Brain cells oxidize glucose to**
 - a. Lactate.
 - b. Ketone bodies.
 - c. carbon dioxide.
 - d. carbon dioxide and water.
6. **Glucose 6-phosphate is utilized in which pathway in brain cells?**
 - a. Glycolysis.
 - b. Pentose phosphate pathway
 - c. Both (a) and (b).
 - d. Neither (a) nor (b)
7. **What is the importance of HMP pathway in the brain cells?**
 - a. Nucleic acid synthesis.
 - b. Antioxidant.
 - c. Gluconeogenesis.
 - d. Mitochondrial transfer of reducing equivalents.



1. **In carcinoid syndrome the argentaffin tissue of the abdominal cavity overproduce:**
 - a. Serotonin
 - b. Histamine
 - c. Tryptamine
 - d. Tyrosine
2. **Tryptophan could be considered as precursor of:**
 - a. Melatonin
 - b. Thyroid hormones
 - c. Melanin
 - d. Epinephrine
3. **The rate limiting step in the biosynthesis of catecholamines is:**
 - a. Decarboxylation of dihydroxyphenylalanine
 - b. Hydroxylation of phenylalanine
 - c. Hydroxylation of tyrosine
 - d. Oxidation of dopamine
4. **Which of the following amino acids may act as a neurotransmitter?**
 - a. Glycine.
 - b. Glutamic acid
 - c. Gamma-amino-butyric acid (GABA).
 - d. All the above.
5. **The enzyme dopamine -oxidase (hydroxylase) which catalyzes conversion of dopamine to norepinephrine requires:**
 - a. Vitamin A
 - b. Vitamin C
 - c. Vitamin E
 - d. Vitamin B12
6. **Histamine is formed from histidine by:**
 - a. Deamination
 - b. Dehydrogenation
 - c. Decarboxylation
 - d. Carboxylation
7. **vasodilating compound is produced by the decarboxylation of the amino acid:**
 - a. Arginine
 - b. Aspartic acid
 - c. Glutamine
 - d. Histidine
8. **Gamma-amino-butyric acid (GABA) is Produced by**
 - a. Decarboxylation of glutamic acid.
 - b. Transamination of gamma-hydroxy- butyric acid.
 - c. a reaction that requires vitamin B6 derived cofactor.
 - d. A and c
9. **Serotonin is synthesized from.**
 - a. Serine
 - b. Phenylalanine
 - c. Tyrosine
 - d. Tryptophan
10. **Which of the following increases in 24 Hour urine with pheochromocytoma (adrenaline-secreting tumor)**
 - a. Noradrenaline.
 - b. Vanillylmandelic acid (VMA),
 - c. 5-Hydroxyindole-acetic acid (5-HIAA)
 - d. Phenylacetyl-glutamine.
11. **Which of the following may be measured in 24-hour urine to diagnose carcinoid syndrome (argentaffinoma, serotonin secreting tumor)?**
 - a. 5-Hydroxytryptamine.
 - b. 5-Hydroxyindole-acetic acid (5-HIAA)
 - c. Vanillylmandelic acid (VMA).
 - d. Phenylacetyl-glutamine.

12. Which of the following may serve as a tumor marker?

- a. Vanillylmandelic acid.
- b. 5-hydroxyindole acetic acid
- c. Phenyl pyruvic acid.
- d. a and b.

13. Which of the following enzymes is Important for synthesis of both epinephrine and melatonin?

- a. Aromatic amino acid decarboxylase.
- b. O-methyltransferase.
- c. Dihydrobiopterin reductase
- d. a and c.

14. Methyl dopa decreases blood pressure by

- a. Inhibiting the synthesis of catecholamines
- b. Antagonizing the action of aldosterone
- c. Stimulating the release of renin
- d. Inhibiting the breakdown of angiotensin

15. Nitric Oxide is a hormone and a neurotransmitter synthesized.

- a. by the action of nitric oxide synthase.
- b. From glycine.
- c. both (a) and (b).
- d. Neither (a) nor (b).

16. GABA (gamma amino butyric acid) is:

- a. Post-synaptic excitatory transmitter
- b. Post-synaptic inhibitor transmitter
- c. activator of glia-cell function
- d. Inhibitor of glia-cell function

17. Mono Amine Oxidase (MAO) catalyzes the conversion of serotonin to:

- a. Dopamine.
- b. 5-hydroxyindole-acetic acid
- c. Epinephrine.
- d. Vanillylmandelic acid
- e. Melatonin.

18. L-Dopa (Dihydroxyphenylalanine) is.

- a. produced by decarboxylation of Noradrenaline Adrenaline Dopamine.
- b. converted to tyrosine by a hydroxylase Enzyme.
- c. both (a) and (b).
- d. Neither (a) nor (b).

19. Decarboxylation of 5-hydroxytryptophan results in formation of

- a. Tryptamine.
- b. Serotonin.
- c. melatonin.
- d. Tyramine.

20. Serotonin is.

- a. 5-hydroxytryptamine.
- b. A tyrosine derivative.
- c. produced by N-acetylation and methylation of melatonin.
- d. and c.

21. Serotonin is catabolized by.

- a. Monoamine oxidase (MAO).
- b. catechol O-methyltransferase (COMT).
- c. both (a) and (b).
- d. Neither (a) nor (b).

22. Melatonin is synthesized from which of the following?

- a. Calcitonin.
- b. Melanin.
- c. Tyrosine,
- d. Tryptophan.
- e. Melanocyte stimulating hormone.

23. Catecholamines

- a. include dopamine, (norepinephrine) (epinephrine).
- b. are Both hormones And Neurotransmitters.
- c. both (a) and (b).
- d. Neither (a) nor (b).

24. Catecholamines

- a. Contain the catechol (3,4-dihydroxyphenyl) ring in their structure.
- b. Are synthesized from the amino acid Glycine.
- c. both (a) and (b).
- d. neither (a) nor (b).

25. Which of the following is measured in the urine to confirm the presence of an Adrenaline secreting tumor?

- a. Noradrenaline.
- b. 5-hydroxy-indole-acetate.
- c. Vanillomandelic acid.
- d. Monoamine oxidase.
- e. Epinephrine.

26. Norepinephrine is produced by.

- a. Hydroxylation of dopamine.
- b. methylation of epinephrine.
- c. decarboxylation of dopamine.
- d. oxidation of epinephrine.

27. Production Of Norepinephrine from Dopamine requires.

- a. copper-containing oxidase
- b. vitamin C.
- c. tetrahydrobiopterin.
- d. a + b.

28. Hormones and neurotransmitters

- a. Are signal molecules used for cell-cell communication.
- b. Are mostly polypeptides or nitrogenous Compounds, both derived from amino-acids.
- c. include non-nitrogenous compounds too.
- d. All the above

29. Vitamin B6 is important for which of the following regarding brain metabolism?

- a. Production of Important neurotransmitters.
- b. Detoxification of ammonia by formation of glutamine.
- c. Energy production from acetyl CoA.
- d. Energy storage as creatine Phosphate.
- e. Catabolism of catecholamines.

30. The end product of adrenaline and Noradrenaline catabolism is.

- a. Vanillylmandelic acid (VMA).
- b. homovanillic acid.
- c. phenyl acetic acid.
- d. Phenyl pyruvic acid.

31. To increase the synthesis of catechol-Amines in the brain, a patient with Parkinsonism may be given.

- a. L-dopa.
- b. α-methyldopa.
- c. MAO inhibitors
- d. Tyramine.

32. Which of the following coenzymes is required for the synthesis of GABA, serotonin, and histamine from their respective amino acid precursors?

- a. Tetrahydrofolate.
- b. Pyridoxal phosphate.
- c. Thiamine pyrophosphate.
- d. Methyl cobalamin.
- e. Biotin.

33. Synthesis of γ-aminobutyric acid (GABA) occurs by a

- a. carboxylation reaction requiring vitamin K.
- b. transamination reaction requiring vitamin B6.
- c. hydroxylation reaction requiring vitamin C.
- d. Decarboxylation reaction requiring vitamin B6.
- e. Oxidation reaction requiring niacin.

34. Production of Epinephrine from Nor-epinephrine requires.

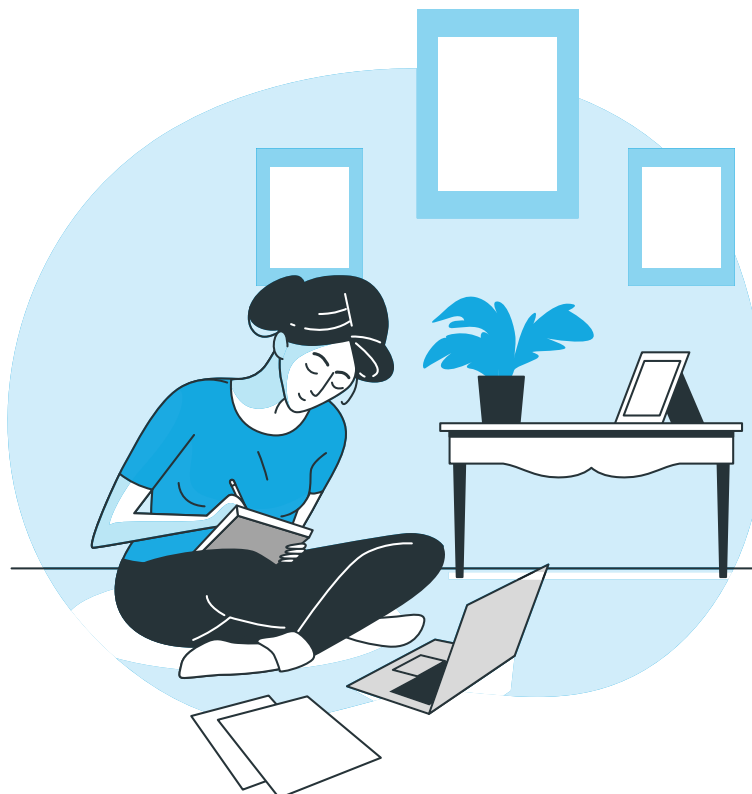
- a. N-methyltransferase.
- b. S-adenosylmethionine.
- c. both (a) and (b).
- d. Neither (a) nor (b)

35. Nitric oxide synthase

- a. acts on arginine and needs oxygen and NADPH.
- b. produces citrulline and nitric oxide.
- c. has three isozymes: neuronal, endothelial and phagocytic.
- d. all the above.

36. Which of the following is a substrate for nitric oxide synthase?

- a. Guanosine triphosphate.
- b. Sodium nitrate.
- c. Nitric oxide.
- d. Arginine.
- e. Glycine.





1. A ketogenic amino acid is:

- a. Valine
- b. Cysteine
- c. Leucine
- d. Threonine

2. In metabolic point of view, amino acids are classified as

- a. Glycogenic
- b. Ketogenic
- c. Glycogenic or Ketogenic
- d. All of these

3. The main sites for oxidative deamination are:

- a. Liver and kidney
- b. Skin and pancreas
- c. Intestine and mammary gland
- d. Lung and spleen

4. Alanine can be synthesized from:

- a. Glutamate and α -ketoglutarate
- b. Pyruvate and glutamate
- c. Pyruvate and α -ketoglutarate
- d. Aspartate and α -ketoglutarate

5. All of the following are required for synthesis of alanine except:

- a. Pyruvate
- b. α -ketoglutarate
- c. Glutamate
- d. Pyridoxal phosphate

6. All of the following are required for synthesis of glutamine except:

- a. Glutamate
- b. Ammonia
- c. Pyridoxal phosphate
- d. ATP

7. The amino acid that undergoes oxidative deamination at Significant rate is:

- a. Alanine
- b. Aspartate
- c. Glutamate
- d. Glutamine

8. Allosteric inhibitor of glutamate dehydrogenase is.

- a. ATP
- b. ADP
- c. AMP
- d. GMP

9. Allosteric activator of glutamate dehydrogenase is.

- a. ATP
- b. GTP
- c. ADP
- d. AMP and GMP

10. Free ammonia is released during:

- a. Oxidative deamination of glutamate
- b. Catabolism of purines
- c. Catabolism of pyrimidines
- d. All of these

11. Pyruvic acid can be obtained by transamination of alanine with

- a. α -keto glutaric acid
- b. Acetoacetic acid
- c. β -OH butyric acid
- d. Phosphonyl Pyruvic acid

12. All followings are ketogenic Amino acids except:

- a. Leucine
- b. Isoleucine
- c. Phenylalanine
- d. Glycine

13. The transaminase Activity needs the coenzyme:

- a. ATP
- b. B6-PO₄.
- c. FAD.
- d. NAD

14. Transamination is a

- a. Irreversible process
- b. Reversible process
- c. Both (A) and (B)
- d. None of these

15. Oxidative conversion of many amino acids to their corresponding ketoacids occurs in mammalian:

- a. Liver and kidney
- b. Adipose tissue
- c. Pancreas
- d. Intestine

16. An elevated plasma alanine transaminase activity signifies which of the following?

- a. Higher turnover of muscle protein.
- b. Increased gluconeogenesis in fasting states.
- c. High protein in a recent meal.
- d. Lowered functional capacity of liver cells.
- e. Active liver disease.

17. Which of the following enzymes produces pyruvic acid?

- a. Alanine transaminase.
- b. Aspartate transaminase.
- c. Asparaginase.
- d. Pyruvate carboxylase.
- e. L-amino acid decarboxylase.

18. Which of the following is the precursor for a cofactor needed by transaminases?

- a. Vitamin C.
- b. Vitamin B1.
- c. Vitamin B6.
- d. Vitamin B12.

19. Transamination reactions are characterized by which of the following?

- a. They can be used for the synthesis of amino acids.
- b. They are reversible.
- c. Pyridoxal phosphate is a form of vitamin B6 used in transamination.
- d. All of the above.

20. Alanine is most probably produced by which of the following?

- a. α -ketoglutarate by transamination.
- b. Digestion of dietary phospholipids.
- c. Action of L-amino acid oxidase.
- d. Aspartate transaminase reaction.
- e. Proteolysis in the muscle cells.

21. What are the sources of blood ammonia?

- a. Deamination of amino acids and other nitrogenous compounds is the main source.
- b. Action of bacterial urease in the intestine on urea diffused from the Blood.
- c. Bacterial putrefaction in the large Intestine.
- d. All the above.



1. Urea cycle is a pathway that.

- a. Takes place mainly in the liver cells.
- b. produces ammonia from urea.
- c. both (a) and (b).
- d. neither (a) nor (b).

2. In the urea cycle, urea is produced from.

- a. arginine by arginase enzyme.
- b. fumarate by fumarase.
- c. arginino-succinate by arginino-succinase.
- d. carbon dioxide and ammonium ions by carbamoyl phosphate synthetase.

3. Although arginine is formed in the urea cycle, it is considered an essential amino-acid. Why?

- a. It must be broken down by arginase enzyme.
- b. Its rate of synthesis is not enough for growing children.
- c. It is formed in the cytosol; therefore, it cannot be used.
- d. It is formed in the D-configuration.

4. The main site of urea synthesis in mammals is:

- a. Liver
- b. Skin
- c. Intestine
- d. Kidney

5. The enzymes of urea synthesis are found in:

- a. Mitochondria only
- b. Cytosol only
- c. Both mitochondria and cytosol
- d. Nucleus

6. The number of ATP required for urea synthesis is:

- a. 0
- b. 1
- c. 2
- d. 3

7. Two reactions of the urea cycle take place in the mitochondria. These are.

- a. Carbamoyl phosphate synthetase and ornithine transcarbamoylase.
- b. Arginino-succinate synthetase and argininosuccinase.
- c. argininosuccinase and fumarase.
- d. fumarase and arginase.

8. Which of the following is a high energy compound formed in the urea cycle?

- a. Ornithine
- b. Carbamoyl phosphate.
- c. Citrulline.
- d. Arginino-succinate.

9. Which of the following amino acids is formed in the urea cycle and used in protein synthesis?

- a. Ornithine.
- b. Citrulline,
- c. Arginino-succinate.
- d. Arginine.

10. In urea synthesis, the amino acid functioning solely as an enzyme activator:

- a. N-acetyl glutamate
- b. Ornithine
- c. Citrulline
- d. Arginine

11. Control of urea cycle involves the enzyme:

- a. Carbamoyl phosphate synthetase
- b. Ornithine transcarbamoylase
- c. Argininosuccinase
- d. Arginase

12. In the urea cycle, which amino acids should be transported between the mitochondria and the cytosol in either direction?

- a. Ornithine and citrulline.
- b. Citrulline and arginine.
- c. arginine and ornithine.
- d. Citrulline and arginino-succinate.

13. The 2 nitrogen atoms in urea are contributed by:

- a. Ammonia and glutamate
- b. Glutamine and glutamate
- c. Ammonia and aspartate
- d. Ammonia and alanine

14. Which of the following is produced by urea cycle with the production of urea?

- a. Ammonia.
- b. Fumarate.
- c. Arginine.
- d. Aspartate.

15. The urea cycle is active in cases of

- a. carbohydrate intake.
- b. Protein intake.
- c. fasting.
- d. B and c

16. How many carbon and nitrogen atoms are there in a urea molecule?

- a. One carbon and one nitrogen.
- b. One carbon and two nitrogens.
- c. Two carbon and tow nitrogen.
- d. Two carbon and one nitrogen.

17. Which of the following is the source of urea carbon?

- a. Methyl group of methionine.
- b. Methylenetetrahydrofolate.
- c. Carbon dioxide.
- d. Carbonyl carbon of pyruvate

18. Which of the following gives one nitrogen atom of the urea molecule?

- a. Aspartate.
- b. Glutamate.
- c. Fumarate.
- d. Ornithine.

19. Which of the following is a source of urea nitrogen?

- a. Ammonium ion formed from ammonia.
- b. Aspartate.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

20. Glutamate can be the source of both nitrogen atoms of urea through which reactions?

- a. Glutaminase.
- b. Glutamate dehydrogenase
- c. Glutamate oxaloacetate transaminase.
- d. B and c

21. Ammonia is introduced in the urea cycle by.

- a. reacting as ammonium ion with carbon Dioxide and ATP.
- b. Forming ornithine.
- c. making a part of N-acetyl glutamate.
- d. The arginase reaction.

22. How is carbamoyl phosphate synthetase I different from synthetase II? Carbamoyl phosphate

- a. I is mitochondrial while II is cytosolic.
- b. I is used for pyrimidine synthesis but II is for urea synthesis.
- c. I does not need N-acetyl glutamate as allosteric activator, unlike II.
- d. I uses glutamine not ammonia as a Nitrogen donor.

23. The major site of urea synthesis is:

- a. Brain
- b. Kidneys
- c. Liver
- d. Muscles

24. Carbamoyl phosphate required for urea synthesis is formed in.

- a. Cytosol
- b. Mitochondria
- c. Both (A) and (B)
- d. None of these

25. The following enzyme of urea cycle is present in cytosol:

- a. Argininosuccinic acid synthetase
- b. Argininosuccinase
- c. Arginase
- d. All of these

26. ATP is required in following reactions of urea cycle:

- a. Synthesis of carbamoyl phosphate and citrulline B
- b. Synthesis of citrulline and arginosuccinate
- c. Synthesis of arginosuccinate and arginine
- d. Synthesis of carbamoyl phosphate and arginosuccinate

27. Which of the following is an energy-producing pathway?

- a. Citrate cycle.
- b. Urea cycle
- c. Both (a) and (b).
- d. Neither (a) nor (b).

28. ATP is consumed in which of the following reactions?

- a. Carbamoyl phosphate synthetase.
- b. Ornithine transcarbamoylase
- c. Arginino-succinate synthetase.
- d. A and c

29. The major end product of protein nitrogen metabolism in man is:

- a. Glycine
- b. Uric acid
- c. Urea
- d. NH_3

30. An amino acid not involved in urea cycle is:

- a. Arginine
- b. Histidine
- c. Ornithine
- d. Citrulline

31. NH_3 is detoxified in brain chiefly as:

- a. Urea
- b. Uric acid
- c. Creatinine
- d. Glutamine

32. In humans, NH_3 is detoxified in liver as

- a. Creatinine
- b. Uric acid
- c. Urea
- d. Uronic acid

33. What is the use of glutamate for the urea. Cycle?

- a. Production of ammonia by glutamate dehydrogenase reaction.
- b. Regeneration of aspartate by the glutamate: oxaloacetate transaminase Reaction.
- c. Formation the cycle activator: N-acetyl-Glutamate.
- d. All the above.

34. How is citric acid cycle useful for urea cycle?

- a. Providing energy.
- b. Regeneration of aspartate from Fumarate, through malate and oxalo- acetate.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

35. Which reaction of the urea cycle is the primary control reaction of the pathway?

- a. Carbamoyl phosphate synthetase.
- b. Ornithine transcarbamoylase.
- c. Arginino-succinate synthetase.
- d. Arginase.

36. The integrator between the TCA cycle and urea cycle is.

- a. Fumarate
- b. Malate
- c. Pyruvate
- d. Citrate

37. The rate of the urea cycle is controlled by.

- a. substrate availability.
- b. Induction of the cycle enzyme.
- c. arginine, which activates the synthesis of the allosteric activator.
- d. All the above.

38. In the synthesis of 1 molecule of urea, the number of AMPs formed is

- a. 1
- b. 2
- c. 3
- d. 4

39. The following 4 amino acids are required for completion of urea Cycle except

- a. Aspartic acid
- b. Arginine
- c. Ornithine
- d. Glycine

40. A normal blood urea concentration increases with

- a. Protein intake.
- b. obesity.
- c. pregnancy.
- d. Childhood.

41. Abnormal increase in blood urea concentration is seen with.

- a. liver cell failure.
- b. renal failure.
- c. high-protein diet.
- d. all the above.

42. From dietary protein as well as from the urea present in fluids secreted into the gastrointestinal tract intestinal bacteria produce

- a. Carbon dioxide
- b. Ammonia
- c. Ammonium sulphate
- d. Creatine

43. Which of the following may be considered a physiological ammonia-fixing reaction?

- a. glutamate dehydrogenase.
- b. glutamine synthetase.
- c. carbamoyl phosphate synthetase I.
- d. b and c.

44. Synthesis of glutamine is accompanied by the hydrolysis of

- a. ATP
- b. ADP
- c. TTP
- d. Creatin phosphate

45. Carbamoyl phosphate synthetase structure is markedly change in the presence of

- a. N-Acetyl glutamate
- b. N-Acetyl Aspartate
- c. Neuraminic acid
- d. Oxalate

46. A genetic defect of an enzyme of urea Cycle leads to

- a. low blood ammonia.
- b. High blood urea.
- c. high blood glutamine.
- d. all the above.

47. A genetic defect of N-acetyl glutamate production leads to a high level of

- a. Urea.
- b. Arginine.
- c. glutamine.
- d. citrulline.

48. Arginine may stimulate the urea cycle by

- a. activation of glutamate N-acetylation.
- b. providing ornithine for ornithine transcarbamoylase reaction.
- c. both (a) and (b).
- d. neither (a) nor (b).

49. If there were complete absence of carbamoyl phosphate synthetase, what would be the source of urea in the blood?

- a. Non-enzymatic synthesis.
- b. Arginase action on arginine.
- c. bacterial action on ammonia.
- d. production by the renal tubules.

50. N-acetyl glutamate is an essential allosteric activator of:

- a. glutamine synthetase
- b. glutaminase.
- c. ornithine transcarbamoylase.
- d. Arginase
- e. Carbamoyl phosphate synthetase

51. Which of the following increases with deficiency of any enzyme of urea cycle?

- a. Glutamine.
- b. Citrulline.
- c. Both (a) and (b)
- d. Neither (a) nor (b).

52. Which of the following may be true about the glutamine synthetase reaction in the brain?

- a. It normally helps detoxify ammonia.
- b. At high ammonia concentration, it can deplete glutamate, with consequent decrease of GABA.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

53. Ammonia is normally used for synthesis of which of the following?

- a. Glutamine.
- b. Alanine.
- c. Glutamate
- d. Aspartate.
- e. Purines and pyrimidines

1. Ammonia toxicity to the brain is Probably through which of the Following mechanisms?

- a. Inhibition of energy production in brain Cells.
- b. Production of increased amounts of the inhibitory neurotransmitter GABA.
- c. Acidifying effect on the intra- and Extracellular fluids.
- d. Conversion to urea, increasing the Osmotic pressure.
- e. Inability of the brain cells to utilize any of the amino acids.

2. Hyperammonemia may be caused by

- a. genetic defect of any enzyme of urea cycle (primary hyperammonemia).
- b. liver cell failure (secondary hyperammonemia).
- c. either (a) or (b).
- d. Neither (a) nor (b).

3. The symptom of ammonia intoxication includes.

- a. Blurring of vision
- b. Constipation
- c. Mental confusion
- d. Diarrhea

4. Ammonia intoxication symptoms occur when brain ammonia Levels are.

- a. Slightly diminished.
- b. Highly diminished
- c. Increased
- d. All of these

5. Primary hyperammonemia may be treated by

- a. Low-protein diet.
- b. agents that increase excretion of Conjugated amino acids, e.g., benzoic acid and phenylacetate.
- c. acidification of the intestinal lumen to help get rid of ammonia as ammonium ions.
- d. All the above.

6. Deficiency of urea cycle enzymes results into accumulation of citrulline arginosuccinate arginine in the liver resulting in Increasing concentration of In the blood.

- a. Calcium
- b. Sodium
- c. Ammonia
- d. Lipid

7. The toxic effect of ammonia on the brain is probably due to

- a. Reversal reaction of glutamate dehydrogenase with depletion of ketoglutarate and cessation of citrate cycle and energy production.
- b. depletion of glutamate by glutamine Synthetase reaction, depriving the Brain from the two neurotransmitters: Glutamate and gamma-aminobutyric Acid (GABA).
- c. increased permeability of brain cells to potassium and chloride ions, leading to disturbance of brain electricity.
- d. all the above.

8. Phenyl acetate is used in treatment of hyperammonemia because:

- a. it directly binds ammonia facilitating its excretion in urine.
- b. it converts ammonia to a nontoxic compound.
- c. it binds glutamine forming a urinary excretable complex.
- d. It breaks down ammonia to nitrogen and hydrogen.
- e. it inhibits production of ammonia from amino acid catabolism.

9. Which of the following is a probable mechanism for ammonia toxicity?

- a. Depletion of α -ketoglutarate in brain cells.
- b. Increased urea concentration in the plasma and liver cells
- c. Lowering of the pH of plasma & cerebrospinal fluid.
- d. Inhibition of the glutaminase reaction in the kidney.

10. Hyperammonemia is a feature of

- a. Renal failure.
- b. Liver cell failure
- c. Brain infarction.
- d. High protein diet
- e. Low protein diet.

11. Regarding liver function tests, which of the following may be a marker of liver cell failure?

- a. Hyper-ammonemia.
- b. Hypo-albuminemia.
- c. Both (a) and (b).
- d. Neither (a) nor (b)

12. Hyperammonemia following bleeding from esophageal varices may be attributed to which of the following?

- a. Inhibition of cytosolic carbamoyl phosphate synthetase.
- b. An X-linked genetic disease.
- c. Digestion of blood proteins
- d. acidification of the intestinal content.

13. What could be the cause of hepatic coma following bleeding from esophageal varices?

- a. Escape of blood to the stomach, constituting a protein load to the gastrointestinal tract.
- b. Inability of the diseased liver to detoxify the ammonia resulting from this protein load.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

14. What measures are usually taken for a patient with liver cell failure to protect against ammonia toxicity?

- a. A low-protein diet.
- b. Antibiotics to kill intestinal bacteria.
- c. Acidification of intestinal lumen to help get rid of ammonia as ammonium ions.
- d. All the above.

15. What measures may be taken following bleeding from esophageal varices to prevent hepatic coma?

- a. Gastric lavage and enema.
- b. Acidification of the intestine.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

16. An organ which is extremely sensitive to ammonia toxicity is:

- a. Liver
- b. Brain
- c. Kidney
- d. Heart



1. Maple syrup urine diseases is an inborn error of metabolism of:

- a. Sulphur-containing amino acids
- b. Aromatic amino acids
- c. Branched chain amino acids.
- d. Dicarboxylic amino acids

2. Hydroxylation of phenylalanine requires all of the following except:

- a. Phenylalanine hydroxylase
- b. Tetrahydrobiopterin
- c. NADH
- d. Molecular oxygen

3. All the following statements about phenylketonuria are correct Except:

- a. Phenylalanine cannot be converted into tyrosine.
- b. Urinary excretion of phenylpyruvate and phenyl lactate is increased.
- c. It can be controlled by giving a low phenylalanine diet.
- d. It leads to increased synthesis of thyroid hormones, catecholamines and melanin.

4. The amino acid which synthesizes many hormones:

- a. Valine
- b. Phenylalanine
- c. Alanine
- d. Histidine

5. A genetic defect of an enzyme of amino-acid metabolism can lead to

- a. Minor or no observed abnormality.
- b. serious disease.
- c. fatal disturbance of metabolism.
- d. any of the above.

6. Organic acidemia is observed in which of the following conditions?

- a. Maple syrup urine disease.
- b. Glycinuria.
- c. Lysine-cystinuria.
- d. Melanocyte tyrosinase deficiency.
- e. Vitamin B12 excess

7. Which of the following may result from deficient tetrahydrobiopterin metabolism?

- a. Pellagra
- b. Tyrosinemi
- c. Phenylalaninemia and phenylketonuria.
- d. Deficient formation of nicotinic acid.

8. Black urine may result from disturbed metabolism of which of the following?

- a. Tryptophan.
- b. Histidine.
- c. Homocysteine.
- d. Phenylalanine.
- e. Melanin.

9. Phenylketonuria is caused by deficiency of

- a. Conversion of phenylalanine to a phenylketoacid.
- b. Production of phenylalanine from a phenylketoacid.
- c. Reaction of phenylalanine with an α -ketoacid.
- d. Conversion of phenylalanine to another aminoacid.

10. Milk may be harmful to patients having which of the following?

- a. Phenylalanine hydroxylase deficiency
- b. Glucose 6-phosphate dehydrogenase deficiency.
- c. Gluten sensitivity.
- d. Alkaptonuria. E
- e. Albinism.

11. Lactic acidosis that may be seen in a case of maple syrup urine disease is probably due to which of the following?

- a. Lactate dehydrogenase deficiency.
- b. Defective enzyme subunit common to pyruvate dehydrogenase and branched chain α -Ketoacid dehydrogenase.
- c. Conversion of branched-chain α -ketoacids to lactic acid.
- d. Failure of glucose production from leucine.
- e. Inability to metabolize pyruvate by alanine aminotransferase.

12. Phenylketonuria is due to defect in:

- a. Phenyl alanine hydroxylase enzyme.
- b. Tyrosine hydroxylase enzyme.
- c. Tryptophan hydroxylase enzyme.
- d. All of the above.

13. Conversion of Phenylalanine to tyrosine is catalyzed by phenylalanine hydroxylase which requires.

- a. NAD
- b. FAD
- c. Tetrahydrofolate
- d. Tetrahydrobiopterin

14. The phenylalanine hydroxylase reaction results in formation of

- a. Phenylalanine.
- b. Tyrosine.
- c. Dopa.
- d. Tetrahydrobiopterin.

15. Which of the following is not a characteristic of untreated phenylketonuria?

- a. Hyper-phenylalaninemia.
- b. Black urine.
- c. Light-colored skin and hair
- d. Mental deficiency.

16. Which of the following passes in the urine of a patient with phenylketonuria?

- a. Phenyl-pyruvic acid.
- b. Phenyl-lactic acid.
- c. phenyl-acetic acid & phenyl-acetyl- glutamine.
- d. All the above.

17. Maple syrup urine disease

- a. varies in severity according to the type of genetic abnormality.
- b. often leads to mental retardation.
- c. may be fatal at an early age.
- d. All the above.

18. Management of phenylketonuria necessitates.

- a. Early detection by routine neonatal screening.
- b. Low phenylalanine feeding formula (milk substitute).
- c. starting treatment before the age of 1 month.
- d. All the above.

19. The clinical picture of phenylketonuria may result from

- a. Deficiency of tyrosine.
- b. Interference of phenylalanine with Normal tyrosine metabolism.
- c. both (a) and (b).
- d. Neither (a) nor (b).

20. Which of the following would you give to patients with deficient tetrahydrobiopterin Metabolism?

- a. High doses of folic acid.
- b. Vitamin Bs.
- c. Phenylalanine hydroxylase enzyme.
- d. Tyrosine.

21. The clinical picture of classic phenylketonuria results from deficiency of

- a. Catecholamines.
- b. melanin.
- c. melatonin.
- d. A and b.

22. What is the non-essential aminoacid that becomes essential in treated phenylketonuria patients?

- a. Phenylalanine.
- b. Alanine.
- c. tyrosine.
- d. Glutamine.

1. **A fully oxidized carbon atom makes a**
 - a. Carbon dioxide molecule.
 - b. Methane molecule.
 - c. methyl group.
 - d. Formyl group.
2. **A one-carbon group (moiety) transferred in chemical reactions may be.**
 - a. Methyl ($-\text{CH}_3$).
 - b. Methylene ($-\text{CH}_2-$).
 - c. Methenyl ($=\text{CH}-$), formyl ($\text{O}-\text{CH}-$), or One carbon unit & formimino vitamins s
 - d. Any of the Ascending oxidation state.
3. **One-carbon group is carried on.**
 - a. N5 and N10 of folic acid.
 - b. tetrahydrofolate.
 - c. Tetrahydrobiopterin
 - d. either (b) or (c).
4. **What is the major contributor to the one-carbon pool?**
 - a. Serine.
 - b. Proline.
 - c. Threonine.
 - d. Methionine
5. **Which of the following aminoacids can give a methylene group to tetrahydro- folate?**
 - a. Serine.
 - b. Glycine.
 - c. Methionine.
 - d. A and b
6. **What is the source of the methyl group on tetrahydrofolate?**
 - a. Labile methyl of S-adenosylmethionine.
 - b. Reduction of methylene group on tetrahydrofolate.
 - c. Both (a) and (b).
 - d. Neither (a) nor (b).
7. **What is the importance of one-carbon groups on tetrahydrofolate?**
 - a. Nucleotides and nucleic acid Thesis.
 - b. Maintenance of active methionine cycle.
 - c. Both (a) and (b).
 - d. Neither (a) nor (b).
8. **The 5-methyl group of thymidylates is supplied by.**
 - a. S-adenosylmethionine.
 - b. N-methyltetrahydrofolate.
 - c. either (a) or (b).
 - d. Neither (a) nor (b).
9. **What is meant by methyl trap of folic acid?**
 - a. Folic acid traps vitamin B12 as methyl-cobalamin.
 - b. Vitamin B12 traps folic acid as methyl-tetrahydrofolate.
 - c. Folic acid is trapped as methyltetrahydrofolate in absence of vitamin B12.
 - d. Methyl group is trapped on tetrahydro-Folate with the help of vitamin B12.
10. **Which of the following is not a recognized reaction of the one-carbon pool?**
 - a. Formation of serine from glycine.
 - b. Formation of formimino-glutamate.
 - c. Formation of thymidylate.
 - d. Formation of methionine.
11. **If serine, labeled allover with carbon-14, was given to an individual, the label will appear in**
 - a. S-adenosylmethionine.
 - b. 5-methyltetrahydrofolate.
 - c. Thymidylate.
 - d. All the above.

12. Methionine is synthesized in human body from:

- a. Cysteine and homoserine
- b. Homocysteine and serine
- c. Cysteine and serine
- d. None of these

13. A coenzyme required for the synthesis of glycine from serine is:

- a. ATP
- b. Pyridoxal phosphate
- c. Tetrahydrofolate
- d. NAD

14. Sources of the formimino group on tetrahydrofolate include.

- a. formimino-glutamate (FIGlu) formed from histidine.
- b. oxidation of formyl group on Tetrahydrofolate.
- c. both (a) and (b).
- d. neither (a) nor (b).

15. Sources of the formyl group on tetrahydrofolate include.

- a. Formate from tryptophan and other sources.
- b. Methenyl group on tetrahydrofolate Without oxidation or reduction.
- c. both (a) and (b).
- d. neither (a) nor (b).

16. Serine hydroxymethyltransferase reaction may generate.

- a. N5, N10-methylenetetrahydrofolate.
- b. Glycine.
- c. serine.
- d. All the above.

17. In activated methionine cycle, methionine is regenerated by reaction of

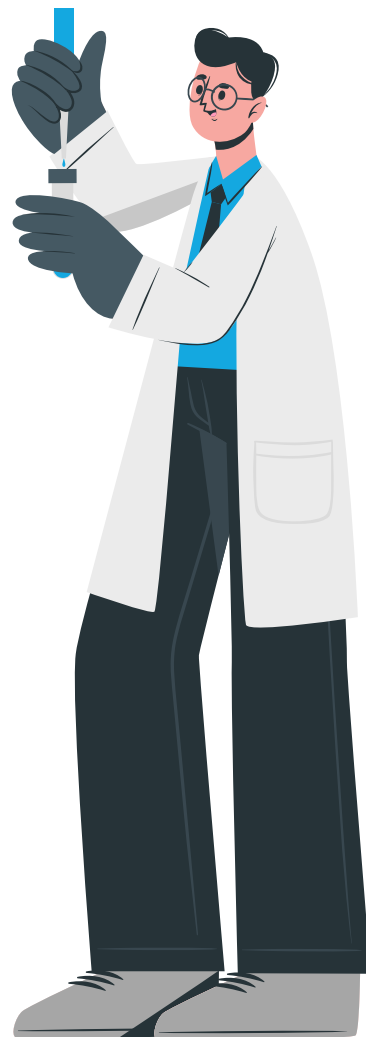
- a. Cysteine with serine.
- b. Homocysteine with serine.
- c. homocysteine with methyl tetrahydro- folate.
- d. Tetrahydrofolate methionine with S-adenosyl-

18. Methionine synthase needs which co-enzyme?

- a. Methylcobalamin.
- b. Pyridoxal phosphate.
- c. NAD*.
- d. NADPH.

19. Which of the following has a high group transfer potential?

- a. S-Adenosylmethionine.
- b. S-Adenosylhomocysteine.
- c. Methionine.
- d. Homocysteine.



1. Hyper-homocysteinemia may result from deficiency of

- a. Pyridoxine (vitamin B6).
- b. Cobalamin (vitamin B12).
- c. folic acid.
- d. Any of the above.

2. Folic acid may be measured to investigate a case of:

- a. Anemia
- b. Jaundice.
- c. Renal failure.
- d. Obesity
- e. Atherosclerosis.

3. Which of the following may be required for clearance of homocysteine?

- a. Folic acid, cobalamin, and pyridoxine.
- b. Coenzyme A and biotin.
- c. Proteins of high biological value.
- d. Antioxidant vitamins
- e. High fiber diet.

4. Why is it important to give the pregnant folic acid supplement?

- a. Folic acid deficiency occurs with increased demand as in pregnancy.
- b. Folate deficiency causes megaloblastic anemia due to defective synthesis of DNA.
- c. Folate deficiency causes fetal neural tube defects, e.g., spina bifida.
- d. All the above.

5. Beriberi is caused by a deficiency of Which of the following vitamins?

- a. Folic acid.
- b. Cobalamine.
- c. Thiamine.
- d. Vitamin E.
- e. Vitamin C.

6. Which of the following may be measured in the urine as an indicator of folic acid deficiency?

- a. Glycine.
- b. Serine.
- c. Histidine.
- d. Formimino-glutamate (FIGlu).

7. Folate antagonists like methotrexate and trimethoprim inhibit DNA synthesis by inhibiting.

- a. Folic acid absorption.
- b. Dihydrofolate reductase.
- c. folic acid synthesis.
- d. methylcobalamin synthesis.

8. Which of the following is true about folic acid and vitamin B12?

- a. They are both fat-soluble vitamins.
- b. They are only important for production of healthy red blood cells.
- c. Their deficiency leads to hematological and neurological pathology.
- d. None of the above.

9. Cobalamin (vitamin B12)

- a. has a complex structure.
- b. contains a tetrapyrrole corrin ring (similar to porphyrin ring)
- c. has a red color due to presence of cobalt ion at the center of the corrin ring.
- d. All the above.

10. Which of the following statements is correct?

- a. Folic acid is not a vitamin in bacterial Cells.
- b. Sulfonamides are analogues of PABA, which can inhibit folate synthesis in Mammalian cells.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

11. Which of the following statements is correct?

- a. Sulfonamides and trimethoprim are used as antibacterial agents.
- b. Methotrexate is used for treatment of Cancer and as immunosuppressant.
- c. Both (a) and (b).
- d. Neither (a) nor (b).

12. Which of the following statements is not Correct?

- a. Folic acid is important for synthesis of DNA.
- b. Folate deficiency results in megaloblastic anemia.
- c. Methotrexate can be used for treatment of folate deficiency anemia.
- d. Folic acid can alleviate vitamin B deficiency anemia, but not the neurological symptoms.

13. Vitamin B12 is stored in the liver with an additional group attached to the cobalt ion forming.

- a. Hydroxycobalamin.
- b. Methylcobalamin.
- c. 5'-deoxyadenosylcobalamin.
- d. All the above.

14. Active B12 coenzymes are needed for.

- a. Methylmalonyl CoA mutase.
- b. Methionine synthase.
- c. both (a) and (b).
- d. Neither (a) nor (b).

15. Which of the following is true about folic acid activation?

- a. The active form of folic acid is dihydro-folate.
- b. Activation of folic acid is catalyzed by dihydrofolate reductase.
- c. Reduction of folic acid requires FADH_2 .
- d. All the above.

16. Folic acid deficiency may be seen.

- a. In pregnant women, who need double the normal daily requirements of this vitamin.
- b. With vitamin B12 deficiency, which leads to the methyl trap of folic acid.
- c. both (a) and (b).
- d. Neither (a) nor (b).

17. Folic acid deficiency leads to

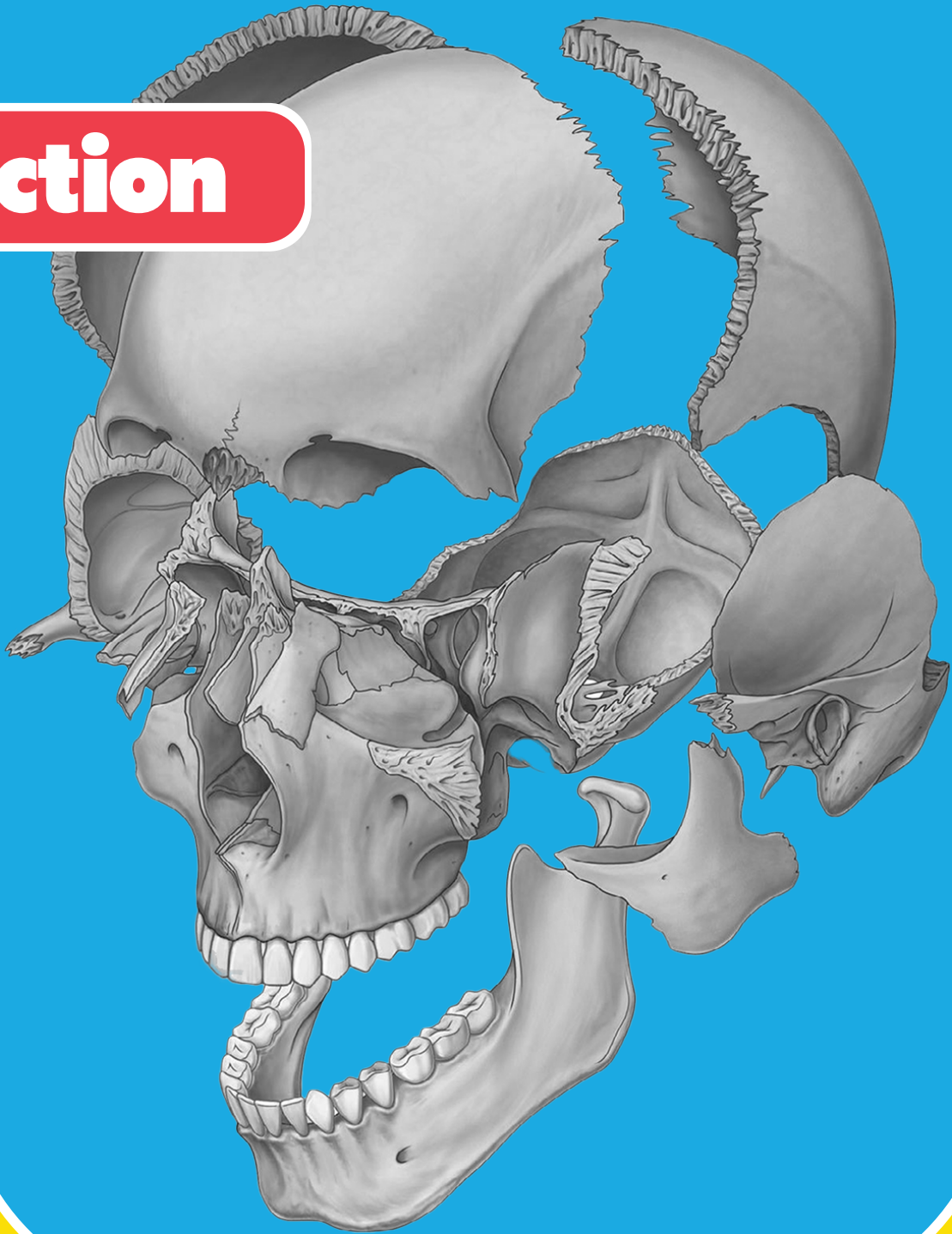
- a. megaloblastic anemia due to defective DNA synthesis.
- b. Decrease of the methionine-dependent choline synthesis needed for myelin phospholipids.
- c. fetal anomalies, e.g., neural tube defect (spina bifida) due to folate deficiency during pregnancy.
- d. all the above.



Short Essay

1

Section



Anatomy

Chapter 1 ?

Spinal cord

1. Mention the 2 enlargements present in the spinal cord

- Cervical enlargement & lumbar enlargement

2. Compare between the spinal cord of the adult and the newborn

- **Adult:** C1 till L1
- **Newborn:** C1 till L3

3. Mention the 3 ligaments supporting the spinal cord.

- Filum terminale, ligamentum denticulatum, subarachnoid spetum

4. Complete:

Dura mater and arachnoid mater end opposite which vertebrae **S2**

Chapter 2 ?

Scalp

1. What is the extent of the scalp?

- **Anteriorly:** supra orbital margin
- **Posteriorly:** external occipital protuberance and highest nuchal line
- **Laterally:** superior temporal line

2. What are the effects of the dense connective tissue in the scalp?

- Infection remains localized, wounds don't gap, and a small wound causes profuse bleeding

3. Compare between frontal belly and occipital belly (according to the nerve supply)

- **Frontal belly:** Temporal branch of facial nerve
- **Occipital belly:** Posterior auricular branch of facial nerve

4. Explain: Loose connective tissue is considered as the dangerous layer of the scalp.

- Because its emissary veins can spread infection from the scalp to the cranial cavity.

5. Explain the sensory nerve supply of the scalp.

- **In front of the auricle:** Supratrochlear, supraorbital, zygomaticotemporal, auriculotemporal
- **Behind the auricle:** Great auricular, lesser occipital, greater occipital, third occipital

Chapter 3 ?

Face

1. What are the boundaries of the face?

- **Superiorly:** Hair line
- **Inferiorly:** lower border of mandible
- **Sideways:** from one auricle to the other

2. Facial artery terminates by anastomosing with which artery?

- Dorsal nasal artery

3. Mention 2 branches of the facial artery in the face.

- Superior labial and Inferior labial

4. What are the deep communications of the anterior facial vein?

- Facial vein is connected to superior ophthalmic vein which drains into cavernous sinus
- Facial vein is connected to deep facial vein which drains in pterygoid venous plexus; pterygoid venous plexus is connected to cavernous sinus via 3 emissary veins

5. What is the dangerous area of the face?

- It's a triangle that includes upper lip, nose, and medial angles of eye. If the area becomes infected, the infection would spread to the cavernous sinus causing cavernous sinus thrombosis.

6. Mention 4 branches of ophthalmic nerve.

- Supratrochlear, supraorbital, palpebral branch of lacrimal, infratrochlear

7. Mention 3 branches of maxillary nerve.

- Zygomaticotemporal, zygomaticofacial, infraorbital

8. Mention 3 branches of infraorbital nerve.

- Palpebral, nasal, labial

9. What are the branches of mandibular nerve?

- Auriculotemporal, buccal, and mental

10. Complete: Muscle of facial expression are developed from which pharyngeal arch: 2nd arch

Chapter 4 ?

Cerebral Hemisphere: Sensory Cortex

1. What does lesion of first somatosensory area cause?

- Contralateral hemianesthesia

2. What is the site of first somatosensory area?

- Postcentral gyrus and posterior part of paracentral lobule

3. What is the site of second somatosensory area?

- Superior lip of post ramus of lateral sulcus behind central sulcus

4. What is the site of area 43?

- Parietal operculum extending into anterior part of insula

5. What does lesion of Areas 5 & 7 cause?

- Astereognosis

6. What is the site of inferior parietal lobule?

- Supramarginal gyrus and angular gyrus

7. What does lesion of inferior parietal lobule cause?

- Receptive aphasia

8. What are the 3 speech centers?

- Broca's, Wernicke's & Supplementary motor area

Chapter 5 ? Cerebral Hemisphere: Motor Cortex

1. What is the site of primary motor area?

- Precentral gyrus and anterior part of paracentral lobule

2. What is the body representation of premotor area?

- Upside down, contralateral, and representation is proportionate to skill.

3. What does lesion of Area 4 cause?

- Contralateral hemiplegia especially fine skilled movements

4. What is the site of premotor area?

- Infront of area 4 (in sup, middle, and inf frontal gyri) and medial surface of cerebral hemisphere

5. What does lesion of area 6 cause?

- Apraxia, Muscles spasticity and reappearance of grasp reflex

6. What is the site of frontal eye field?

- Infront of area 6 (in sup & middle frontal gyri)

7. What does lesion of Area 8 cause?

- Ipsilateral deviation of both eyes and inability to turn eyes to opposite side

8. What is the site of Broca's area?

- Pars triangularis and pars opercularis of the inferior frontal gyrus

9. What does lesion of Areas 44 and 45 cause?

- Expressive aphasia

10. What is the site of supplementary motor area?

- Medial frontal gyrus

11. What does lesion of MII cause?

- Akinetic mutism

Chapter 6 ? Ascending tracts

1. What are the 3 neurons that form the pathway of pain and temperature from the body?

- **First neuron:** DRG
- **Second neuron:** Fast pain: lamina 1, 4, Slow pain: lamina 2(SGR)
- **Third neuron:** VPLN of thalamus

2. What are the 3 neurons that form the pathway of pain and temperature from the face?

- **First neuron:** Trigeminal Ganglion
- **Second neuron:** Spinal nucleus of trigeminal
- **Third neuron:** VPMN of thalamus

3. What are the 3 neurons that form the pathway of crude touch and pressure?

- **First neuron:** DRG
- **Second neuron:** Lamina 4/5 = Nucleus proprius
- **Third neuron:** VPLN of thalamus

Chapter 7 ?

Dorsal column tracts

1. What are the sensations carried by dorsal column tracts?

- Fine touch, proprioception, vibration, deep pressure, and stereognosis

2. What are the 3 neurons that for the pathway of conscious proprioception and fine touch?

- **First neuron:** DRG
- **Second neuron:** Gracile and Cuneate nuclei of medulla oblongata
- **Third neuron:** VPLN

3. What are the four tracts that carry unconscious proprioception to the cerebellum?

- Posterior spinocerebellar tract, anterior spinocerebellar tract, spino olivary tract, and cuneocerebellar tract

4. Which tract forms the external arcuate fibers?

- Cuneocerebellar tract

5. List the tracts that carry unconscious proprioception from the lower limb.

- Posterior spinocerebellar
- Anterior spinocerebellar
- Spino olivary

Chapter 8 ?

Descending tracts

1. Mention 2 extrapyramidal tracts.

- Tecto spinal tract and rubro spinal tract

2. What is the origin of olivo spinal tract?

- Inferior olivary nucleus of medulla

3. What is the lamination of of fibers in the lateral corticospinal tract in the spinal cord?

- Cervial fibers are medial while sacral fibers are lateral

4. What is the lamination of pyramidal tract in the internal capsule?

- Fibers for face muscles are in the genu; upper limb area lies most anterior and lower limb area lies most posterior in the posterior limb of internal capsule

5. Which tracts end on alpha and gamma neurons of extensor muscle?

- Medial and lateral vestibulo spinal tracts

6. Complete: Tecto spinal tract forms the dorsal tegmental decussation which the rubro spinal tract forms the ventral tegmental decussation

Chapter 9



Blood supply of spinal cord

1. What are the 3 arteries that supply the spinal cord?

- Anterior spinal artery, posterior spinal artery, and segmental arteries

2. Which artery supplies the medial part of medulla oblongata and the anterior 2/3 of the spinal cord?

- Anterior spinal artery

3. Which artery supplies the posterior 1/3 of the spinal cord?

- Posterior spinal artery

4. What is the origin of anterior spinal artery?

- It's a single artery formed by the union of two anterior spinal arteries, each is a branch of the vertebral artery.

5. What is the origin of posterior spinal artery?

- Vertebral artery or posterior inferior cerebellar artery (PICA)

6. What is the origin of radicular arteries?

- Vertebral, ascending cervical, posterior intercostal, and 1st lumbar artery

7. What are the 6 veins that drain the spinal cord?

- Anteromedian vein, posteromedian vein, and four veins that run on either side of the ventral dorsal roots

Chapter 10



Spinal cord lesions

1. What is the effect of spinal cord complete transverse section?

- **Above C5:** death
- **Between C5 T1:** Quadriplegia
- **Below T1:** Paraplegia

2. What is the effect of spinal cord hemisection?

- **At the same level of the lesion:** Ipsilateral loss of all types of sensation, Ipsilateral LMNL
- **Below the level of the lesion (Ipsilateral):** Loss of proprioception, UMNL
- **Below the level of the lesion (contralateral):** Loss of pain and temperature sensations

3. What is the effect of anterior spinal artery occlusion?

- Bilateral UMNL paralysis below the lesion
- Bilateral loss of pain and temperature sensations below the lesion with preservation of proprioception and touch

Chapter 11?

Basal Ganglia

1. What are the parts of neostriatum?

- Caudate and Putamen

2. What are the parts of paleostriatum?

- Globus pallidus internal and external

3. What is the blood supply of basal ganglia?

- Central branches of anterior and middle cerebral arteries

4. What is the venous drainage of basal ganglia?

- Veins that drain into thalamostriate vein

5. Complete: The head of caudate bulges into the anterior horn of lateral ventricle, body lies in the floor of body of lateral ventricle, tail lies in the roof of inferior horn of lateral ventricle

Chapter 12?

Cerebellum

1. What are the 3 lobes of the cerebellum (Functional classification)?

- Archicerebellum, paleocerebellum, and neocerebellum.

2. What is the archicerebellum formed of?

- 2 flocculi and nodule(part of inferior vermis)

3. What is the paleocerebellum formed of?

- Vermal part and 2 paravermal parts of the cerebellar hemisphere

4. What is the neocerebellum formed of?

- Lateral parts of cerebellar hemisphere

5. Mention the 4 cerebellar nuclei.

- Dentate, Emboliform, Globose, and Fastigial

6. What are the arteries that supply the cerebellum?

- Superior cerebellar artery, anterior inferior cerebellar artery, and posterior inferior cerebellar artery

7. What are the veins that drain the cerebellum?

- Superior cerebellar veins and inferior cerebellar veins

8. Mention 4 afferent fibers of the ICP.

- Posterior spinocerebellar tracts
- Dorsal external arcuate fibers
- Ventral external arcuate fibers
- Olivo cerebellar fibers

9. Mention the efferent fibers of the ICP.

- Cerebello vestibular fibers
- Cerebello reticular fibers
- Cerebello olivary fibers

10. Mention the afferent fiber of the MCP.

- Cortico ponto cerebellar fibers

11. Mention the afferent fibers of the SCP.

- Anterior spinocerebellar tract
- Tecto cerebellar tract

12. Mention the efferent fibers of the SCP.

- Fastigio reticular fibers
- Cerebello rubral fibers
- Dentato thalamic fibers and dentato rubro thalamic fibers

1. Mention 3 external features of the ventral surface of the medulla oblongata.

- Pyramid, Olive, ICP

2. Mention the cranial nerves present in the ventral surface of the medulla oblongata.

- Glossopharyngeal (9)
- Vagus (10)
- Accessory (11)
- Hypoglossal (12)

3. Mention 3 external features of the ventral surface of the pons.

- MCP, Basilar artery sulcus, and transverse grooves between bundles of transverse pontine fibers

4. Mention the cranial nerves present in the ventral surface of the pons.

- Trigeminal
- Facial
- Abducent
- Vestibulo cochlear

5. Which nerves are affected by the cerebellopontine angle tumor?

- Facial and Vestibulo cochlear

6. What are the 4 structures that wind around the cerebral peduncle?

- Optic tract
- Trochlear nerve
- Posterior cerebral artery
- Basal vein

7. What are the boundaries of the interpeduncular fossa?

- **Anteriorly:** Optic chiasma
- **Anterolaterally:** Optic tract
- **Posterolaterally:** Cerebral peduncle
- **Posteriorly:** Upper border of pons

8. What are the contents of interpeduncular fossa?

- Tuber cinereum
- Mamillary bodies
- Posterior perforated substance
- Oculomotor nerve

9. Mention the 3 elevations of the closed medulla (lower part of the back of medulla oblongata).

- Gracile, Cuneate, ICP

10. What are the borders of the open medulla?

- **Base:** Medullary stria
- **Apex:** continuous with central canal
- **Floor:** Hypoglossal trigone, vagal trigone, and vestibular trigone

11. What are the borders of the back of pons?

- **Base:** Medullary stria
- **Apex:** Continuous with cerebral aqueduct of sylvius
- **Floor:** Medial eminence, facial colliculus, and vestibular area

12. What are the external features of the back of midbrain?

- 4 colliculi: 2 superior and 2 inferior

Chapter 14?

Fourth Ventricle

1. What forms the roof of the fourth ventricle?

- **Superiorly:** Superior cerebellar peduncle and superior medullary velum
- **Inferiorly:** Inferior medullary velum

2. What forms the floor of the fourth ventricle?

- **Pontine part:** medial eminence, facial colliculus, and vestibular area
- **Medullary part:** hypoglossal trigone, vagal trigone, and vestibular trigone

3. What are the lateral boundaries of the fourth ventricle?

- **Superiorly:** Superior cerebellar peduncle
- **Inferiorly:** Inferior cerebellar peduncle+ gracile and cuneate tracts
- **Lateral recess**

4. Where do choroid plexus of the fourth ventricle lie?

- Lower part of roof of fourth ventricle

Chapter 15?

Columns of cranial nerve nuclei

1. Mention the cranial nerve nuclei belonging to GSE column.

- Nucleus of 3
- Nucleus of 6
- Nucleus of 4
- Nucleus of 12

2. Mention the cranial nerve nuclei belonging to SVE column and mention their distribution.

- **Motor nucleus of 5:** supply muscles of 1st arch
- **Motor nucleus of 7:** supply muscles of 2nd arch
- **Nucleus Ambiguous: 9:** stylopharyngeus, 10 and 11: muscles of palate, pharynx, and larynx

3. Mention the cranial nerve nuclei belonging to GVE column.

- Edinger Westphal nucleus of 3
- Superior salivatory nucleus of 7
- Inferior salivatory nucleus of 9
- Dorsal vagal nucleus of 10

4. What does the spinal part of accessory nerve supply?

- Sternomastoid and Trapezius

5. Mention the origin of spinal part of accessory nerve.

- Spinal nucleus located in the ventral horn of upper 5 cervical segments of the spinal cord

6. Mention the muscles that are supplied by motor nucleus of trigeminal.

- 4 muscles of mastication (Temporalis, masseter, medial pterygoid, lateral pterygoid)
- Tensor palate
- Tensor tympani
- Anterior belly of digastric
- Mylohyoid

1. Mention the arteries that supply the medulla.

- Anterior spinal artery
- Posterior inferior cerebellar artery
- Medullary branches of vertebral artery
- Posterior spinal artery

2. Mention the arteries that supply the pons.

- Branches of basilar artery

3. What does pontine hemorrhage cause?

- Pin point pupil
- Hyperpyrexia
- Quadriplegia

4. Mention the arteries that supply the midbrain.

- Branches of posterior cerebral and superior cerebellar arteries

5. What does the medial medullary syndrome cause?

- Contralateral hemiplegia(flaccid paralysis)
- Contralateral loss of proprioception and fine touch
- Ipsilateral LMNL paralysis of tongue muscles

6. What does the lateral medullary syndrome cause?

- Cerebellar ataxia
- Ipsilateral loss of pain and temperature from the face
- Contralateral loss of pain and temperature from the body
- Ipsilateral paralysis of palate, pharynx & larynx
- Loss of taste sensation
- Ipsilateral horner's syndrome



1. What does fracture of anterior cranial fossa cause?

- Bleeding and cerebrospinal rhinorrhea
- Hemorrhage into orbital cavity and beneath conjunctiva

2. What does fracture of middle cranial fossa cause?

- Bleeding & CSF discharge through the ear
- Cranial nerves 7 and 8 could be involved

3. What does fracture of posterior cranial fossa cause?

- Bleeding into pharynx
- Blood escapes into the nape of neck and collects in the posterior triangle

4. Attachments of falx cerebri.

- **Apex:** crista galli and frontal crest
- **Base:**
attached to upper surface of tentorium cerebelli
- **Upper border:** to lips of sagittal sulcus
- **Lower free border:**
encloses inferior sagittal sinus

5. Attachments of falx cerebelli.

- **Base:** attached to tentorium cerebelli
- **Apex:**
attached to margins of foramen magnum
- **Posterior border:**
attached to internal occipital crest
- **Anterior border:** free

6. What's the nerve supply of dura?

- **Anterior cranial fossa:** anterior ethmoidal nerve+maxillary nerve
- **Middle cranial fossa:** maxillary nerve+mandibular nerve
- **Posterior cranial fossa:** meningeal branches of 10 and 12+ recurrent branches of C1,2,and 3

7. What's the venous drainage of dura?

- Meningeal veins and diploic veins

8. What's the difference between extradural hemorrhage and subdural hemorrhage?

- **Extradural hemorrhage:** usually of arterial origin, due to injury of middle meningeal vessels
- **Subdural hemorrhage:** usually of venous origin

9. List the 4 single sinuses.

- Superior sagittal
- Inferior sagittal
- Straight
- Occipital

10. List the 6 paired sinuses.

- Sphenoparietal
- Cavernous
- Superior petrosal
- Inferior petrosal
- Sigmoid
- Transverse

11. Mention the tributaries of superior sagittal sinus.

- Superior cerebral veins, arachnoid villi and granulations

12. Mention the external relations of the cavernous sinus.

- **Anteriorly:** Superior orbital fissure and apex of orbit
- **Posteriorly:** Apex of petrous temporal bone and crus cerebri of midbrain
- **Medially:** Pituitary gland and sphenoidal air sinus
- **Laterally:** Temporal lobe
- **Superiorly:** Optic tract and ICA
- **Inferiorly:** Foramen lacerum

13. Mention the structures present inside the cavernous sinus.

- Oculomotor, trochlear, ophthalmic, maxillary, abducent nerves
- ICA

14. Mention 4 tributaries of the cavernous sinus

- Superior ophthalmic vein
- Inferior ophthalmic vein
- Central vein of retina
- Superficial middle cerebral vein

15. What is the drainage of the cavernous sinus

- Superior and inferior petrosal sinus

16. What are the communications of the cavernous sinus?

- Pterygoid venous plexus via emissary vein
- Facial vein via superior ophthalmic vein
- Opposite side via 3 intracavernous sinus

17. Which nerve pierces the cribriform plate of ethmoid?

- Olfactory nerve

18. Which nerve passes through the optic canal?

- Optic nerve

19. Which nerve passes in front of the point of crossing of the free margin and the attached margin of tentorium cerebelli?

- Oculomotor nerve

20. Which nerve passes at the point of crossing of the free margin and the attached margin of the tentorium cerebelli?

- Trochlear nerve

21. Which nerve passes through the hypoglossal canal?

- Hypoglossal nerve

22. What are the branches of the cerebral part of ICA?

- Ophthalmic
- Posterior communicating
- Anterior choroidal branches
- Anterior and middle cerebral

23. What are the boundaries of temporal fossa?

- **Superiorly:** superior temporal line
- **Inferiorly:** zygomatic arch

24. What are the boundaries of infratemporal fossa?

- **Laterally:** Ramus of mandible
- **Medially:** Lateral pterygoid plate
- **Anteriorly:** Posterior surface of maxilla
- **Posteriorly:** Styloid process
- **Superiorly:** Greater wing of sphenoid

25. What are the communications of infratemporal fossa?

- With temporal fossa through the gap deep to zygomatic arch
- With middle cranial fossa through foramen ovale and foramen spinosum
- With the orbit through inferior orbital fissure
- With pterygopalatine fossa through pterygo maxillary fissure

26. Mention the nerve supply of muscles of mastication.

- All are supplied by anterior division of mandibular nerve except medial pterygoid by trunk of mandibular nerve

27. Mention the actions of muscles of mastication.

- All protrude the mandible except Temporalis retracts the mandible
- All elevate the mandible except Lateral pterygoid depresses
- Medial and Lateral pterygoid perform side to side movement

28. Mention the lateral relations of lateral pterygoid.

- Ramus of mandible, superficial head of med pterygoid, tendon of temporalis, maxillary artery, and pterygoid plexus of veins

29. Mention the medial relations of lateral pterygoid.

- Deep head of medial pterygoid, tensor palati, MMA, maxillary artery, mandibular nerve trunk, otic ganglion, sphenomandibular ligament.

30. Mention the 3 ligaments of Tempromandibular Joint.

- Lateral ligament
- Stylomandibular ligament
- Spheno mandibular ligament

31. Mention the branches of posterior division of mandibular nerve.

- Auriculotemporal nerve, Lingual nerve, Inferior alveolar nerve

32. Mention the branches of inferior alveolar nerve.

- Mylohyoid nerve, mental nerve, incisive nerve

33. Mention 4 branches of maxillary nerve.

- Meningeal nerve, two ganglionic branches, zygomatic nerve, posterior superior alveolar nerve

34. Mention the boundaries of pterygopalatine fossa.

- **Anterior:** Posterior surface of maxilla
- **Posterior:** Pterygoid process
- **Medial:** palatine bone
- **Lateral:** opens into infratemporal fossa

35. Mention the contents of pterygopalatine fossa

- 3rd part of maxillary artery
- Maxillary nerve
- Pterygopalatine ganglion

36. Mention the branches of third part of maxillary artery.

- Posterior superior alveolar
- Infraorbital
- Greater and lesser palatine
- Sphenopalatine
- Pharyngeal artery
- Artery of pterygoid canal

37. Mention the drainage of pterygoid plexus of veins.

- Maxillary vein

38. Mention the communications of pterygoid plexus of veins

- With facial vein via deep facial vein
- With cavernous sinus via emissary veins
- With inferior ophthalmic vein via a vein that passes through inferior orbital fissure

39. Mention the site, sympathetic root, parasympathetic root, and distribution of otic ganglion.

- **Site:** below foramen ovale
- **Sympathetic root:** around middle meningeal artery
- **Parasympathetic root:** inferior salivatory nucleus > tympanic branch of 9 > tympanic plexus > lesser petrosal nerve > otic ganglion
- **Distribution:** Postganglionic fibers to parotid gland

40. Compare between otic ganglion, submandibular ganglion, sphenopalatine ganglion (according to its sympathetic root)

- **Otic ganglion:** around MMA
- **Submandibular ganglion:** around Facial artery
- **Sphenopalatine ganglion:** around ICA (deep petrosal nerve)



1. What are the tributaries of external jugular vein?

- Anterior jugular vein
- Transverse cervical vein
- Suprascapular vein
- Posterior external jugular vein

2. What forms the floor of the posterior triangle?

- Semispinalis capitis
- Splenius capitis
- Levator scapulae
- Scalenus medius
- Scalenus anterior

3. Mention 5 contents of the posterior triangle.

- Spinal accessory nerve
- Roots and trunks of brachial plexus
- Cutaneous nerves of cervical plexus
- Subclavian vein
- External jugular vein

4. Mention the cutaneous branches of the cervical plexus.

- Great auricular
- Lesser occipital
- Transverse cervical
- Supraclavicular

5. Mention the boundaries of the anterior triangle.

- **Anteriorly:** Anterior median line of neck
- **Posteriorly:** Anterior border of sternomastoid
- **Superiorly:** Inferior border of mandible and a line joining angle of mandible to mastoid process

6. Mention the boundaries of submental triangle.

- **Base:** Hyoid bone
- **Apex:** Symphysis menti
- **On each side:** Anterior belly of digastric
- **Floor:** Two mylohyoid muscles

7. Mention the contents of submental triangle.

- Submental lymph nodes, submental artery and vein, small veins that unite to form anterior jugular vein, mylohyoid nerve and vessels

8. Mention the boundaries of digastric triangle.

- **Anteriorly:** anterior belly of digastric
- **Posteriorly:** posterior belly of digastric and stylohyoid
- **Superiorly:** inferior border of mandible and a line drawn from angle of mandible to mastoid process
- **Roof:** Skin, superficial fascia, deep fascia
- **Floor:** Mylohyoid and hyoglossus

9. Mention 5 contents of digastric triangle.

- Submandibular gland
- Submandibular LN
- Submandibular ganglion
- Facial artery
- Facial vein

10. Mention the boundaries of carotid triangle.

- **Superiorly:** Posterior belly of digastric
- **Anteriorly:** Superior belly of omohyoid
- **Posteriorly:** Anterior border of sternomastoid
- **Roof:** skin, superficial fascia, investing layer of deep cervical fascia
- **Floor:** Hyoglossus, thyrohyoid, middle and inferior constrictors of the pharynx

11. Mention 3 contents of carotid triangle.

- CCA, ICA, ECA

12. Mention the contents of the muscular triangle.

- Infrahyoid muscles

13. What is the nerve supply of infrahyoid muscles?

- All are supplied by ansa cervicalis except thyrohyoid which is supplied by C1 fibers that accompany hypoglossal nerve

14. What is the nerve supply of suprahyoid muscles?

- Ant belly of digastric and mylohyoid by nerve to mylohyoid
- Post belly of digastric and stylohyoid by facial nerve
- Geniohyoid by C1 via hypoglossal nerve

15. What is the structure of ansa cervicalis?

- It's formed of 2 limbs:
 - » **Superior limb:** fibers from C1 that join hypoglossal nerve then leave it and descend in front of carotid sheath
 - » **Inferior limb:** fibers from C2 and C3

16. Compare between the cervical sympathetic ganglion(site)

- Superior cervical ganglion: C2,3
- Middle cervical ganglion: C6,7
- Inferior cervical ganglion: C7

17. Mention the branches of glossopharyngeal nerve.

- Tympanic branch
- Carotid branch
- Pharyngeal
- Muscular
- Tonsillar
- Lingual

18. Mention the branches of the vagal nerve from the superior vagal ganglion.

- Meningeal nerve and Auricular nerve



Chapter 19?

Diencephalon

1. What's the arterial supply of thalamus?

- **Medial and anterior regions** by posteromedial group of posterior cerebral artery
- **Lateral and posterior regions** by posterolateral group of posterior cerebral artery (thalamogeniculate artery)

2. What's the venous drainage of thalamus?

- Thalamostriate vein

Chapter 20?

Third ventricle

1. Mention the boundaries of lateral ventricle.

- **Lateral wall:** thalamus and hypothalamus
- **Roof:** Layer of ependyma. Choroid plexus hangs from its roof.
- **Floor:** optic chiasma, infundibulum, tuber cinereum, mamillary bodies, posterior perforated substance, tegmentum of midbrain
- **Anterior wall:** lamina terminalis, column of fornix, anterior commissure
- **Posterior wall:** posterior commissure, pineal body, suprapineal recess

2. Mention the recesses of third ventricle.

- Optic recess
- Infundibular recess
- Pineal recess
- Suprapineal recess

Chapter 21?

Limbic system

1. What are the connections of septal areas?

- **Medial forebrain bundle:** connects septal areas and amygdala with the hypothalamus
- **Stria medullaris thalami:** connects septal areas to the habenular nuclei

2. Mention 1 connection carried out by the fornix.

- Hippocampus and mamillary bodies

3. What does lesion of hippocampus or papez circuit cause?

- Anterograde amnesia(loss of recent memory and preservation of old memory)

4. What does the limbic lobe consist of?

- Subcallosal area (paraolfactory gyrus)
- Cingulate gyrus
- Parahippocampal gyrus
- Uncus
- Isthmus

Chapter 22?

Cerebral white matter

1. Enumerate the long association fibers.

- Superior longitudinal bundle, inferior longitudinal bundle, fronto occipital bundle, uncinate bundle, cingulum

2. Mention the blood supply of internal capsule.

- Upper half of anterior limb and upper half of posterior limb: MCA
- Lower ½ of anterior limb and genu: ACA
- Lower ½ of posterior limb: Post communicating and anterior choroidal artery
- Retrolentiform and Sublentiform: Anterior choroidal artery

3. Mention the relations of corpus callosum

- **Superiorly:** Falx cerebri and Anterior cerebral arteries
- **Inferiorly:** Fornix and septum pellucidum

Chapter 23?

Lateral ventricle, Cerebral meninges & CSF circulation

1. Mention the subarachnoid cisterns

- Cerebello medullary cistern
- Cistern of great cerebral vein
- Cisterna pontis
- Interpeduncular cistern
- Cistern of lateral fossa

2. How is the CSF formed?

- By choroid plexus of ventricles
- A little amount is formed around cerebral vessels and from ependyma cells

3. Mention the circulation of CSF

- Lateral ventricles > interventricular foramen > third ventricle > aqueduct of Sylvius > fourth ventricle > Median foramen of Magendie and two lateral foramina of Luschka > subarachnoid space

4. Mention the absorption of CSF

- By arachnoid villi and granulations into dural venous sinuses

1. Mention the branches of vertebral artery

- Meningeal, Posterior inferior cerebellar, posterior spinal, anterior spinal, medullary branches

2. Mention the beginning and termination of vertebral artery.

- Begins as a branch of first part of subclavian artery
- Ends at lower border of pons by joining the other vertebral artery to form basilar artery

3. Mention the beginning and termination of basilar artery.

- It's formed by union of right and left vertebral arteries at lower border of pons
- Ends at upper border of pons by giving its two terminal branches "right and left posterior cerebral arteries"

4. Mention the branches of basilar artery

- Pontine branches, labyrinthine, anterior inferior cerebellar, superior cerebellar, posterior cerebellar

5. Mention the arteries that form the circle of Willis

- ACA, anterior communicating artery, ICA, PCA, posterior communicating

6. List the cortical areas supplied by anterior cerebral artery.

- **Medial surface:** All except occipital lobe
- **Lateral surface:** One inch strip superiorly
- **Inferior surface:** Medial part of orbital surface

7. List the cortical areas supplied by middle cerebral artery.

- **Lateral surface:** large area except occipital lobe+ a strip along superior border+ a strip along inferior border
- **Inferior surface:** lateral part of orbital surface+ temporal pole

8. List the cortical areas supplied by posterior cerebral artery.

- **Medial surface:** occipital lobe
- **Lateral surface:** occipital lobe+ strip along lower border of hemisphere
- **Inferior surface:** tentorial part except temporal pole



1. What does the mesenchyme around Meckle's cartilage give?

- Mandible, Sphenomandibular ligament, Spine of sphenoid, Anterior ligament of malleus

2. Compare between 2nd and 3rd arch. (according to aortic arch)

- 2nd arch: disappear leaving stapedial artery
- 3rd arch: CCA, proximal part of ICA, and ECA arises as a sprout

3. What is the derivative of 3rd pharyngeal pouch?

- Inferior parathyroid gland and epithelium of thymus

4. Mention the 3 derivatives of intermaxillary segment.

- Philtrum of upper lip
- Part of upper jaw that carries the 4 incisor teeth
- 1st palate

5. Mention the 5 secondary brain vesicles and their derivatives in adult brain.

- Telencephalon: Cerebral hemisphere
- Diencephalon: Thalamus, hypothalamus, epithalamus, subthalamus
- Mesencephalon: Midbrain
- Metencephalon: Pons
- Myelencephalon: Medulla

6. Mention the derivatives of neural crest.

- Neuroblast, Spongioblast, Melanocyte, Chromaffin cells, Odontoblasts and cartilaginous elements of the branchial arches

7. What do chromaffin cells give?

- Adrenal medulla, carotid body, and paraganglia



2

Section



Physiology

Section 1

Sensory

1. Mention functions of the nervous system

- Maintains homeostasis
- Control of muscles & glands
- Control of mental function
- Receive sensory impulses
- Processing & integrating information

2. Define synapse

- It is the site of contact between 2 neurons with separate protoplasm (functional junction)

3. Enumerate functions of synapses

- Transmission of impulses
- Convergence of impulse:
Collection of Impulses – intensity of impulses (intensification)
- Divergence of impulses:
spreading of impulses
- The site of action of drugs

4. Compare between chemical synapse & electric synapse regarding Def. & Characteristics

	Chemical synapse	Electric synapse
Def.	Occur by release of neurotransmitter from presynaptic terminal (terminal button) traverse the synaptic cleft & affect postsynaptic membrane	Occur by passage of ions & small molecules through gap junction between 2 adjacent cells
Characteristics	• More common • Unidirectional • Less fast • Drug sensitive • Excitatory or inhibitory	• Less common • Bidirectional • Faster • No drug sensitivity • Excitatory only

5. Illustrate the mechanism of chemical neurotransmission in synapses

- Release of neurotransmitter from presynaptic knob
- NT crosses the cleft.
- Combines with its specific receptor on postsynaptic membrane.
- Changes the RMP of postsynaptic membrane by opening or closing channels.
- Removal of NT.

6. List the ways of removal of NT

- Active Reuptake by the presynaptic membrane
- Inactivation by enzyme on postsynaptic membrane.
- Passive diffusion away.

7. Define:

• Neurohormone

A chemical substance secreted by a neuron into blood

• Neuromodulator

A slowly inactivated (long acting) chemical substance secreted

• Neurotransmitter

A chemical substance secreted by neurons to stimulate or inhibit other neurons

8. Mention the sites of release of Acetyl choline

- Brain (cortex, thalamus)
- NM junction
- All preganglionic neurons
- All postganglionic parasympathetic neurons
- Some postganglionic sympathetic as sweat glands and blood vessels of skeletal muscles.

9. Mention effects of Norepinephrine

- Stimulatory on α_1 receptors Inhibitory on β_2 receptors

10. What is the effect of GABA?

- Inhibitory, by increasing the influx of Cl^- causing hyperpolarization

11. Define sensory receptors

- Specialized structures present at the peripheral end of afferent nerves.

12. Mention the function of sensory receptors

- Detect changes in environment.
- Transducers of stimulus to potential receptor.
- Generate action potential in afferent nerve.
- Inform the CNS about changes in the environment.

13. Classify receptors according to specificity of receptors to stimuli

- Mechanoreceptors
- Thermoreceptors
- Chemoreceptors
- Electromagnetic receptors
- Nociceptors (Pain receptors)

14. Regarding adaptation, Touch receptors are rapidly adapting receptors while pain receptors are slowly adapting receptors

15. Enumerate properties of receptors

- Specificity "Muller's law"
- Excitability
- Weber-Fechner law
- Adaptability

16. Illustrate mechanism of receptors adaptation

- Gradual closure of Na^+ channels.
- Gradual decrease excitability of 1st node of Ranvier
- Loss of stimulus energy
- Accommodation of afferent nerve to generator potential
- Readjustment & remodeling of receptor

17. Define sensory code

- It is the ability of CNS to recognize modality & locality & intensity of sensation

18. Define sensory unit

- Single sensory axon and all its peripheral branches that vary in number

19. Define the receptive field

- The spatial distribution from which a stimulus produces a response in that unit

20. Explain paradoxical cold sensation

- Rise of temp. above 40°C leads to sudden & rapid discharge of impulses from cold receptors

21. Compare between crude touch sensation & thermal sensation regarding (Def. receptors & afferent neuron type)

	Crude touch sensation	Thermal sensation
Def	the ability of the body to detect contact with Objects	The ability of the body to feel different degrees of hot & cold water
Receptors	Free nerve endings Hair end organ	Free nerve ending (hot sensation) Krause end bulb (cold sensation)
Afferent neuron type	Ad & C	C for warm sensation Ad & C for cold sensation

Enumerate causes of cutaneous pain

- Injury of skin
- Inflammation of skin
- Irritation of dorsal roots
- Referred pain to the skin from deep & visceral diseased tissues

22. Mention the mechanism of stimulation of pain receptors.

- Tissue damage causing release of proteolytic enzymes, kinins, K^+ , substance P, histamine causing stimulation of chemosensitive pain receptors and facilitation of Mechanosensitive pain receptors and thermosensitive pain receptors.

23. Compare between fast cutaneous pain & slow cutaneous pain regarding (Afferent pathway receptors)

	Fast cutaneous pain	Slow cutaneous pain
Afferent	Ad fibers	C fibers
pathway	Neo-Lateral spinothalamic tract	Paleo-Lateral spinothalamic tract
Receptors	Mechansensitive & thermosensitive pain receptors	All types of pain receptors especially the chemosensitive

24. Mention causes of deep pain

- Injury (trauma) of deep structures
- Inflammation of deep structures
- Ischemia of deep structures
- Spasm of muscle

25. Mention causes of visceral pain

- Injury of viscera
- Inflammation of peritoneal covering of viscera
- Irritation of viscera (e.g. irritation by HCl in peptic ulcer)
- Ischemia of viscera
- Spasm of viscera (e.g. ureter spasm)
- Overdistention of viscera (e.g. bladder distention by accumulated urine)

26. Compare between 1^{ry} hyperalgesia & 2^{ry} hyperalgesia regarding (site pain sensation pain threshold)

	1 ^{ry} hyperalgesia	2 ^{ry} hyperalgesia
Site	Inflamed skin around an injury	Healthy skin
Pain sensation	A non-painful stimulus becomes painful	A painful stimulus become more painful
pain threshold	Decreased by chemicals, mainly by kinins	Normal

27. Explain convergence-projection theory.

- The afferent pain fibers from skin and diseased viscera converge on the same cell of SGR and activate the same cortical neuron. Whatever the source of pain the cortex will project the pain to the skin because it is the most common source of pain.

28. Explain the convergence-facilitation theory.

- Afferent fibers of skin and viscera converge on SGR. 2nd order neuron of the skin pathway is facilitated by visceral afferents. So, pain is felt in the skin due to activation of its pathway from SGR to cortex.

29. Define headache

- pain felt at the head, It is usually referred from other structures

30. Enumerate extracranial causes of headache

- Spasm of neck & scalp muscles
- Sinusitis (irritation of nasal sinuses)
- Otitis media
- Toothache
- Visual causes
- Errors of refraction
- Excessive exposure to ultraviolet irradiation

31. Mention causes of intracranial headache

- Meningeal irritation
- Low CSF pressure
- Alcohol
- Hypertension
- Migraine

32. Periventricular area (PVA) in hypothalamus Send neurons that secrete β -endorphins to stimulate Periaquiductal gray area (PAGA)

33. Mention 3 types of opioid peptides

- Enkephalins
- Endorphins
- Dynorphins

34. Mention sensations carried by dorsal column tract

- Fine touch
- Proprioception
- Vibration
- deep pressure
- Stereognosis.

35. Compare between Fine touch & vibration sensation regarding receptors

	Fine touch	Vibration sensation
Receptors	Meissner's corpuscles, Merckle's discs, Pacinian corpuscles & Ruffini end organs	Pacinian corpuscles (respond to frequency 30-80 cycles/sec) Meissner's corpuscles (respond to frequency 2-80 cycles/sec)

36. Enumerate requirements of stereognosis

- Intact all sensations
- Intact cortical association areas (areas 5,7)
- Intact memory

37. Mention functions of somatic sensory area 1

- It perceives:
- Fine touch
- Fast pain
- Fine degrees of temperature
- Vibration
- Proprioceptive sensations (position & movement)
- Discrimination of various weights (deep pressure)
- Stereognosis

38. Mention the effect of somatic sensory area-1 lesion

- Pain & temperature are poorly affected
- Crude localization (crude touch) is intact
- The person loses the sensations perceived by S1

39. Define Neglect syndrome

- Partial neglect & inattention to tactile, visual and proprioceptive stimuli of the contralateral side of the body

Section 2

Motor

1. Compare between reflex movements & voluntary movements regarding requirements

	Reflex movements	Voluntary movements
Requirements	Intact muscles Intact lower motor neuron (LMN) Intact pathway of a reflex action	Intact muscles Intact lower motor neuron (LMN) Intact upper motor neuron (UMN)

2. Define motor unit

- It is the functional unit used by the motor system to perform a movement.

3. Define motor neuron pool

- The total α motor neurons that innervate a muscle

4. Mention importance of interneurons

- Lateral inhibition helps in sharpening of motor actions
- Recurrent inhibition helps in prevents over excitation in CNS, and limits the discharge of a motor neurons

5. Enumerate characters of stretch reflex

- The fastest reflex in the body
- Restricted to the stretched muscle (Local character)
- Its more apparent in the antigravity muscles
- It shows reciprocal innervation

6. Mention the clinical importance of stretch reflex

- Examination of static stretch reflex activity (muscle tone)
- Examination of dynamic stretch reflex (deep reflex)
- Diagnosis of site and cause of neurological diseases.

7. Compare between nuclear bag fibers & nuclear chain fibers

Nuclear bag fibers	Nuclear chain fibers
Central part contains large number of nuclei like a bag	Central part contains nuclei spreading like a chain
1-3 fibers in each spindle	3-9 fibers in each spindle Each 1-3 fibers are attached to the sides of one nuclear bag fiber
Detect the velocity of change of muscle length (dynamic response)	Detect the degree of change of muscle length (static response)
Phasic receptor	Tonic receptor

8. Compare between primary afferent endings & secondary afferent endings

Primary afferent endings	Secondary afferent endings
Annulo-spiral	Flower spray
Type 1a fibers	Type II fibers
Around central parts of nuclear bag & chain fibers	Around central parts of nuclear chain fibers only
Rapidly conducting	Slowly conducting
Acts with dynamic & static stimulation of receptor	Acts only with static stimulation of the receptor

9. Mention properties of stretch reflex

- It's a spinal, deep reflex
- It's a monosynaptic reflex
- It has shortest reflex time
- It has no recruitment or after discharge
- It shows reciprocal innervation
- Its activity is controlled by higher centers
- It's a semi quantitative reflex

10. Compare between dynamic stretch reflex & static stretch reflex regarding (Def. example & response)

	Dynamic stretch reflex	Static stretch reflex
Def.	Its reflex sudden tetanic muscle contraction in response to sudden stretch	It's a reflex maintained sub-tetanic muscle contraction in response to maintained weak stretch
Example	Tendon jerk	Muscle tone
Response	Dynamic response Rapid tetanic contraction followed by rapid relaxation	Static response continuous sub-tetanic contraction

11. Mention function of muscle tone

- Maintains the body posture against the effect of gravity.
- Gives postural background needed to support the body during voluntary movements.
- Maintains heat production thus regulating body temperature.
- Maintains venous return and lymphatic drainage especially in muscles of lower limbs.
- Keeps the viscera in position by tone of anterior abdominal wall muscles

12. Explain: Muscle tone is not accompanied by fatigue

- It is due to alternate contraction of different muscle fibers.
- Contraction is sub-tetanic.
- Muscle fibers involved in muscle tone are the red, slow, oxidative muscle fibers, which are rich in mitochondria, myoglobin and capillaries They contract slowly and can sustain force in muscle for along time

13. Compare between facilitatory centers & inhibitory centers

Facilitatory centers	Inhibitory centers
Motor area 4	Cortical suppressor area (4s,6)
Facilitatory (pontine) reticular formation, it has an intrinsic activity, it sends its impulses through the (ventral reticulo-spinal tract).	- Inhibitory (medullary) reticular formation, it has no intrinsic activity, so it is activated by other centers, it sends its impulses through lateral reticulospinal tract
Vestibular nucleus through vestibulospinal tract	- Red nucleus through rubrospinal tract
Caudate nucleus of basal ganglia through pontine reticular formation	- Lentiform nucleus of basal ganglion through medullary reticular formation.

14. Mention stimulus of golgi tendon organ reflex

- Increase in muscle tension by:
- Increase active tension by isometrically contracted muscle.
- Increase passive tension by excessive stretch of the muscle beyond physiologic limit.

15. Mention the aim of the inverse stretch reflex.

- Prevents marked increase in muscle tension and avulsion of tendon, equalizes the contraction force among muscle fibers during normal muscle activity

16. Mention properties of flexor withdrawal reflex

- It is initiated by injurious stimuli
- Impulses are transmitted by AB & C fibers.
- Polysynaptic reflex (long central delay of conduction)
- Irradiation
- Reverberating circuit between the interneurons results in prolonged after discharge to maintain muscle contraction for some time after removal of the stimulus.
- Reciprocal innervation

17. Mention Function of primary motor cortex (area4)

- Initiation of fine movements of distal limb muscles of opposite side of the body
- Shares in initiation of gross movements of axial & proximal limb muscles of opposite side of the body
- Stimulatory to muscle tone
- Stimulatory to reflexes & tendon jerks
- Stimulatory to superficial reflexes
- Essential for normal plantar reflex

18. What are the areas to which area 6 is connected?

- Area4
- Paleocerebellum
- Basal ganglia
- Supplemental motor area
- area5,7
- Spinal cord

19. Mention the function of supplemental area.

- Shares in the control of postural background for fine movements
- planning of complex bimanual bilateral movements

20. Mention the components motor hierarchy

- Higher motor level: Association areas in the cortex
- Middle motor level: Motor cortex, Basal ganglia & Cerebellum:
The site of planning & programming
- Lower motor level: Spinal cord motor neurons & Brain stem nuclei
The site of performance

21. Compare between medial motor system & lateral motor system regarding (site of termination – muscles innervated – function – tracts involved)

	Medial motor system	Lateral motor system
Site of termination	Medial or ventral portion of anterior horn	Lateral portion of anterior horn
Muscles innervated	Trunk muscles Proximal limb muscles	Distal limb muscles
Function	• Controls gross movements • Controls body posture • Provides stable background during performance of fine movements by distal limb muscles	Controls fine, skilled voluntary movements
Tracts involved	• Ventral corticospinal tract • Reticulospinal tracts • Vestibulospinal tracts • Tectospinal tract	• Lateral corticospinal tract • Rubrospinal tract

22. Mention the effect of anterior spinal artery occlusion

- Bilateral UMNL paralysis below the lesion
- Bilateral loss of pain & temperature sensations below the lesion with preservation of proprioception & touch

23. Mention the effect on muscle state in LMNL

- Marked atrophy (degeneration, wasting) of the affected muscle occur due to:
- Loss of voluntary and reflex activity of the muscle.
- Loss of trophic effects on metabolism of the affected muscle due to damage of their nerve supply.

24. Define fasciculation

- Jerky visible contractions of groups of muscle fibers. Can be felt and seen

25. Define fibrillations

- Fine, irregular contractions of individual fibers, detectable electro-physiologically, but invisible to the eye. Felt only

26. Compare between chronic UMNL & LMNL regarding (character – muscle state – superficial reflexes – deep reflexes –clonus)

	Chronic UMNL	LMNL
Character	Spastic paralysis	Flaccid Paralysis
Muscle state	Disuse Atrophy on long term	Immediate atrophy Muscle → fibrous tissue
Superficial reflexes	All Lost except +ve Babinski	All are lost
Deep reflexes	Hyperreflexia	Hyporeflexia
Clonus	Present	Absent

27. Define spinal shock

- Complete loss of all reflexes below the level of the lesion.

28. Mention cause of spinal shock

- Sudden withdrawal of the facilitatory corticospinal, reticulospinal, vestibulospinal impulses. This result in decrease excitability of spinal motor neurons, and loss of their response to stimulation by afferents in reflex pathway.

29. Mention causes of reflex recovery in spinal shock

- The development of denervation hypersensitivity by the remaining spinal neurons, and so they can now respond to afferent stimulation.
- Another possibility is the increase of collaterals from existing neurons, with the formation of more excitatory endings on interneuron's and motor neurons.

30. Enumerate immediate effects of thalamic syndrome

- Crossed hemi-anesthesia
- Secondary hyperalgesia due to release or rebound from gate inhibitory fibers
- Sensory ataxia

31. Enumerate delayed effects of thalamic syndrome

- Recovery of crude sensation but become more poorly localized & with high threshold
- Fine sensation never recover
- Secondary hyperalgesia
- Paroxysmal attacks of pain due to release from gate inhibition
- Thalamic hyperpathia
- Emotional disturbances
- Motor ataxia

32. Compare between sensory ataxia & motor ataxia regarding (cause – coordination test – romberg's sign – gate)

	Sensory ataxia	Motor ataxia
Cause	Incoordination of movement due to loss of proprioception sensations	Incoordination of movements due to cerebellar lesion
Coordination test	Like finger to nose test can be done only with eyes opened Finger move before & after the nose	Like finger to nose test shows hypermetria even eyes are opened
Romberg's sign	+ve	-ve
Gate	Stamping gate	Zigzag like gate

33. Compare between the caudate & putamen circuits regarding function

	Caudate circuit	Putamen circuit
Function	cognitive (thinking) control of sequences of motor pattern. (Cognitive control of motor activity determines subconsciously, and within seconds, which patterns of movement will be used together to achieve a complex goal that might itself last for many seconds).	control complex pattern of skilled learned movements

34. Enumerate neurotransmitters in basal ganglia

- Fibers from cerebral cortex to corpus striatum release glutamate.
- Intrastriatal interneurons (caudate-putamen circuit) release acetyl choline.
- Fibers from Substantia nigra to corpus striatum release dopamine.
- Fibers from corpus striatum to substantia nigra release GABA.
- Fibers from basal ganglia to brain stem release noradrenalin, encephalin and serotonin

35. What happen if degeneration in the dopamine releasing neurons occurs?

- The activity in the direct pathway decreases and accordingly there is a decrease in the cerebral cortex activity. This appears clearly in the hypokinesia and slow Voluntary movements with a disease called parkinsonism. Additionally, when the activity of the indirect Pathway increases (which is inhibitory to cortex) ...helping in hypokinesia

36. What is the function of cerebellum

- Planning & programming of movements

37. Mention function of climbing fibers of cerebellum

- Motor learning (training) & cognitive function

38. Mention function of cerebellum

- Regulation of balance, posture & equilibrium
- Coordination of voluntary movements
- Control of stretch reflex & muscle tone

39. Compare between spasticity & rigidity

	Spasticity	Rigidity
Tremors	Absent	Static tremors
Cause	Chronic UMNL	Lesion in nigro-striatal connections
Active motor neurons	Y motor neurons	A & Y motor neurons
Affected muscles	Antigravity muscle	All muscles especially flexors
Type	Clasp knife	Cog wheel Lead pipe
Clonus	Present	Absent
Reflexes	hyperreflexia	Normoreflexia
Gait	circumduction	shuffling



Section 3

Higher Brain Function

Chapter 1 ?

Reticular formation

1. What is the reticular formation (RF)?

- A complex network of interconnected neurons in the brain stem.

2. What are the two parts of the reticular formation (RF)?

- Sensory neurons
- Motor neurons

3. What are the subdivision of motor neurons in the reticular formation (RF)?

- They are divided into two antagonistic groups: Pontine (Excitatory) RF & Medullary RF.

4. What is the function of the ascending reticular activating system (ARAS)?

- It is excitatory to all parts of the cerebral cortex, promoting arousal.

5. True or False: The ascending reticular activating system (ARAS) is directly excitatory to motor neurons of antigravity muscles.

- True

6. What inhibits the ascending reticular activating system (ARAS)?

- Inhibitory reticular formation
- Inhibitory areas of the cerebral cortex
- Paleocerebellum
- Basal ganglia
- Red nucleus.

7. What is the function of the medullary reticular formation (RF)?

- It is inhibitory to motor neurons of antigravity muscles, leading to a decrease in muscle tone.
- Can inhibit the Excitatory RF and decrease brain activity.

8. What is the reticulo-thalamo-cortical pathway?

- It is a polysynaptic, multineuronal pathway that originates from the excitatory reticular formation (RF)

9. What factors can affect the reticular activating system (RAS)?

- Sensory impulses (especially pain)
- Cortical impulses (emotions, motor, thinking)
- Damage of RAS
- Analeptic drugs
- Sympathomimetics, stress, thyroid hormones, and caffeine.

10. What are the functions of the reticular formation (RF)?

- Link between higher brain centers and spinal cord centers for integration.
- Support of body and maintenance of posture during standing.
- Non-specific activation of the cerebral cortex for wakefulness and alert state.
- Regulation of vital functions.
- Pain modulation.
- Center of rapid eye movement (REM) sleep.



1. What are the types of evoked potentials?

- Primary (localized) evoked potentials: characterized by a positive wave followed by a negative wave.
- Secondary (diffuse) evoked potentials: characterized by a large positive wave due to the activation of non-specific thalamic nuclei.

2. What is an electroencephalogram (EEG)?

- A recording of spontaneous brain electrical activity in a conscious person.

3. What is the definition of evoked potentials?

- Recording of electrical activity of cerebral cortex after receptor stimulation in an anaesthetized person

4. What are the uses of evoked potentials?

- Mapping of brain
- Diagnosis of disease

5. In what conditions is the alpha (α) brain wave recorded?

- Adult, awake, relaxed, closed eyes

6. What is the character of alpha (α) brain wave?

- Regular synchronized

7. In what conditions is the beta (β) brain wave recorded?

- Awake, mental activity, opened eyes

8. What is the character of beta (β) brain wave?

- Irregular desynchronized

9. In what conditions is the theta (θ) brain wave recorded?

- Children, light sleep, emotional stress (frustrations)

10. What is the character of theta (θ) brain wave?

- Regular synchronized

11. In what conditions is the delta (δ) brain wave recorded?

- Infants, deep sleep, deep coma, brain damage

12. What is the character of delta (δ) brain wave?

- Regular synchronized

13. In what conditions is the gamma (γ) brain wave recorded?

- Hyper-aroused, high mental activity

14. What is the character of gamma (γ) brain wave?

- Irregular desynchronized

15. What is alpha block?

- When a relaxed person, with closed eyes (with a rhythm recorded from him) is focused on something or open his eyes, so a rhythm is replaced by β rhythm.

16. What is the value of EEG?

- A non-invasive test to study:
 - Brain electrical activity
 - Brain death ($\rightarrow$ flat EEG)
 - Site of brain tumors
 - Sleep stages & disorders
 - Epilepsy



1. What is the definition of consciousness?

- Awareness of self, surroundings, thoughts & feelings

2. What is the mechanism of consciousness?

- Non-specific excitation to the whole cortex by RAS (brainstem) & may be hypothalamus.

3. What happens after hours of consciousness?

- Fatigue of neurons which decrease positive feedback Activation of sleep centers and so inhibition of RAS

4. What is coma?

- Unconsciousness from which the person cannot be aroused

5. What is sleep?

- Physiological and spontaneous unconsciousness from which a person can be aroused by stimuli

6. What are the functions of sleep?

- Restoration of physiological & biochemical processes (Cognition, metabolic, thermal, caloric & immune processes)
- Helps learning & memory
- Neural plasticity (maturation)

7. What are the stages and EEG characteristics of very light sleep?

- Theta waves

8. What are the stages and EEG characteristics of light sleep?

- Theta waves + sleep spindles (Large α : 12-14 Hz) & K complexes (Large & slow)

9. What are the stages and EEG characteristics of moderate sleep?

- Delta (3-3.5 Hz)

10. What are the stages and EEG characteristics of deep sleep?

- Low frequency Delta (0.5-1.5 Hz)

11. What are the characteristics of Non-Rapid Eye Movement (Non-REM) sleep?

- Slow wave sleep with absence of dreams and penile erection

12. What are the EEG characteristics of Rapid Eye Movement (REM) sleep?

- Rapid, irregular, low voltage, desynchronized β waves with presence of dreams and penile erection

13. What happens to sleep after the first cycle?

- Sleep lightens
- non-REM time decreases
- stages 3 and 4 decrease
- REM time increases

14. What are the three theories of sleep?

- Passive theory
- Metabolic theory
- Active theory

15. What is the passive theory of sleep?

- Fatigue and decreased activity of Reticular Activating System (RAS) lead to a decrease in positive feedback of cortical excitation

16. What is the metabolic theory of sleep?

- Prolonged wakefulness leads to the accumulation of Sleep Inducing Substances (SIS) in the cerebrospinal fluid or brain stem, resulting in inhibition of RAS

17. What are examples of Sleep Inducing Substances (SIS)?

- Lipid and Muramyl peptide

18. What is the active theory of sleep?

- The suprachiasmatic nucleus of hypothalamus synchronizes sleep/wakefulness rhythm with the dark/light cycle, and sleep occurs through stimulation of sleep centers

19. Which brain regions and neurotransmitters are involved in arousal?

- Arousal involves the Reticular Activating System (RAS) and posterior hypothalamus
- norepinephrine, serotonin, histamine

20. What are the neurochemical mechanisms of sleep and arousal?

- The sleep/wakefulness circadian rhythm is controlled by brain stem nuclei and hypothalamic nuclei
- orexin produced in the hypothalamus regulates the activity of brain stem nuclei which lead to switch between sleep and wakefulness

21. Which neurotransmitters are increased in wakefulness?

- Norepinephrine, serotonin, histamine

22. Which neurotransmitters are increased in non-REM sleep?

- Norepinephrine, serotonin, GABA and acetylcholine

23. Which neurotransmitters are associated with REM sleep?

- Acetylcholine

24. Which neurotransmitter is decreased during wakefulness?

- Acetylcholine
- GABA

25. Which neurotransmitter is decreased during non-REM sleep?

- Histamine

26. Which neurotransmitter is decreased during REM sleep?

- Norepinephrine
- Serotonin

27. What are the effects of acetylcholine, GABA, norepinephrine, and serotonin in wakefulness?

- Acetylcholine and GABA decrease, while norepinephrine and serotonin increase in wakefulness.

28. What are the effects of histamine in wakefulness and non-REM sleep?

- Histamine increases in wakefulness and decreases in non-REM sleep



1. What is the function of the left hemisphere?

- Language (speech), logical, analytical, sequential & verbal tasks

2. What is the function of the right hemisphere?

- Visuospatial relation, recognition of faces and objects, musical & artistic talents

3. True or False: The prefrontal association area is responsible for self control. (True)

4. What is the function of the parieto-occipito-temporal association area?

- Analysis of spatial coordination of the body & its surrounding
- Contains Wernicke's area for Language comprehension
- Contains Angular gyrus area For interpretation of visual Language (Reading)
- Contain area for naming objects

5. What is the function of the prefrontal association area?

- High intellectual functions
- problem solving
- personality traits
- Contains Broca's area
- Creativity
- Decision making
- Elaboration of thoughts
- Emotional behavior
- Ordering of events
- Operation of working memory
- Planning of voluntary Movements

6. What is the function of the limbic association area?

- Behavior, emotions, motivation

7. What is the function of the visual association area?

- Understand the meaning of written words

8. What is the function of the auditory association area?

- Understand the meaning of spoken words

9. What is the function of Broca's area?

- Word formation
- Stores programs of Vocalization
- Motor speech center

10. What is the function of Exner's area?

- Hand skills area
- stores programs of coordinated hand movements
- Where are the programs of vocalization and hand movements stored?
- Specialized areas of premotor cortex (area 6)

Chapter 5 ?

Speech

1. What are the components of speech?

- Sensory (input, Receptive): vision & hearing
- Motor (output, Expressive): vocalization & writing

2. What is the sensory component of speech responsible for?

- Understand the meaning of written words
- Understand the meaning of spoken words
- Determination of thoughts
- Choice of proper words

3. What is the motor component of speech responsible for?

- Formation of words
- Stores vocalization programs
- Stores programs of coordinated hand movements

4. Where is the visual association area located?

- Occipital lobe

5. Where is the auditory association area located?

- Temporal lobe

6. Where is Wernick's area located?

- Superior temporal gyrus

7. Where is Broca's area located?

- Frontal lobe

8. Where is Exner's area located?

- Frontal lobe

9. What is the primary auditory area responsible for?

- Perception of spoken words

10. What is the primary visual area responsible for?

- Perception of written words

Chapter 6 ?

Speech disorders

1. What is aphasia?

- Defects in speech due to a lesion in speech centers in the categorical hemisphere.

2. What are the two types of sensory aphasia?

- Visual aphasia (word blindness)
- Auditory aphasia (word deafness).

3. What is Wernicke's aphasia?

- Aphasia caused by a lesion in Wernicke's area
- Characterized by excessive talking with neologism
- But inability to understand thoughts of written or spoken words.

4. What is Broca's aphasia?

- Expressive, non-fluent aphasia caused by a lesion in Broca's area, limiting speech to 2-3 words.

5. What is agraphia?

- The inability to express oneself through written words, caused by a lesion in Exner's area.

6. What is global aphasia?

- Aphasia caused by extensive brain lesion
- Affecting both receptive and expressive language skills
- Leading to non-fluent speech.

7. What is conductive aphasia?

- Aphasia caused by a lesion in or near the auditory cortex
- resulting in the inability to put parts of words together or repeat words correctly.

8. What is anomic aphasia?

- Aphasia caused by a lesion in the angular gyrus in the categorical hemisphere
- leading to the failure of processing visual language and the inability to understand or give names for seen words and pictures.

9. What is dysarthria?

- Abnormal speech due to a defect in articulation.

10. What are the types of dysarthria?

- Flaccid dysarthria
- spastic dysarthria
- ataxic dysarthria
- hypokinetic dysarthria
- hyperkinetic dysarthria
- mixed dysarthria.

Chapter 7 ?

Memory and learning

1. What is explicit memory?

- declarative, non-reflexive memory, which is Consciously recalled

2. Where is explicit memory stored?

- Anterior & Lateral Temporal cortex
- Hippocampus
- Medial temporal lobe
- Neocortex

3. What is implicit memory?

- Reflexive, non-declarative memory, which subconsciously recalled

4. What is the duration of short-term memory (STM)?

- Lasts for seconds to hours

5. What is the duration of long-term memory (LTM)?

- Lasts for days to years

6. What is the site of storage for long-term memory (LTM)?

- Neocortex

7. What is the mechanism of long-term memory (LTM)?

- Permanent structural and functional changes
- Increase in presynaptic terminals, number of vesicles, area of transmitter release and protein synthesis
- Changes in structure of dendritic Spine
- Formation of new synapses
- Transmission of stronger impulses

8. What is consolidation of memory?

- Conversion of STM into LTM
- Or The time need for the structural & functional synaptic changes to occur

9. How long does the consolidation of memory take?

- Begins in 5-10 mins, becomes completed in 1-2 hrs

10. What can enhance the consolidation of memory?

- Attractive information
- Rehearsal
- Mental rest
- Quiet sleep

11. What can impair the consolidation of memory?

- Anesthesia
- Brain concussion
- Mental fatigue
- Electroshock therapy

12. What is working memory?

- A type of STM that keeps information available for some time

13. Where is working memory integrated?

- Prefrontal association area

14. What is the role of the amygdala in memory?

- Processing and remembering the emotional component of memories

15. What is synaptic plasticity?

- The ability to change the function of synapse according to demand

16. Can synaptic transmission increase or decrease for short periods?

- Yes

17. What is post-tetanic facilitation?

- Continuation of postsynaptic response even after removal of the stimulus up to 60 sec due to stimulation of presynaptic neuron by rapid tetanizing stimulus

18. What's the mechanism of Post-tetanic facilitation?

- The tetanizing stimulation causes accumulation of Ca^{++} in presynaptic neuron lead to continuation of release of transmitter.

19. What is long-term potentiation (LTP)?

- Persistent enhancement of the post-synaptic response after rapid repeated brief stimulation of the presynaptic neuron

20. Where does long-term potentiation (LTP) occur?

- Hippocampus and other areas

21. What is the value of Long-term potentiation (LTP)?

- It is the basis of long-term memory

22. What is long-term depression (LTD)?

- Decrease in the strength of synaptic transmission after prolonged, slow, low-frequency stimulation of the presynaptic neuron

23. What neurotransmitter is involved in Long-term potentiation (LTP)?

- Glutamate

24. What does Long-term potentiation (LTP) depend on ?

- Glutamate neurotransmitter
- May depend on NMDA receptors
- Increase intracellular Ca^{++}
- Increase protein synthesis
- Increase growth and connections of pre & post-synaptic neurons

25. What is the value of long-term depression (LTD)?

- It's the basis of cerebellar learning

26. What is amnesia?

- Partial or complete loss of memory

27. What is anterograde amnesia?

- Inability to form new long-term memories

28. What causes anterograde amnesia?

- Old age, atherosclerosis, lesions in hippocampus or medial temporal lobe

29. What is retrograde amnesia?

- Inability to recall recent events that occur shortly before brain trauma or injury

30. What causes retrograde amnesia?

- Concussion, anesthesia, electroshock therapy

31. What is global amnesia?

- Inability to recall recent and remote events

32. What causes global amnesia?

- Extensive brain damage

33. What is simple non-associative learning?

- The process of modifying behavior through repeated stimulus.

34. What is habituation?

- The gradual loss of response to a repeated benign stimulus.

35. What is sensitization?

- The process where a noxious stimulus leads to an augmented response when a benign stimulus occurs with it

36. What is the mechanism of sensitization?

- Presynaptic facilitation
- Release of serotonin by the neuron that transmits the noxious Stimulus, increases adenylyl cyclase enzyme, which increases cAMP in the Presynaptic neuron of the habituated synapse, block of K^+ Channels leads to depolarization and release of the Transmitter leads to prolonged augmented response.

37. What neurotransmitter is released in the process of sensitization?

- Serotonin

38. What is the value of sensitization?

- It's the basis of Short-Term Memory (STM) and Long-Term Memory (LTM)

39. What is classical conditioning?

- Pairing of an unconditioned stimulus that naturally produces a response with a conditioned neutral stimulus that doesn't produce a response.

40. What is operant conditioning?

- Teaching a person to perform tasks to obtain a reward or avoid punishment

1. What are the functions of the limbic system?

- Perception and discrimination of olfactory sense
- Control of autonomic responses
- Control of feeding and appetite
- Control of maternal and sexual behavior
- Encoding and consolidation of STM
- Control of emotions
- Control of motivation

2. What is the role of the piriform cortex in smell?

- Perception and discrimination of odor

3. What is the role of the amygdala in smell?

- Emotional response to odors

4. What is the role of the enterorhinal cortex in smell?

- Olfactory memories

5. What is responsible for the conscious discrimination of odors?

- Frontal and orbitofrontal neocortex

6. What is the role of normal limbic system in maternal & sexual behavior?

- Normal maternal behavior
- Emotional expression during sexual act

7. What factors can modify sexual behavior?

- Prefrontal cortex
- social factors
- psychic factors

8. What happens with a bilateral lesion in the hippocampus?

- Anterograde amnesia

9. How does a lesion in the amygdala affect feeding and appetite?

- Hyperphagia
- amniophagia

10. Which parts of the limbic system are responsible for rage and aggression?

- Amygdala & lateral hypothalamus

11. Which parts of the limbic system are responsible for fear?

- Amygdala & periventricular nucleus

12. Which part of the limbic system is responsible for placidity?

- Ventromedial nucleus of hypothalamus

13. Which parts of the brain are involved in the reward (approach) system?

- Lateral & VM hypothalamus, septum, Amygdala, thalamus & basal ganglia

14. Which parts of the brain are involved in the punishment (avoidance, aversion) system?

- PVN & PAGA, Amygdala & hippocampus

15. What does cocaine do to the reward system?

- Gives the sensation of wellbeing by increasing dopaminergic neurons

1. What is the main function of the hypothalamus?

- Control of internal homeostasis:
 - Control of endocrine functions
 - Control of Autonomic responses
 - Regulation of food intake and appetite
 - Regulation of body temperature
 - Regulation of H₂O balance & ECF volume
 - Control of sleep & arousal
 - Control of Circadian rhythm
 - Role in Learning & motivation
 - Role in emotion & sexual behavior

2. How does the hypothalamus control endocrine functions Through nervous control?

- Control of posterior pituitary gland leads to secret oxytocin & vasopressin
- Control of Adrenal medulla leads to secret catecholamines

3. How does the hypothalamus control endocrine functions Through hormonal control?

- Control of Anterior pituitary gland, which controls most body endocrine glands ,through:
 - arcuate nucleus synthesize dopamine and growth hormone releasing hormone (GHRH)
 - preoptic nucleus synthesize gonadotropin releasing hormone (GnRH)
 - paraventricular nucleus synthesizes thyrotropin releasing hormone (TRH), corticotropin releasing hormone (CRH) & somatostatin (SST)

4. Which parts of the hypothalamus control autonomic responses?

- Postero-lateral hypothalamus (sympathetic effects)
- Anterior hypothalamus (parasympathetic effects)

5. Which part of the hypothalamus is the feeding center?

- Lateral hypothalamus

6. Which part of the hypothalamus is the satiety center?

- VM hypothalamus

7. What's the characteristics of feeding center?

- Located in Lateral hypothalamus
- Has intrinsic activity
- Inhibited by satiety center
- Lesion in it leads to Anorexia

8. What stimulates the satiety center in the VM hypothalamus?

- Glucose, in the presence of insulin (insulin-dependent glucostat)

9. What happens when there is a lesion in the VM hypothalamus?

- Hyperphagia

10. Which part of the hypothalamus controls heat gain?

- Posterior hypothalamus

11. Which part of the hypothalamus controls heat loss?

- Anterior hypothalamus

12. Which part of the hypothalamus controls the thirst center?

- Lateral hypothalamus

13. Which part of the hypothalamus controls the release of ADH (Vasopressin)?

- Supraoptic nucleus

14. What hormone is released from the hypothalamus to switch between sleep and arousal?

- Orexin Hormone

15. What part of the brain controls the circadian rhythm?

- Suprachiasmatic nucleus

16. What is the role of the suprachiasmatic nucleus in control of circadian rhythm?

- Periodic, diurnal changes in many body physiological processes according to Light-Dark cycles

17. What is the role of the hypothalamus in emotion and sexual behavior?

- Through its role with the Limbic system, stimulate sexual drive

18. What is the part of hypothalamus associated with Sexual drive?

- Most anterior and most posterior parts of the hypothalamus

19. What happens when there is a lesion in the anterior hypothalamus?

- Loss of sexual interest

20. What is the Papez Circuit coarse?

- Hippocampus → Fornix → Mammillary bodies → Anterior Nuclei of thalamus → Cingulate gyrus → Hippocampus

21. How is the Papez Circuit characterized?

- Long after discharge
- reverberating circuits
- has a major role in emotions



Notes

3

Section



Histology



1. Mention the cutaneous unencapsulated receptors?

- Free nerve endings
- Peritrichial nerve endings
- Merkel's tactile disc

2. Mention the cutaneous encapsulated receptors?

- Meissner's corpuscles
- Pacinian corpuscles
- Ruffini corpuscles
- Krause's end bulbs

3. Describe the microscopic structure of free nerve endings.

- The afferent nerve fibers arise from the subepidermal plexus of nerves then lose their myelin and neurolemmal sheath. The naked nerve fibers penetrate the epidermis, run between the epidermal cells and terminate by small swellings or knobs upon the surface of cells. In cornea, they reach to the superficial layers.

4. Describe the microscopic structure of peritrichial nerve endings.

- The afferent nerve fibers arise from the subepidermal plexus of nerves then lose their sheath and form a basket like arborization around the connective sheath of the hair follicle, just below its sebaceous gland.

5. Describe the microscopic structure of Merckle's cell.

- **L.M.:** each cell has lobulated nucleus and cytoplasmic secretory granules close to the disc.
- **E.M.:** the Merkel's cells have cytoplasmic projections that extend between the keratinocytes, to which they are joined by desmosomes.

6. Mention the site of free nerve ending in unencapsulated receptors?

- The widely distributed in the epidermis of the skin, cornea, mucous membrane of visceral organs, serous membrane, muscle, deep fascia. They are more numerous in the skin.

7. In the muscle spindles the intrafusal fibers are characterized by?

- Small diameter
- Indistinct striations
- Non-contractile central portions

8. Compare between nuclear bag & nuclear chain fibers

	Nuclear bag	Nuclear chain
Shape	Thick and long.	Thin and short.
Number	Fewer in number.	Numerous in number.
The nuclei	Central part is expanded and contain numerous nuclei.	Central part is narrow and contains a chain of nuclei.

9. Compare between Meissner's corpuscles and Pacinian corpuscles?(according to the site, microscopic structure and function)

	Meissner's corpuscles	Pacinian corpuscles
Site	Dermal papillae of skin of palms, fingertips, soles, lips, nipples, eyelids and external genitalia.	Dermis and hypodermis Pancreas, mesentery, breasts, external genitalia, and the wall of the urinary bladder Periosteum, tendons, and joint capsule
Microscopic structure	The corpuscles lie in the dermal papilla with its long axis perpendicular to the surface. It's formed of transversely arranged stacks of flattened modified schwann cells. The unmyelinated afferent fiber branch and run in a spiral course between the flattened schwann cells.	The corpuscle is supplied by a myelinated afferent nerve fiber that loses its myelin sheath inside the corpuscle. It runs axially and terminates by a club like termination. The afferent fiber is surrounded by numerous concentric layers of schwann cells
Function	Respond to crude touch and two tactile discriminations. They are mechanoreceptors.	Respond to deep pressure and vibration. They are mechanoreceptors.

10. Compare between muscle spindle and Golgi tendon organ.

	Muscle spindle	Golgi tendon organs
Site	In skeletal muscle, numerous in the muscles of the extremities and muscles of fine control	At the junction of the muscles with their tendon
Microscopic structure	Fusiform in shape, longitudinally arranged in parallel. Each spindle contains 2-12 intrafusal muscle fibers.	These are encapsulated receptors. Each has a thick lamellated capsule. Primary afferent fibers enter the capsule at its equatorial region dividing into many unmyelinated branches that surround the collagen fibrils

11. Complete:

- krause's end bulb are present in the conjunctiva, tongue, mouth, external genitalia, where they are known as **genital corpuscles**.
- The adequate stimulus for the muscle spindle is change in muscle **length**, while the adequate stimulus for the Golgi tendons is change in muscle **tension**.

Chapter 2 ?

Spinal cord internal structure

1. Complete the following: comparison between different segments of spinal cord

	Cervical	Thoracic	Lumbar
Posterior horn	Thin	Thin	Thick
Anterior horn	Thick	Thin	Thick

2. Mention the nerve cells in the posterior horn and intermediate grey of the spinal cord.

- **Golgi type 1** : include cells with long axons that form long ascending tracts
- **Golgi type 2**: include cells with short axons that terminate in the gray matter near their origin

3. Mention the 5 nuclei in the lateral nuclear group in the cervical and lumbar segments.

- Anterior nucleus
- Antro-lateral nucleus
- Postro-lateral nucleus
- Retropostro-lateral nucleus
- Central nucleus

4. List the short tract present in the white matter of the spinal cord?

- Comma shaped tract
- Septomarginal tract
- Lissauer's tract
- Fasciculus proprius tract

5. Compare between lateral corticospinal tract and ventral corticospinal tract?

Lateral corticospinal tract	Ventral corticospinal tract
Descends in the lateral column.	Descends in the anterior column.
Extend to the most caudal part of the cord and progressively diminishes in size.	Extend only to the upper thoracic spinal cord levels and innervates the axial muscles of the upper extremities and neck.

6. Compare between the comma shaped tract and Septomarginal tract:

Comma shaped tract	Septomarginal tract
Lies between gracile and cuneate tracts. Present only in the upper half of the spinal cord.	Lies close to the dorsomedian septum in the lower half of the spinal cord between gracile tracts.
It represents the descending fibers from cuneate tract	It represents the descending fibers from the gracile tract.

7. Complete:

- Histologically, the main difference between the gray matter and white matter of the spinal cord is the presence of **nerve** Cells and **Unmyelinated** fibers in the gray matter.
- all sensory nuclei of the posterior horn are found in all level of spinal cord, except clarcke's nucleus which extend from the **8th cervical** segment to the **3rd lumbar segment**.
- Medial longitudinal bundles: it descends in the **ventral white column**, where it forms a well-defined bundle in the **cervical region** only.

1. What are the fibers of the molecular layer in cerebellar cortex?

- Dendritic arborizations and axons of its own cell
- The dendrites of cells in the deeper layers (purkinje cells and Golgi type 2 cells)
- Climbing fibers, the afferent to the cerebellum
- Axons of granule cells
- Recurrent axons of purkinje cells

2. The cerebellar islands (glomerulus) are formed of?

- Mossy fiber terminal
- Dendrites of numerous granule cells
- Axons of Golgi type 2

3. Compare between climbing fibers and mossy fibers.

Climbing fibers	Mossy fibers
They are olivocerebellar fibers.	Include all afferent fibers except the olivocerebellar fibers.
Pass from the white matter through the granular and purkinje cell layer to reach dendrites of purkinje cells.	These fibers enter the white matter of the cerebellum, then they enter the granular layer of the cortex.

4. Compare between outer stellate & basket cell.

Outer stellate	Basket cell
Found in the superficial part of the molecular layer.	Found in the deep part of the molecular layer.
Have small cell bodies with short dendrites located in the same layer.	Have dendrites in the same layer.
Their axon passes parallel to the surface of the cerebellum, to synapse with the dendrites of purkinje cells.	Their axon form basket-like terminal arborizations around purkinje cell bodies, where a single cell makes synapse with many cells up to 10 cells.

5. Compare between granule cell and Golgi type 2 cell

Granule cells	Golgi type 2 cell
Small cells have large rounded nuclei and thin rim of cytoplasm with Nissl's bodies.	Found in the upper parts of the granular layer.
Dendrites show claw like ending that terminate in the granulus.	Their dendrites extend to all layers of the cortex.
Their axon ascends to the molecular layer.	Their axon end within the cerebellar glomeruli.

6. Mention the 4 deep cerebellar nuclei

- Dentate nucleus
- Globose nucleus
- Emboliform nucleus
- Fastigial nucleus

7. Mention the histological appearance of purkinje cell layer:

- The middle layer of the cerebellar cortex. It is formed of purkinje cell, arranged uniformly in a single plane along the upper margin of granular layer.
- Has a large pyriform flask shaped cell body, vesicular central nucleus, coarse cytoplasmic Nissl's granules.
- Their dendrites form an elaborate fan-like flattened dendritic tree, at right angles to the long axis of folia. These dendrites branch to give rise to 1^{ry}, 2^{ry}, & 3^{ry} branches. Both 1^{ry} & 2^{ry} are smooth, while 3^{ry} branches show spines.
- Purkinje cell axons are myelinated passing through the granular layer & white matter to synapse with deep cerebellar nuclei.

8. Complete:

the **purkinje cells** Are the key cells in the cerebellar cortex. They receive 2 types of afferent fibers: **climbing fibers** Directly from the olivary nuclei and **mossy fibers** Which include all other cerebellar afferents which relay first on granule cells.



1. Mention the cerebellar relay nuclei in the medulla?

- Accessory cuneate nucleus
- Arcuate nucleus
- Nuclei of reticular formation
- Olivary nuclear complex

2. Describe the site of the vestibular nuclei.

- In the floor of 4th ventricle:
 - The superior and lateral nuclei are at the inferior level of pons.
 - The inferior and medial nuclei are at the level of open medulla.

3. What are the alternating layers of white and gray matter in superior colliculus?

- Stratum zonale
- Stratum cinereum
- Stratum opticum
- Stratum lemnisci

4. What are the connections of the motor nucleus of oculomotor nerve?

- Bilateral corticonuclear
- Vestibular nuclei
- Abducent nerve nucleus

5. What are the decussation of the tegmentum of the midbrain?

- At superior level:
 - Dorsal tegmental decussation
 - Ventral tegmental decussation
- At inferior level:
 - Decussation of the superior cerebellar peduncle
 - Decussation of the trochlear nerve

6. What is the structure of the red nucleus?

- Caudal part: which is magnocellular (motor function).
- Rostral part: which is parvicellular (motor learning).

7. Mention 3 of the ascending and descending tracts in the medulla.

- Medial lemniscus
- Spinothalamic tracts
- Medial longitudinal fasciculus

8. Complete:

- The fibers of oculomotor nerve (III) pass ventrally traversing the red nucleus and pass in the interpeduncular fossa.
- Parts of midbrain: tectum, tegmentum, Crus cerebri.

1. List the cells of the cerebral cortex:

- Pyramidal cells
- Granule cells
- Fusiform cells
- Cells of martinotti
- Horizontal cells of cajal

2. List layers of the cerebral cortex:

- Molecular layer
- External granular layer
- External pyramidal layer
- Inner granular layer
- Inner pyramidal layer
- Pleomorphic or multiform layer

3. Compare between the radial and horizontal fibers of the cerebral cortex:

- **Radial fibers:** afferent fibers of thalamo-cortical radiation efferent fibers of corticospinal, corticoreticular, and corticopontine fibers.
- **Horizontal fibers:** arranged in tangential fibers running in the first layer, and outer and inner band of billarger are terminal branches of afferent fibers that bend horizontally to synapse with cortical cells.

4. Mention the difference between the motor cortex and sensory cortex:

Motor cortex	Sensory cortex
Thickest zone	Thinner than cortex
Small and medium-sized pyramidal cell in layer III	Pyramidal cells are ill defined
Giant pyramidal cell in fifth layer	Contain closely packed granule cells
Granule cells are rare or absent	

5. What are the types of astrocytes?

- Fibrous astrocytes: found in white matter and have long, thin processes.
- Protoplasmic astrocytes: found in grey matter and have short extensive processes.

6. What is the blood brain barrier formed of?

- Endothelial cells of the continuous blood capillaries which are tightly sealed together by tight junction (zonula occludens).
- The continuous basement membrane of the endothelial cells of the capillaries.
- The pericytes of the blood capillaries with perivascular macrophages (microglia).
- The limiting layer of perivascular astrocytic feet (foot processes of astrocytes) that completely encircles the basal lamina of the capillaries.

7. The blood brain barrier lacks in:

- Endocrine regions as posterior pituitary and pineal body to release hormones.
- Region of hypothalamus, where plasma components are monitored.
- The choroid plexus, where CSF is produced.

8. What is blood-CSF barrier formed of?

- Endothelial cells of the fenestrated blood capillaries.
- The basement membrane of the endothelial capillaries.
- The connective tissue on which the ependymal cells rest.
- The ependymal cells that are connected apically by junctional complex.

9. What is brain-CSF formed of?

- Ependymal cells that are connected apically by junctional complex.
- The connective tissue on which the ependymal cells rest.
- The foot processes of the astrocytes.

10. What is the function of the blood-brain barrier?

- It protects the brain tissue against toxic drugs and other harmful substance as bacteria that may present in the blood.

11. Mention the types of neuroglia in the CNS(central glia) and in the PNS?

- **In the CNS:**
 - Oligodendrocytes
 - Astrocytes
 - Microglia
 - Ependymal cells
- **In the PNS:**
 - Schwann cells
 - Satellite cells

12. Mention the functions of oligodendroglia?

- They form the myelin sheath of nerve fibers in CNS. Each cell can form myelin sheath of several internodal segments in the same axon or different axons.
- Support the neurons.
- Insulation of nerve impulses.

13. Compare between the light and dark cells of oligodendrocytes.

- **Light cells:** Predominant in CNS of fetus and neonate. They have large euchromatic nuclei, pale cytoplasm capable of cell division.
- **Dark cells:** Predominant in CNS of adults. They have heterochromatic nuclei and dense cytoplasm.

14. Mention the microscopical structure of ependymal cells lining the third and fourth ventricle.

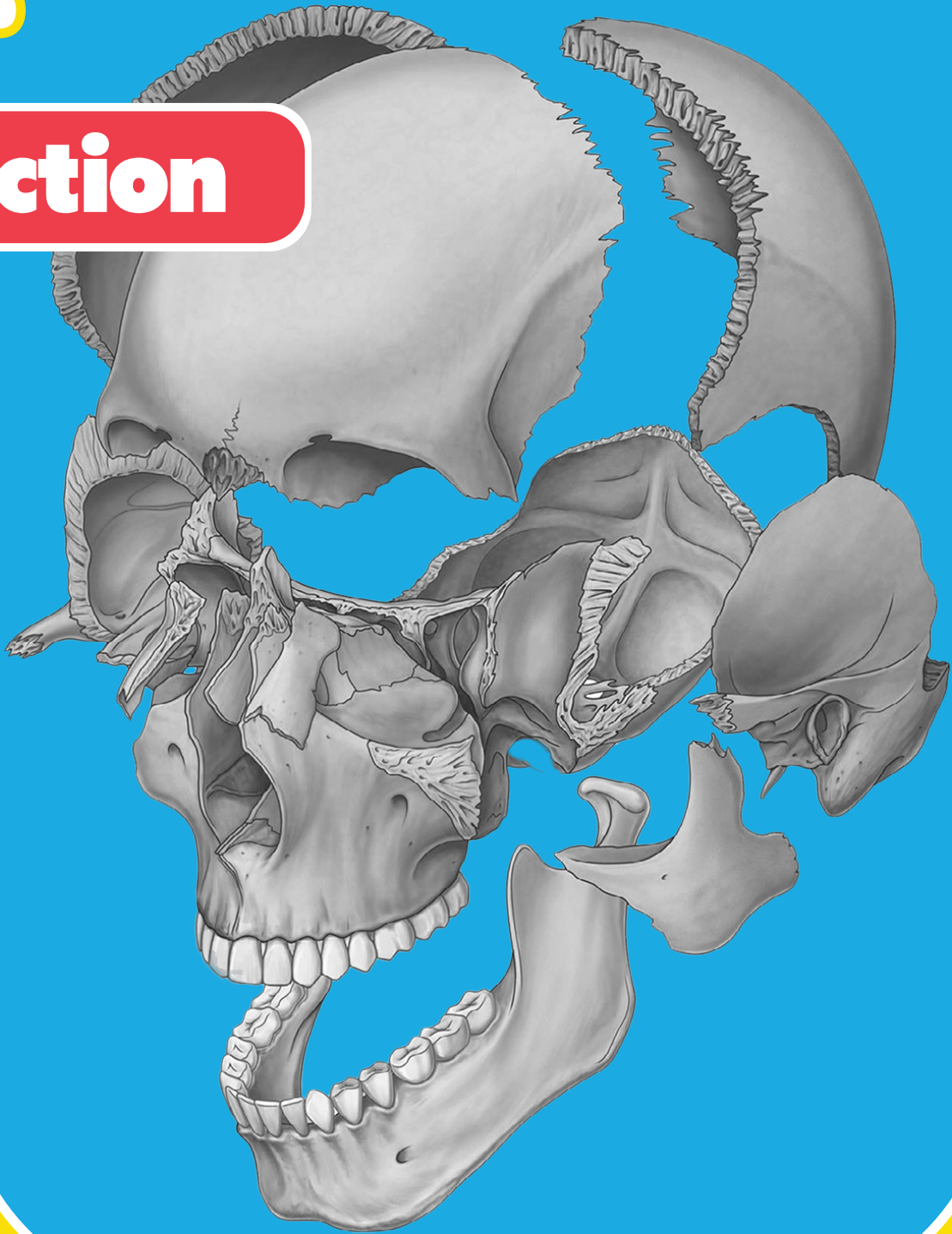
- Apical surfaces have numerous microvilli but rarely ciliated, numerous mitochondria, flattened basal nuclei, joined by tight junction and desmosomes.

15. Complete:

- The inner surface of dura mater is lined by **Simple squamous epithelium**.
- White and grey matter contain a pool of **adult progenitor cells**, which can differentiate to oligodendrocytes that might remyelinate axons in pathologically demyelinated regions.
- There are three barriers in the brain: **the blood brain barriers, the blood CSF barrier, the brain CSF barrier**.
- Outer band of baillarger is present in **inner granular layer** while inner band of baillarger is present in **inner pyramidal layer**.

4

Section



Biochemistry

1. What is the main fuel of the brain?

- Glucose

2. Name the types of brain fuels.

- Glucose
- Ketone bodies
- Mono carboxylic acid (Lactate, Pyruvate & Acetate)

3. What are the sources of brain fuels during the well-fed state?

- Dietary carbohydrates giving glucose

4. What are the sources of brain fuels during the fasting state?

- Liver glycogenolysis (early fasting)
- gluconeogenesis in the liver (late fasting)

5. What does ketolysis produce?

- Acetyl CoA that is oxidized by the TCA Cycle to produce energy

6. How is lactic acid produced by astrocytes shuttled to neurons?

- Through Monocarboxylic transporter (MCT) 1 & 2

7. What enzyme is absent in neurons that results in slow glycolysis?

- 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase-3 (Pfkfb3)

8. What does NADPH from the HMP pathway maintain?

- Glutathione in its reduced form

9. What controls astrocytes glycogen content?

- Neurotransmitters
- Neurohormonal agents
- Local energy state

10. What stimulates glycogenesis in astrocytes?

- Decreased neuron activity (sleeping or anesthesia)

11. What stimulates glycogenolysis in astrocytes?

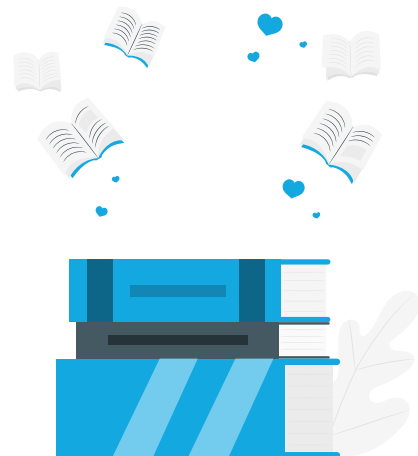
- Increased neuron activity (stimulated by adrenaline & serotonin)

12. What is the importance of brain energy?

- Maintenance of ion gradients involved in signaling processes
- Uptake and recycling of neurotransmitters
- Brain function, including learning, memory, emotions, and cognitive support

13. What happens when brain energy metabolism is impaired?

- Perturbation of neurological functions
- loss of consciousness, and coma



1. What are neurotransmitters?

- Endogenous substances that act as chemical messengers used by neurons to communicate

2. What is the rate limiting enzyme in catecholamine synthesis?

- Tyrosine hydroxylase

3. What is the fate of catecholamines?

- Active uptake by sympathetic nerve endings
- Enzymatic hydrolysis

4. What enzymes cause the oxidative deamination and inactivation of catecholamines?

- Monoamine oxidase (MAO)

5. What is the primary end product of dopamine inactivation?

- Homovanillic acid (HVA)

6. What is the primary end product of epinephrine and norepinephrine enzymatic inactivation?

- Vanillylmandelic acid (VMA)

7. What is the neurodegenerative movement disorder characterized by insufficient dopamine production?

- Parkinson Disease

8. What is the immediate metabolic precursor of dopamine that can cross the blood-brain barrier?

- L-DOPA

9. What is the inhibitor of dopa-decarboxylase used in the treatment of Parkinson Disease?

- Carbidopa

10. What tumor is characterized by excessive production of catecholamines? And what is the treatment of it?

- Pheochromocytoma
- α -methyl tyrosine

11. What enzyme is inhibited by α -methyl tyrosine in the treatment of pheochromocytoma?

- Tyrosine hydroxylase

12. What is the neurotransmitter produced in the CNS, gastrointestinal tract & platelets?

- Serotonin (5-HT)

13. What is the end product of serotonin degradation?

- 5-hydroxyindolacetic acid (5-HIAA)

14. What tumor is characterized by increased production of serotonin and decreased production of nicotinic acid?

- Carcinoid tumor (Argentaffinoma)

15. What is the hormone involved in regulating sleep patterns and follows a circadian rhythm? And how it is formed?

- Melatonin
- From serotonin in pineal body by acetylation and methylation

16. What happens in Vitamin B6 deficiency?

- Reduction in GABA levels with signs of hyperactivity and convulsions

17. What factors can lead to melatonin deficiency?

- Cigarettes
- Antidepressants
- Caffeine
- β -blocker
- Frequent stress

18. How is histamine formed?

- Histamine is formed by decarboxylation of histidine in a reaction requiring PLP

19. What is the role of histamine?

- Histamine mediates a wide range of cellular responses, including allergic and inflammatory reactions, gastric acid secretion
- Act as a Neurotransmitter in CNS
- Powerful vasodilator

20. What is the function of histamine in the brain?

- Histamine acts as a Neurotransmitter in parts of the brain by mediating arousal.

21. What are the neurotransmitters that act as vasodilators?

- Histamine
- Nitric oxide

22. How is nitric oxide synthesized?

- Nitric oxide is synthesized from L-arginine in a reaction catalyzed by nitric oxide synthase enzyme

23. What are the functions of nitric oxide?

- Vasodilatation in cerebral, coronary & muscle arteries
- Relaxation of GIT & urinary tract
- Stimulate release of hormones from hypothalamus
- Produced from macrophage, monocytes & neutrophils to kill microorganisms
- Pharmacological functions: Nitroglycerin - treat angina; Inhalation of NO - treat pulmonary hypertension & pulmonary edema without decrease systemic blood pressure

24. Which neurotransmitter is derived from tyrosine?

- Catecholamines (Dopamine, Norepinephrine, Epinephrine)

25. Where are dopamine and norepinephrine synthesized?

- In the brain and function as neurotransmitters

26. Epinephrine is synthesized in..... and the primary end product in the adrenal medulla of newborns is

- In the adrenal medulla
- Norepinephrine

27. What happens to epinephrine with advancing age?

- Dramatically raised, depending upon cortisol levels

28. What is histamine synthesized from? And what is the enzyme required for histamine synthesis?

- Histidine
- Histidine decarboxylase

29. The coenzyme required for histamine synthesis is..... and cells secrete it is.....

- PLP
- Mast cells

30. What are the major functions of histamine? (Name two)

- Mediating allergic and inflammatory reactions.
- gastric acid secretion.
- as a neurotransmitter in parts of the brain.

31. Is histamine a vasodilator? If true, what else? And if false, why?

- True
- Nitric oxide

32. How is nitric oxide synthesized? And What enzyme catalyzes the synthesis of nitric oxide?

- From L-arginine
- Nitric oxide synthase

33. What is glycine synthesized from? And what is the enzyme catalyzes its formation?

- Serine
- Serine hydroxy-methyltransferase

34. How is glycine removed from the synapse?

- Reuptake through high-affinity transporter

35. What are the biogenic amines neurotransmitters?

- Acetylcholine
- Monoamines

36. What are the effects of acetylcholine?

- Mostly excitatory effects

37. What is Myasthenia gravis?

- A chronic autoimmune neuromuscular disease causing weakness in the skeletal muscles

38. What is the primary end product of catecholamine synthesis in the adrenal medulla?

- Norepinephrine (NE)

39. What happens when there are elevated levels of excitatory amino acids?

- Neurotoxicity resulting in neuronal death or injury

40. What is the principal inhibitory neurotransmitter?

- GABA (Gamma Aminobutyric Acid)

41. What is GABA derived from?

- Glutamic acid

42. What are the classification of neurotransmitters based on functions?

- Excitatory
- Inhibitory

43. What are examples of excitatory amino acid neurotransmitters?

- Glutamate
- Aspartate

44. What is the principal excitatory neurotransmitter?

- Glutamate

45. Where are neurotransmitters stored?

- In secretory vesicles (synaptic vesicles) before release into the synaptic cleft

1. What are the substrates for glutamate transaminase?

- α -ketoglutarate
- L-aspartate

2. How is ammonia utilized in the conversion of glutamate to glutamine?

- Ammonia is used by glutamine synthetase to convert glutamate to glutamine

3. Which amino acids can produce both glucose and ketone bodies?

- Tyrosine
- phenylalanine
- tryptophan
- threonine
- isoleucine

4. What is the main pathway of deamination for amino acids?

- L-glutamate dehydrogenase

5. Which enzyme is responsible for the oxidative deamination of D-AAs and glycine?

- D-AA oxidase flavoprotein

6. What is the function of D-AA oxidase flavoprotein?

- Conversion of D-amino acids to the α -keto acid will remove the asymmetry of the α -carbon atom.

7. What is the product of the transdeamination of glutamine?

- Glutamate.

8. What is the role of glutamate dehydrogenase?

- Conversion of glutamate to α -ketoglutarate.

9. What is the major pathway of oxidative deamination?

- L-glutamate dehydrogenase

10. Which organs mainly carry out oxidative deamination?

- Liver, kidney
- heart, brain, and muscle

11. What coenzyme is required for L-glutamate dehydrogenase?

- NAD
- NADP

12. True or False: NAD^+ is used in oxidative deamination and NADP^+ in reductive amination.

- True

13. What stimulates L-glutamate dehydrogenase?

- ADP
- GDP

14. What inhibits L-glutamate dehydrogenase?

- ATP
- GTP
- NADH

15. What is the coenzyme of the enzyme D-AA oxidase ?

- FAD

16. In which organs is D-AA oxidase found?

- Liver and kidney

17. What is the enzyme responsible for non-oxidative deamination of glutamine?

- Deamidases

18. What substrates does Ketoglutarate-D-AA oxidase flavoprotein act on?

- D-AAs
- glycine.

19. What happens to glutamine in the liver?

- Cleaved by glutaminase enzyme to produce ammonia

20. What amino acids can't go through transamination?

- Lysine
- Threonine
- Proline
- Hydroxyproline

21. What enzyme converts pyruvate to alanine in muscles?

- ALT enzyme

22. What is the function of L-AA oxidase?

- Oxidative deamination of L-amino acids.

23. What is the process of non-oxidative deamination for glutamine and asparagine?

- Deamidases release NH_3 which enters urea cycle in the liver, or passes in urine in the kidney.

24. Where does NH_3 released by deamidases in the liver go?

- Urea cycle

25. Where does NH_3 released by deamidases in the kidney go?

- Urine

26. What are the sources of ammonia NH_3 ?

- Catabolism of amino acids
- catabolism of purine and pyrimidine
- glutaminase enzyme
- Catabolism of amines
- intestinal bacteria

27. What is the synthetic fates of ammonia?

- Synthesis of non essential amino acids
- Synthesis of glutamine

28. What is the major mechanism of ammonia excretion in the liver?

- Synthesis of urea



1. How is hepatic encephalopathy diagnosed?

- Exclusion of other causes of encephalopathy
- identification of the precipitating cause
- rapid response to empiric treatment

2. How does lactulose help in the reduction of ammonia production?

- Inhibits intestinal ammonia production by reducing colonic bacterial load and acidifying the gut lumen

3. What is the function of neomycin in the treatment of hepatic encephalopathy?

- Kills intestinal bacteria

4. Name three substances used to increase ammonia clearance.

- Sodium benzoate
- sodium phenylbutyrate
- sodium phenylacetate

5. How does sodium benzoate contribute to ammonia clearance?

- Interacts with glycine to form Hippurate, which is then excreted in the urine

6. What is the role of L-ornithine-L-aspartate in hepatic encephalopathy treatment?

- Stimulates the urea cycle, leading to the loss of ammonia

7. What is the effect of hyperammonia on brain function?

- Decrease in Krebs' cycle and ATP production
- decrease in GABA (neurotransmitter)
- interference with electric activity in the brain
- cerebral edema, and increased serotonin production

8. What are the causes of hyperammonemia?

- Acquired hyperammonemia due to liver diseases
- congenital hyperammonemia due to deficiencies in urea cycle enzymes, membrane-associated ornithine transporter, or N-acetyl glutamate synthase

9. What is the first step of the urea cycle?

- Carbamoyl phosphate synthesis

10. What is the enzyme responsible for carbamoyl phosphate synthesis?

- Carbamoyl phosphate synthetase I

11. What is the enzyme catalyzing the second step of the urea cycle?

- Ornithine transcarbamoylase

12. What is the second step of the urea cycle? And its enzyme?

- Citrulline synthesis

13. What is the third step of the urea cycle? And its enzyme?

- Argininosuccinate synthesis
- Argininosuccinate synthetase

14. What is the fourth step of the urea cycle? And its enzyme?

- Cleavage of argininosuccinate
- Argininosuccinate lyase

15. What is the final step of the urea cycle? And its enzyme?

- Cleavage of arginine
- Arginase

16. What is the rate-limiting enzyme of the urea cycle?

- Carbamoyl Phosphate Synthetase I (CPS I)

17. Synthesis of urea occurs in **the liver** and excretion by **the kidney****18. Which enzyme is present in the cytosol and used in pyrimidine synthesis? And what is its other mitochondrial enzyme used in urea cycle?**

- CPS II
- CPS I

19. How many ATP molecules and high energy bonds are utilized in the urea cycle?

- 3 ATP
- 4 High energy bonds

20. What is the main fate of urea?

- Excreted in urine

21. What activates carbamoyl phosphate synthetase I (CPS I)?

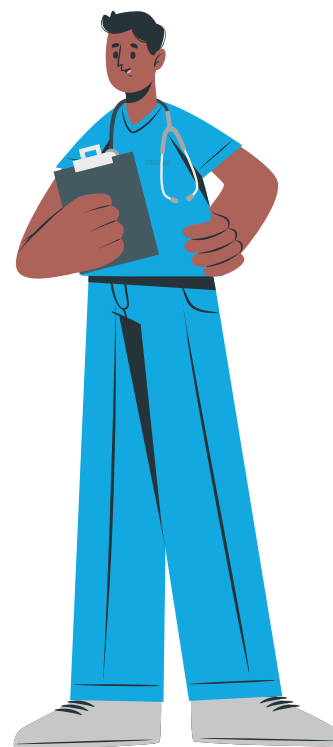
- N-acetylglutamate

22. What is the relationship between the urea cycle and the citric acid cycle?

- CO_2 needed for urea synthesis is mostly produced from the citric acid cycle
- Ammonia is derived from glutamic acid by glutamate dehydrogenase enzyme that give α -ketoglutarate. "intermediate in krebs' Cycle"
- Fumarate produced by urea cycle can be converted into oxaloacetic acid by citric acid cycle.
- Aspartic acid used in urea cycle is formed from oxaloacetate by transamination with AST (GOT) enzyme.
- ATP needed for urea cycle is derived from citric acid cycle.

23. What is hepatic encephalopathy?

- A syndrome observed in patients with cirrhosis
- It's characterized by personality changes, intellectual impairment, and a depressed level of consciousness



Chapter 5 ? Inborn error of amino acids metabolism

1. What is phenylketonuria?

- Phenylketonuria is a condition caused by a deficiency of
 - phenylalanine hydroxylase
 - dihydrobiopterin reductase
 - dihydrobiopterin biosynthesis.

2. What are the clinical pictures of phenylketonuria?

- Mental retardation
- Depigmentation
- Urine with a bad mousy odor.

3. How is phenylketonuria treated?

- A diet low in phenylalanine, supplementation of tyrosine
- Giving DOPA and serotonin if there is dihydrobiopterin deficiency.

4. What is maple syrup urine disease?

- Maple syrup urine disease is a condition resulting from a deficiency of the dehydrogenase complex involved in branched-chain amino acid catabolism.

5. What are the effects of maple syrup urine disease?

- Maple syrup urine disease causes mental retardation and may lead to early death.
- The urine has a sweet smell-like maple syrup or burned sugar.

6. What is the treatment for maple syrup urine disease?

- The treatment for maple syrup urine disease is a diet low in branched-chain amino acids.

7. What are the effects of homocystinuria?

- Deep vein thrombosis
- Stroke
- Atherosclerosis.
- Marfan-like habitus
- Mental retardation
- Joint contractures

8. What is alkaptonuria?

- Alkaptonuria is a condition characterized by
 - Dark urine
 - Ochronosis
 - Arthritis.



1. What is Vitamin B9 also known as? And can it be synthesized by humans?

- Folic Acid
- No

2. What is the chemical composition of Vitamin B9?

- Consists of pteridine ring + poly glutamic amino acid 2-7 + PABA

3. What is the best source of Vitamin B9?

- Liver
- Green leafy vegetables

4. Can Vitamin B9 be synthesized by intestinal bacteria?

- Yes

5. What is the active form of Vitamin B9?

- Tetra hydro folate (THF)

6. What is the daily requirement of Vitamin B9?

- 400 microgram / day
- 800 microgram / day in pregnancy and lactation

7. What is the function of Vitamin B9?

- Carrier of one carbon unit for:
 - Synthesis of DNA (Purine & thymine)
 - Synthesis of serine & glycine
 - Synthesis of Methionine, choline

8. Which processes does Vitamin B9 assist in DNA synthesis?

- Synthesis of Purine & Thymine

9. Which amino acids does Vitamin B9 help in synthesizing?

- Serine, Glycine, Methionine, Choline

10. What causes folate deficiency?

- Increase rate of cell division (e.g., pregnancy, anemia)
- poor absorption
- administration of methotrexate
- Vit. B12 deficiency

11. What are the effects of folate deficiency?

- Defect cell proliferation
- megaloblastic anemia

12. What are the antagonists of folic acid?

- PABA analogs (sulfa-Drugs)
- folic acid analogs (methotrexate)

13. Which CNS diseases are related to Vitamin B9 deficiency?

- Homocystinemia,
- Megaloblastic (macrocytic) anemia,
- Neural tube defects in fetus

14. Which vitamin forms the coenzyme thiamine pyrophosphate (TPP)?

- Thiamin (Vitamin B1)

15. What are the plant sources of thiamin (Vitamin B1)?

- Yeast
- green vegetables
- whole (unrefined) grains

16. What are the animal sources of thiamin (Vitamin B1)?

- Organ meat (liver, kidney), eggs, milk

17. What is the function of thiamin (Vitamin B1)?

- Forms the coenzyme thiamine pyrophosphate (TPP)

18. What is the reaction requiring thiamine pyrophosphate (TPP)?

- Oxidative decarboxylation of α -keto acids
- Transketolase
- Decarboxylation of pyruvate in yeast

19. What is the deficiency disease associated with thiamin (Vitamin B1)?

- Beriberi
- Wernicke Korsakoff syndrome

20. True or False: Thiamin deficiency decreases the activity of pyruvate and ketoglutarate dehydrogenases.

- True

21. What is the deficiency disease associated with thiamin (Vitamin B1) in alcoholics?

- Wernicke-Korsakoff syndrome

22. What is the biochemical test for thiamine deficiency?

- Blood thiamine
- Erythrocyte transketolase activity (ETKA)
- Thiamine pyrophosphate effect (TPPE)
- Urinary thiamine levels
- Blood pyruvate and lactate levels

23. True or False: PDH deficiency leads to lactic acidosis.

- True

24. What are the symptoms of PDH deficiency?

- Lactic acidosis
- Hypotonia, poor feeding
- Lethargy
- Seizure
- neurological abnormalities

25. What is the structure of niacin (Vitamin B3)?

- Contains a pyridine ring

26. What are the sources of niacin (Vitamin B3)?

- Same sources as vitamin B1
- synthesized in the body from tryptophan

27. What are the functions of niacin (Vitamin B3)?

- Forms coenzymes NAD^+ and NADP^+

28. True or False: Nicotinic acid is used therapeutically for lowering plasma cholesterol and triglyceride levels.

- True

29. What are the symptoms of niacin (Vitamin B3) deficiency?

- Dermatitis
- Diarrhea
- Dementia

30. What is the antagonist that prevents folic acid synthesis by bacteria?

- PABA analogs (sulfa-Drugs)

31. What enzyme do PABA analogs inhibit?

- Dihydropteroate synthetase

32. What enzyme is competitively inhibited by folic acid analogs (methotrexate)?

- Dihydrofolate reductase

33. What antibiotic inhibits folate reductase and acts on bacteria only?

- Trimethoprim

1. What are the three ways in which one carbon units can be carried and transferred?

- Tetrahydrofolate (THF)
- S-adenosyl methionine (SAM)
- Biotin

2. Which form of one carbon unit is the most reduced?

- Methyl (CH_3)

3. How can glycine and serine interconvert?

- Through serine hydroxymethyltransferase (glycine hydroxymethyltransferase)

4. How is formyl-THF synthesized?

- Direct by formyl-THF synthetase enzyme from free formate" need ATP"
- From methenyl by hydrolysis.
- From tryptophan when converted to Formyl Kyneurinine to Kyneurinine.

5. How is methylene-THF synthesized?

- From β carbon of serine when converted to glycine.
- By glycine cleavage enzyme

6. How is methyl-THF synthesized?

- Irreversible reduction of methylene by methylene THF reductase (NAD linked)

7. How is form-imino-THF synthesized?

- From hisidine by histidinase

8. How is methenyl-THF synthesized?

- Oxidation from methylene THF by NADP dehydrogensae
- From hisidine by histidinase "give formimino that converted to (CH) by deamination"

9. What is the fate of methylene THF?

- Converted to methenyl (CH) by oxidation
- Converted to methyl (CH) by reduction.
- Synthesis of Thymidylate by Thymidylate Synthetase
- Synthesis of serine from glycine

10. What is the fate of methyl THF?

- Vit.B accept CH_3 From CH_3 -THF to form methyl cobalamine which transfers it to homocysteine to form methionine

11. What is the fate of formo-imino THF?

- Converted to (CH) by deamination.

12. What is the fate of methenyl THF?

- Can be converted to N10 formyl THF by hydrolysis.

13. What is the fate of formyl THF?

- C2 & C8 in purine.

14. What is the role of vitamin B12 in one carbon unit metabolism?

- Accepts CH_3 from CH_3 -THF to form methylcobalamin, which transfers it to homocysteine to form methionine

15. What is the consequence of vitamin B12 deficiency in one carbon unit metabolism?

- Folate trap (methyl trap) due to the failure of transfer of CH_3 from folic acid to B12

Notes

تواصل معنا لتحصل علي
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